CHARLES UKE

DIGITAL FORENSIC EXAMINATION REPORT

SBT-DF205 – CASE STUDY 2
iOS DEVICE EXAMINATION

Report detail	Value
Case reference	SBT-DF205-C2
Examiner	Charles Uke
Examination environment	Kali Linux running in VMware
Case working directory	~/SBT-DF205-Case2
Device identified	Apple iPhone SE
Operating system	iOS 13.4.1
Examination period	3–4 November 2026
Report date	4 November 2026

CONFIDENTIAL – FORENSIC CASE DOCUMENT

1. Executive Summary

This examination involved the forensic analysis of supplied iOS evidence associated with an Apple iPhone SE running iOS 13.4.1. The examination was conducted in a Kali Linux virtual environment using preserved working copies of the supplied evidence.

The work included evidence acquisition and inventory, SHA-256 hash verification, archive integrity testing, TAR extraction, examination of device information, and analysis of the Apple CoreDuet knowledgeC.db database. The examination also considered Photos asset metadata and selected application activity records.

The CoreDuet database passed SQLite integrity checking and contained 16 tables. Its examination identified activity metadata associated with Apple Mail, Photos, Maps, Imgur, Safari, Notes, Health, Bluetooth, and charging-related system activity.

The Photos asset metadata output contained 59 records: 58 still-image records and one video record. A consolidated UTC timeline was created containing 17 entries. The timeline passed structural validation, with four columns, no malformed rows, and no blank data rows.

The findings are limited to what the supplied artefacts directly establish. Application activity metadata does not necessarily prove that a person deliberately performed a particular action. Encoded identifiers, system-generated predictions, and incomplete activity records were not treated as proof of user identity, message content, physical location, or intent.

2. Objectives

1. Preserve and document the supplied evidence.
2. Verify the integrity of the supplied ZIP and TAR evidence using SHA-256.
3. Test the ZIP archive for integrity and extract the supplied TAR archive.
4. Identify relevant iOS device information and system version.
5. Examine the CoreDuet database structure and relationships.
6. Analyse selected application activity and associated metadata.
7. Examine available Photos asset metadata.
8. Convert relevant timestamps to UTC and consolidate findings into a master timeline.
9. Document findings, limitations, and the commands used during examination.

3. Evidence Received and Working Structure

The case evidence directory was organised into original, working, output, notes, report, and screenshots subdirectories.

Evidence item	Recorded size
ios_13_4_1.zip	9,629,286,492 bytes
13-4-1.tar	17,223,813,120 bytes

The ZIP archive was copied into the case evidence directory. The TAR archive was obtained from the ZIP archive and retained in the original evidence directory. Working extraction was performed under working/filesystem.

The CoreDuet database examined was working/knowledgec/knowledgeC.db. The Photos asset metadata output was output/media/Photos_Asset_Metadata_UTC.txt. The master timeline was output/timeline/Case2_Master_Timeline_UTC.csv.

4. Evidence Handling and Integrity

4.1 Acquisition and preservation

The supplied ZIP archive was located in the Downloads directory and copied into the case's original evidence directory. SHA-256 values were calculated and compared with the recorded values. The ZIP archive integrity test completed without reported errors.

The TAR archive was extracted from the ZIP archive and subsequently extracted into the working filesystem directory. During TAR extraction, SCHILY.fflags metadata warnings were displayed. The extraction completed, but it is not described as warning-free.

The TAR archive's SHA-256 value was rechecked and matched its recorded baseline.

4.2 SHA-256 values

Evidence item	SHA-256
ios_13_4_1.zip	761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551
13-4-1.tar	991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb
knowledgeC.db	dfdf31fc6f3399777ad51ae28a345aac3365c53d09be79253ba70aceb1e1cd4b

The ZIP and TAR hash checks matched their recorded baselines. The CoreDuet database hash was recorded during examination. A source-to-working-copy hash comparison for knowledgeC.db is not claimed in this report.

4.3 Handling record

- Environment inspection and available disk capacity.
- Virtual disk expansion and filesystem expansion.
- Evidence archive acquisition and copying.
- ZIP SHA-256 verification.
- ZIP integrity testing.
- TAR extraction.
- TAR SHA-256 re-verification.

The handling log is retained as output/Handling_Log.csv.

5. Examination Environment and Tools

The examination was conducted in Kali Linux within a VMware virtual machine. Recorded tool versions included SQLite 3.46.1, ExifTool 13.25, and Python 3.14.7.

The principal tools and utilities included sha256sum for SHA-256 verification; unzip for archive integrity testing and archive content extraction; tar for filesystem extraction; sqlite3 for CoreDuet database examination; Python CSV functionality for timeline validation; and standard Linux shell utilities for inventory and output inspection.

6. Device Identification

The device information records identify the examined device as an Apple iPhone SE running iOS 13.4.1. Device and operating-system identification was based on the supplied device property and system version records. The report does not infer additional hardware identifiers or device ownership beyond what the records establish.

7. CoreDuet Database Examination

7.1 Database integrity and structure

The CoreDuet database examined was working/knowledgec/knowledgeC.db. The SQLite integrity check returned 'ok'. The database contained 16 tables. Relevant tables included ZOBJECT, ZSOURCE, ZCUSTOMMETADATA, ZSTRUCTUREDMETADATA, Z_4EVENT, and Z_PRIMARYKEY.

The Z_PRIMARYKEY entity mapping identified Event as an entity derived from Object. The database also contained separate entities for CustomMetadata and StructuredMetadata.

7.2 Validated metadata relationships

The validated CustomMetadata-to-Event relationship used the Z_4EVENT linking table:

```
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
```

The direct ZCUSTOMMETADATA object fields were not used as a substitute for this linking table because the examined custom metadata rows had NULL values in those object fields.

The validated StructuredMetadata-to-ZOBJECT relationship was ZOBJECT.ZSTRUCTUREDMETADATA = ZSTRUCTUREDMETADATA.Z_PK. The examination identified 21,652 objects, of which 7,600 were flagged as having structured metadata. Relationship validation identified 7,600 matching metadata records and zero unmatched records.

Exploratory joins that did not conform to the validated database relationships were excluded from the final findings.

7.3 Apple Mail activity

The examination identified 48 Apple Mail mailbox activity metadata records linked to 48 activity objects. Four distinct encoded item identifiers were present. The most frequent identifier accounted for 43 of the 48 records.

The earliest observed activity was 22 March 2020, 01:13:20 UTC. The latest observed activity was 16 April 2020, 12:09:55 UTC.

The identifiers were encoded values. The examined metadata did not establish mailbox names, message contents, senders, recipients, or a specific message action.

7.4 Photos-related CoreDuet metadata

Photos-related metadata was associated with streams including /photos/engagement/0To1Seconds, /photos/engagement/1To2Seconds, /photos/engagement/2To3Seconds, /photos/deletes/all, /photos/deletes/recent, /photos/share/all, /photos/share/airdrop, and /photos/edit/all.

The examined records included 58 links in the 0To1Seconds engagement stream, 18 in 1To2Seconds, and 12 in 2To3Seconds. Observed Photos-related CoreDuet timestamps ranged from 5 April 2020, 11:33:55 UTC to 16 April 2020, 15:56:10 UTC.

Associated object values included PFLDatasetSample-v1.0. These records appear consistent with Photos-related dataset or prediction samples. They do not independently establish that a user deliberately viewed, deleted, shared, or edited a particular photograph.

7.5 Bluetooth metadata

The examination identified 57 event objects associated with the /bluetooth/isConnected stream. The metadata classified the connected device as an Apple audio device.

Product ID	Observed period (UTC)	Event links
8194	27–30 March 2020	26
8206	31 March–13 April 2020	31

The overall observed event period was 27 March 2020, 18:32:42 UTC to 13 April 2020, 16:28:24 UTC. The product ID values were not independently mapped to a specific hardware model. The records do not establish device ownership or the identity of the person using the device.

7.6 Charging related metadata

The examination identified 16 batteryLevel records associated with the /charging/smartTopOffCheckpoint stream. Observed battery-level values included 75%, 77%, 79%, 80%, 82%, 83%, 86%, 98%, 99%, and 100%. The records were observed between 8 and 16 April 2020.

The timeTillTopOff metadata contained 16 numeric values ranging from 900 to 7,924.4. The interpretation of these values as seconds is based on the field name and scale and is not independently verified. The checkpoint metadata contained values 0, 2, 5, and 7.

The pluggedIn metadata contained two records, with values 1 and 0, associated with the /dasd/batterytemperature stream. These values do not establish exact charging start and stop times and should not be treated as a complete charging history.

7.7 Apple Maps activity

The examination identified 100 Apple Maps structured metadata records linked to 100 activity objects and 77 distinct encoded item identifiers. The observed period was 1 April 2020, 17:46:20 UTC to 15 April 2020, 16:17:25 UTC.

The examined records did not contain non-empty title or content-description values. The encoded identifiers did not reveal destinations, search terms, route details, or physical movement.

7.8 Other application activity

The examination identified application activity associated with Imgur, Apple Notes, Safari, and Apple Health. Observed activity included Imgur feed, profile, and post activity; Apple Notes edit-note activity; Safari browsing and bookmark activity; and Apple Health For You and data-view activity.

The activity summary included 16 Notes edit-note records, 12 Imgur feed records, six Imgur profile records, six Imgur post records, six Safari browsing records, and six Safari bookmark records. The examined summary metadata did not establish the content of Notes, the URLs visited in Safari, the contents of Imgur posts, or specific Health measurements.

8. Photos Asset Metadata Examination

The output file output/media/Photos_Asset_Metadata_UTC.txt contained 59 asset records.

Category	Count
Total asset records	59
Still images	58
Videos	1
Favourite records	0
Hidden records	0
PNG filenames with public.jpeg type	2

The asset creation timestamps ranged from 22 March 2020, 13:05:24 UTC to 16 April 2020, 15:54:58 UTC. The modification timestamps ranged from 27 March 2020, 00:38:01 UTC to 16 April 2020, 15:55:43 UTC.

The video record was identified as IMG_0037.MOV, with a duration of 7.74 seconds and dimensions of 720 × 1280 pixels. Its recorded creation time was 12 April 2020, 01:57:09 UTC, and its modification time was 12 April 2020, 08:59:55 UTC.

Two filenames ending in .PNG had the reported media type public.jpeg. This is a filename/type inconsistency and does not, by itself, establish corruption or deliberate alteration. Asset metadata timestamps are not independently proof of when a photograph was captured or viewed by a person.

9. Consolidated UTC Timeline

The consolidated timeline is stored at output/timeline/Case2_Master_Timeline_UTC.csv. It contains 17 data entries and four columns: Date_or_Period_UTC, Artifact, Verified_Observation, and Interpretation_Limit.

The timeline includes the observed periods for Apple Mail, Photos asset metadata, Imgur, Bluetooth, Notes, Safari, Maps, Health, Photos-related CoreDuet metadata, and charging-related metadata.

9.1 Timeline validation

Validation item	Result
Header columns	4
Data entries	17
Rows with incorrect column counts	0
Blank data rows	0
CSV structure	PASS

The validation confirms the CSV's structure. It does not independently prove that every possible event on the device has been recovered.

10. Key Findings

10. The supplied device records identify an Apple iPhone SE running iOS 13.4.1.
11. The ZIP archive integrity test completed without reported errors.
12. The ZIP and TAR SHA-256 values matched their recorded baselines.
13. The TAR extraction completed with SCHILY.fflags metadata warnings.
14. The CoreDuet database passed SQLite integrity checking and contained 16 tables.
15. The validated structured metadata relationship identified 7,600 matching records and no unmatched records.
16. CoreDuet contained activity metadata associated with several Apple and third-party applications.
17. The Photos asset metadata output contained 59 records, consisting of 58 still images and one video.
18. Bluetooth and charging-related metadata provided system activity information but did not establish user identity or exact physical actions.
19. A 17-entry UTC master timeline was created and passed structural validation.

11. Limitations

- The examination was limited to the supplied evidence and available extracted artefacts.
- TAR extraction displayed metadata warnings. The extraction completed, but the warnings are documented.
- A matching archive hash confirms consistency with the recorded baseline; it does not independently prove the provenance of the original evidence before acquisition.
- The CoreDuet database hash was recorded, but a source-to-working-copy hash comparison for that database is not claimed here.
- Encoded identifiers were not decoded into mailbox names, account identities, or physical destinations.
- Application activity records do not necessarily prove deliberate user interaction.
- Photos-related dataset or prediction records do not independently prove intentional photo viewing, deletion, sharing, or editing.
- Bluetooth product IDs were not independently mapped to exact hardware models.
- Charging-related metadata does not provide a complete battery or charging history.

- The timeline contains selected verified findings, not necessarily every event represented in the original device evidence.
- No conclusion is made about the identity, intent, or conduct of a particular person solely from these system-generated artefacts.

12. Conclusion

The examination successfully processed the supplied iOS evidence and documented its integrity checks, extraction, and selected artefact analysis.

The CoreDuet database was structurally sound according to SQLite's integrity check. Its records provided evidence of application-associated activity and system metadata across the observed period. Photos asset metadata provided a record of 59 assets, including one video. The consolidated UTC timeline brought together the principal verified observations.

The findings should be interpreted within the limitations of the supplied logical evidence and the nature of the examined system-generated metadata. The artefacts support the presence of recorded application and system activity, but they do not independently establish the identity or intent of the person using the device.

13. Evidence and Output Files

File	Purpose
output/Handling_Log.csv	Evidence handling record
output/hashes/ZIP_SHA256.txt	ZIP hash record
output/hashes/TAR_SHA256.txt	TAR hash record
output/hashes/TAR_SHA256_Recheck.txt	TAR hash re-verification
output/hashes/CallHistory_Verification.txt	Call-history verification output
output/media/Photos_Asset_Metadata_UTC.txt	Photos asset metadata
output/timeline/Case2_Master_Timeline_UTC.csv	Consolidated UTC timeline
working/knowledgec/knowledgeC.db	Examined CoreDuet database

14. Screenshot Register

Screenshot	Description
SS119	CoreDuet structured metadata field population
SS120	CoreDuet structured metadata value distributions
SS121	CoreDuet event structured metadata schema
SS122	CoreDuet event metadata link counts
SS123	CoreDuet metadata link frequency
SS124	High-frequency metadata details
SS125	Apple Mail metadata records
SS127	CoreDuet ZOBJECT schema
SS129	CoreDuet entity name mapping
SS130	Correct custom metadata event relationship
SS131	Custom metadata event relationship summary
SS132	Photos metadata stream distribution
SS133	Photos identifier event details
SS134	Photos event object field details

SS135	Photos event UTC timestamp range
SS136	PluggedIn event details
SS137	Battery-level event details
SS138	TimeTillTopOff event details
SS139	Charging checkpoint event details
SS140	Bluetooth metadata event details
SS141	Bluetooth event timeline summary
SS142	StructuredMetadata schema
SS143	StructuredMetadata relationship validation
SS144	Apple Mail structured metadata object links
SS145	Apple Mail activity summary
SS146	Apple Mail item identifier distribution
SS147	Apple Maps structured metadata details
SS148	Apple Maps activity summary
SS149	Selected application activity details
SS150	Selected application activity summary
SS152	Original archive and CoreDuet SHA-256
SS153	CoreDuet database integrity check
SS154	Evidence handling log
SS155	Photos asset metadata UTC timeline
SS156	Master timeline creation
SS157	Master timeline validation

APPENDIX A – COMMANDS USED AND RECOVERED

A1. Case directory navigation

```
cd ~/SBT-DF205-Case2
```

A2. Archive hash verification

```
sha256sum original/13-4-1.tar original/ios_13_4_1.zip
```

A3. ZIP archive integrity test

```
unzip -t original/ios_13_4_1.zip
```

A4. TAR archive extraction from ZIP

```
The TAR archive was extracted from the ZIP archive using unzip -p and redirected into the original evidence
directory. The precise original command line was not retained in the available record.
```

A5. TAR extraction

```
tar --numeric-owner --same-permissions -xpf original/13-4-1.tar -C working/filesystem
```

A6. TAR hash re-verification

```
sha256sum original/13-4-1.tar
```

A7. CoreDuet database hash and size

```
sha256sum working/knowledgec/knowledgeC.db
ls -lh working/knowledgec/knowledgeC.db
```

A8. SQLite integrity check

```
sqlite3 working/knowledgec/knowledgeC.db "PRAGMA integrity_check;"
```

A9. CoreDuet table count

```
sqlite3 working/knowledgec/knowledgeC.db "
SELECT COUNT(*) AS Table_Count
FROM sqlite_master
WHERE type = 'table';
"
```

A10. CoreDuet schema and entity examination

```
PRAGMA table_info(Z_4EVENT);
PRAGMA table_info(ZCUSTOMMETADATA);

SELECT Z_ENT, Z_NAME, Z_SUPER, Z_MAX
FROM Z_PRIMARYKEY
ORDER BY Z_ENT;
```

A11. CustomMetadata-to-Event relationship (reconstructed SQL)

```
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZNAME AS Metadata_Name,
    O.Z_PK AS Event_Object_ID,
    O.ZSTARTDATE AS Raw_Start_Date
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT;
```

A12. StructuredMetadata-to-ZOBJECT relationship (reconstructed SQL)

```
SELECT COUNT(*) AS Valid_Metadata_Matches
FROM ZOBJECT O
JOIN ZSTRUCTUREDMETADATA S
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK;
```

A13. Apple Mail structured metadata (reconstructed SQL)

```
SELECT
    S.Z_PK AS Metadata_ID,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    O.Z_PK AS Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZVALUESTRING AS Object_Value
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE =
      'com.apple.mail.mailbox';
```

A14. Photos CoreDuet timestamp conversion

```
datetime(ZSTARTDATE + 978307200, 'unixepoch')
```

A15. Timeline validation – entry count and CSV columns

```
awk 'END {print NR-1}' output/timeline/Case2_Master_Timeline_UTC.csv
awk -F',' 'NR==1 {print NF}' output/timeline/Case2_Master_Timeline_UTC.csv
head -n 6 output/timeline/Case2_Master_Timeline_UTC.csv
tail -n 5 output/timeline/Case2_Master_Timeline_UTC.csv
```

A16. Full CSV structural validation

```
python3 - <<'PY'
import csv

path = "output/timeline/Case2_Master_Timeline_UTC.csv"

with open(path, newline="", encoding="utf-8") as f:
    rows = list(csv.reader(f))

expected = 4
bad_rows = [
    (number, len(row))
    for number, row in enumerate(rows, start=1)
    if len(row) != expected
]

print("===== FULL CSV VALIDATION =====")
print("Header columns:", len(rows[0]))
print("Data entries:", len(rows) - 1)
print("Rows with incorrect column counts:", len(bad_rows))
print("Blank data rows:", sum(
    not any(cell.strip() for cell in row)
    for row in rows[1:]
))

if bad_rows:
    print("Problem rows:", bad_rows)
else:
    print("CSV structure: PASS")
PY
```

APPENDIX B – REPORT COMPLETION CHECKLIST

- ☐ Insert the relevant screenshots at the appropriate report sections.
- ☐ Confirm the screenshot filenames against the screenshot register.
- ☐ Ensure SS126 and SS128 are excluded.
- ☐ Confirm the final report date and examiner details.
- ☐ Review all observations and interpretation limitations.
- ☐ Preserve the original evidence and working-copy structure.
- ☐ Save the final report in the case report directory.
- ☐ Generate a SHA-256 hash of the final report after completion.

END OF REPORT

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ BLUETOOTH EVENT TIMELINE SUMMARY ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.ZNAME AS Metadata_Name,
    C.ZINTEGERVALUE AS Metadata_Value,
    COUNT(*) AS Linked_Event_Count,
    COUNT(DISTINCT O.Z_PK) AS Distinct_Event_Objects,
    MIN(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Earliest_UTC,
    MAX(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Latest_UTC
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME IN (
    '_DKBluetoothMetadataKey-productID',
    '_DKBluetoothMetadataKey-isAppleAudioDevice'
)
GROUP BY C.ZNAME, C.ZINTEGERVALUE
ORDER BY C.ZNAME, C.ZINTEGERVALUE;
"
═══ BLUETOOTH EVENT TIMELINE SUMMARY ═══
Metadata_Name                              Metadata_Value  Linked_Event_Count  Distinct_Event_Objects  Earliest_UTC         Latest_UTC
_DKBluetoothMetadataKey-isAppleAudioDevice  1               57                  57                      2020-03-27 18:32:42  2020-04-13 16:28:24
_DKBluetoothMetadataKey-productID           8194            26                  26                      2020-03-27 18:32:42  2020-03-30 13:14:06
_DKBluetoothMetadataKey-productID           8206            31                  31                      2020-03-31 18:47:44  2020-04-13 16:28:24

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ ZOBJECT STRUCTURED METADATA RELATIONSHIP ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(ZOBJECT);
"
═══ ZOBJECT STRUCTURED METADATA RELATIONSHIP ═══
```

cid	name	type	notnull	dflt_value	pk
0	Z_PK	INTEGER	0		1
1	Z_ENT	INTEGER	0		0
2	Z_OPT	INTEGER	0		0
3	ZUUIDHASH	INTEGER	0		0
4	ZEVENT	INTEGER	0		0
5	ZSOURCE	INTEGER	0		0
6	ZCATEGORYTYPE	INTEGER	0		0
7	ZINTEGERVALUE	INTEGER	0		0
8	ZCOMPATIBILITYVERSION	INTEGER	0		0
9	ZENDDAYOFWEEK	INTEGER	0		0
10	ZENDSECONDOFDAY	INTEGER	0		0
11	ZHASCUSTOMMETADATA	INTEGER	0		0
12	ZHASSTRUCTUREDMETADATA	INTEGER	0		0
13	ZSECONDSFROMGMT	INTEGER	0		0
14	ZSHOULDSYNC	INTEGER	0		0
15	ZSTARTDAYOFWEEK	INTEGER	0		0
16	ZSTARTSECONDOFDAY	INTEGER	0		0
17	ZVALUECLASS	INTEGER	0		0
18	ZVALUEINTEGER	INTEGER	0		0
19	ZVALUETYPECODE	INTEGER	0		0
20	ZSTRUCTUREDMETADATA	INTEGER	0		0
21	ZVALUE	INTEGER	0		0
22	Z9_VALUE	INTEGER	0		0
23	ZIDENTIFIERTYPE	INTEGER	0		0
24	ZQUANTITYTYPE	INTEGER	0		0
25	ZOBJECT	INTEGER	0		0
26	Z9_OBJECT	INTEGER	0		0

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ STRUCTURED METADATA RELATIONSHIP VALIDATION ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(*) AS Total_Objects,
    SUM(CASE WHEN ZHASSTRUCTUREDMETADATA = 1 THEN 1 ELSE 0 END) AS Flagged_Objects,
    SUM(CASE WHEN ZSTRUCTUREDMETADATA IS NOT NULL THEN 1 ELSE 0 END) AS Objects_With_Metadata_ID,
    SUM(CASE WHEN S.Z_PK IS NOT NULL THEN 1 ELSE 0 END) AS Valid_Metadata_Matches,
    SUM(CASE WHEN ZSTRUCTUREDMETADATA IS NOT NULL AND S.Z_PK IS NULL THEN 1 ELSE 0 END) AS Unmatched_Metadata_IDs
FROM ZOBJECT O
LEFT JOIN ZSTRUCTUREDMETADATA S
    ON S.Z_PK = O.ZSTRUCTUREDMETADATA;
"
═══ STRUCTURED METADATA RELATIONSHIP VALIDATION ═══
Total_Objects  Flagged_Objects  Objects_With_Metadata_ID  Valid_Metadata_Matches  Unmatched_Metadata_IDs
─────────────  ──────────────  ────────────────────────  ──────────────────────  ──────────────────────
21652          7600            7600                      7600                    0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ APPLE MAIL STRUCTURED METADATA OBJECT LINKS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    S.Z_PK AS Metadata_ID,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER AS Item_Identifier,
    O.Z_PK AS Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZVALUESTRING AS Object_Value,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC,
    datetime(O.ZENDDATE + 978307200, 'unixepoch') AS End_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.mail.mailbox'
ORDER BY O.ZSTARTDATE, O.Z_PK
LIMIT 50;
"
═══ APPLE MAIL STRUCTURED METADATA OBJECT LINKS ═══
```

Metadata_ID	Activity_Type	Item_Identifier	Object_ID	Stream_Name	Object_Value	Start_UTC	End_UTC
87	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	270	/app/activity	com.apple.mobilemail	2020-03-22 01:13:20	2020-03-22 01:13:20
132	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	423	/app/activity	com.apple.mobilemail	2020-03-22 01:58:15	2020-03-22 01:58:15
134	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	426	/app/activity	com.apple.mobilemail	2020-03-22 01:58:55	2020-03-22 01:58:55
136	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	429	/app/activity	com.apple.mobilemail	2020-03-22 01:59:05	2020-03-22 01:59:05
191	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	800	/app/activity	com.apple.mobilemail	2020-03-22 13:46:35	2020-03-22 13:46:35
422	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	2047	/app/activity	com.apple.mobilemail	2020-03-24 01:01:10	2020-03-24 01:01:10
441	com.apple.mail.mailbox	c3iZ/Ahb5EswfNhFKCspQQ==	2084	/app/activity	com.apple.mobilemail	2020-03-24 01:06:30	2020-03-24 01:06:30
444	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	2087	/app/activity	com.apple.mobilemail	2020-03-24 01:06:30	2020-03-24 01:06:30
446	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	2090	/app/activity	com.apple.mobilemail	2020-03-24 01:07:05	2020-03-24 01:07:05
455	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	2104	/app/activity	com.apple.mobilemail	2020-03-24 01:09:00	2020-03-24 01:09:00
458	com.apple.mail.mailbox	c3iZ/Ahb5EswfNhFKCspQQ==	2110	/app/activity	com.apple.mobilemail	2020-03-24 01:10:30	2020-03-24 01:10:30
460	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	2112	/app/activity	com.apple.mobilemail	2020-03-24 01:10:30	2020-03-24 01:10:30
472	com.apple.mail.mailbox	7Xyg9VDd17nXS4pqHXqfuw==	2135	/app/activity	com.apple.mobilemail	2020-03-24 01:19:10	2020-03-24 01:19:10

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== APPLE MAIL STRUCTURED METADATA SUMMARY ====="

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(DISTINCT S.Z_PK) AS Distinct_Mail_Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Linked_Activity_Objects,
    COUNT(DISTINCT S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER) AS Distinct_Item_Identifiers,
    MIN(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Earliest_UTC,
    MAX(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Latest_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.mail.mailbox';
"
===== APPLE MAIL STRUCTURED METADATA SUMMARY =====
Distinct_Mail_Metadata_Records  Linked_Activity_Objects  Distinct_Item_Identifiers  Earliest_UTC         Latest_UTC
------------------------------  ----------------------  -------------------------  -------------------  -------------------
48                              48                      4                          2020-03-22 01:13:20  2020-04-16 12:09:55

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ APPLE MAIL ITEM IDENTIFIER DISTRIBUTION ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER AS Item_Identifier,
    COUNT(DISTINCT S.Z_PK) AS Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Activity_Objects,
    MIN(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS First_Seen_UTC,
    MAX(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Last_Seen_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.mail.mailbox'
GROUP BY S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER
ORDER BY Metadata_Records DESC;
"
═══ APPLE MAIL ITEM IDENTIFIER DISTRIBUTION ═══
Item_Identifier          Metadata_Records  Activity_Objects  First_Seen_UTC       Last_Seen_UTC
───────────────────────  ────────────────  ────────────────  ───────────────────  ───────────────────
7Xyg9VDd17nXS4pqHXqfuw==  43                43                2020-03-22 01:13:20  2020-04-16 12:09:55
c3iZ/Ahb5EswfNhFKCspQQ==  3                 3                 2020-03-24 01:06:30  2020-03-24 01:34:15
zyVpCfLKlWVg6w8bs3BiVg==  1                 1                 2020-04-12 17:08:50  2020-04-12 17:08:50
z7LWjAx/1I/xkccl2XeUkQ==  1                 1                 2020-03-28 00:22:30  2020-03-28 00:22:30

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== APPLE MAPS STRUCTURED METADATA DETAILS ====="

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    S.Z_PK AS Metadata_ID,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__TITLE AS Title,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__CONTENTDESCRIPTION AS Content_Description,
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER AS Item_Identifier,
    O.Z_PK AS Object_ID,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.Maps'
ORDER BY O.ZSTARTDATE
LIMIT 30;
"
===== APPLE MAPS STRUCTURED METADATA DETAILS =====
```

Metadata_ID	Activity_Type	Title	Content_Description	Item_Identifier	Object_ID	Start_UTC
2767	com.apple.Maps			9Tr/9r5qVzcPyUVMOh6XDg==	9235	2020-04-01 17:46:20
2768	com.apple.Maps			bdIJ7WcYJRefISWQgSk0vQ==	9236	2020-04-01 17:46:20
2770	com.apple.Maps			Zh9h5j9CvV2KZwImtL1MIA==	9239	2020-04-01 17:46:50
2772	com.apple.Maps			bdIJ7WcYJRefISWQgSk0vQ==	9241	2020-04-01 17:46:50
2774	com.apple.Maps			9Tr/9r5qVzcPyUVMOh6XDg==	9243	2020-04-01 17:46:50
2776	com.apple.Maps			RXNkiEBlmZBV7OQuR3xoKg==	9245	2020-04-01 17:46:50
2784	com.apple.Maps			4UAfamWK8YeJ4nL7E9jk4w==	9268	2020-04-01 17:54:20
2785	com.apple.Maps			bdIJ7WcYJRefISWQgSk0vQ==	9269	2020-04-01 17:54:20
2793	com.apple.Maps			bdIJ7WcYJRefISWQgSk0vQ==	9282	2020-04-01 17:58:20
3991	com.apple.Maps			qvKLB22Txa8U152LGicpsg==	12572	2020-04-05 20:32:50
3993	com.apple.Maps			qvKLB22Txa8U152LGicpsg==	12578	2020-04-05 20:33:20
3995	com.apple.Maps			sQjZcTlzwmfL4Arnge566A==	12580	2020-04-05 20:33:20
3997	com.apple.Maps			sQjZcTlzwmfL4Arnge566A==	12582	2020-04-05 20:33:50
4002	com.apple.Maps			qsQadEE6bybghVXLzzt3hg==	12589	2020-04-05 20:35:20

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ STRUCTURED METADATA ACTIVITY TYPE DISTRIBUTION ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    COUNT(*) AS Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Linked_Objects,
    MIN(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Earliest_UTC,
    MAX(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Latest_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
GROUP BY S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE
ORDER BY Metadata_Records DESC;
"
═══ STRUCTURED METADATA ACTIVITY TYPE DISTRIBUTION ═══
```

Activity_Type	Metadata_Records	Linked_Objects	Earliest_UTC	Latest_UTC
	7286	7286	2020-03-21 21:38:56	2020-04-16 17:19:47
com.apple.Maps	100	100	2020-04-01 17:46:20	2020-04-15 16:17:25
com.apple.mail.mailbox	48	48	2020-03-22 01:13:20	2020-04-16 12:09:55
com.apple.mobileslideshow.album	31	31	2020-03-22 13:14:50	2020-04-16 16:03:25
com.apple.mobileslideshow.oneUp	20	20	2020-03-22 13:13:40	2020-04-16 16:02:50
com.apple.health.view.foryou	20	20	2020-04-03 18:23:35	2020-04-07 00:29:45
com.apple.notes.activity.edit-note	16	16	2020-03-28 00:37:25	2020-03-28 00:51:10
com.imgur.feed	12	12	2020-03-22 15:01:45	2020-04-16 12:07:30
com.apple.health.view.data	12	12	2020-04-04 15:47:40	2020-04-07 00:29:45
com.imgur.profile	6	6	2020-03-22 15:01:45	2020-03-24 01:03:30
com.imgur.post	6	6	2020-03-22 15:06:40	2020-03-22 15:14:10
com.apple.health.view.summary	6	6	2020-04-03 18:53:15	2020-04-04 15:47:40
com.apple.Safari.UserActivity.Bookmarks	6	6	2020-03-28 01:32:05	2020-03-28 01:40:05
NSUserActivityTypeBrowsingWeb	6	6	2020-03-28 01:07:00	2020-03-28 01:39:10
com.venmo.story	4	4	2020-03-25 00:35:10	2020-03-25 00:45:00
com.apple.clock.alarm	4	4	2020-04-12 13:23:40	2020-04-16 12:11:25
com.enchantedcloud.photovault.launch	3	3	2020-04-15 00:40:15	2020-04-15 00:42:20

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ APPLE MAPS ACTIVITY SUMMARY ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(DISTINCT S.Z_PK) AS Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Linked_Objects,
    COUNT(DISTINCT S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER) AS Distinct_Item_Identifiers,
    MIN(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Earliest_UTC,
    MAX(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Latest_UTC,
    COUNT(DISTINCT S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__TITLE) AS Distinct_Title_Values,
    COUNT(DISTINCT S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__CONTENTDESCRIPTION) AS Distinct_Description_Values
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.Maps';
"
═══ APPLE MAPS ACTIVITY SUMMARY ═══
Metadata_Records  Linked_Objects  Distinct_Item_Identifiers  Earliest_UTC         Latest_UTC          Distinct_Title_Values  Distinct_Description_Values
────────────────  ──────────────  ─────────────────────────  ───────────────────  ──────────────────  ─────────────────────  ───────────────────────────
100               100             77                         2020-04-01 17:46:20  2020-04-15 16:17:25  0                      0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== SELECTED APPLICATION ACTIVITY DETAILS ====="

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    O.Z_PK AS Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZVALUESTRING AS Object_Value,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC,
    datetime(O.ZENDDATE + 978307200, 'unixepoch') AS End_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE IN (
    'com.apple.notes.activity.edit-note',
    'com.apple.Safari.UserActivity.Bookmarks',
    'NSUserActivityTypeBrowsingWeb',
    'com.imgur.feed',
    'com.imgur.profile',
    'com.imgur.post',
    'com.apple.health.view.foryou',
    'com.apple.health.view.data'
)
ORDER BY O.ZSTARTDATE, O.Z_PK;
"
===== SELECTED APPLICATION ACTIVITY DETAILS =====
Activity_Type                    Object_ID  Stream_Name    Object_Value     Start_UTC            End_UTC
-------------------------------  ---------  -------------  ---------------  -------------------  -------------------
com.imgur.feed                   923        /app/activity  imgurmobile      2020-03-22 15:01:45  2020-03-22 15:01:45
com.imgur.profile                924        /app/activity  imgurmobile      2020-03-22 15:01:45  2020-03-22 15:01:45
com.imgur.post                   932        /app/activity  imgurmobile      2020-03-22 15:06:40  2020-03-22 15:06:40
com.imgur.feed                   933        /app/activity  imgurmobile      2020-03-22 15:07:10  2020-03-22 15:07:10
com.imgur.post                   934        /app/activity  imgurmobile      2020-03-22 15:07:10  2020-03-22 15:07:10
com.imgur.post                   936        /app/activity  imgurmobile      2020-03-22 15:08:45  2020-03-22 15:08:45
com.imgur.feed                   937        /app/activity  imgurmobile      2020-03-22 15:08:45  2020-03-22 15:08:45
```

```
└─$ cd ~/SBT-DF205-Case2

echo "═══ SELECTED APPLICATION ACTIVITY SUMMARY ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    COUNT(DISTINCT S.Z_PK) AS Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Linked_Objects,
    MIN(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Earliest_UTC,
    MAX(datetime(O.ZSTARTDATE + 978307200, 'unixepoch')) AS Latest_UTC
FROM ZSTRUCTUREDMETADATA S
JOIN ZOBJECT O
    ON O.ZSTRUCTUREDMETADATA = S.Z_PK
WHERE S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE IN (
    'com.apple.notes.activity.edit-note',
    'com.apple.Safari.UserActivity.Bookmarks',
    'NSUserActivityTypeBrowsingWeb',
    'com.imgur.feed',
    'com.imgur.profile',
    'com.imgur.post',
    'com.apple.health.view.foryou',
    'com.apple.health.view.data'
)
GROUP BY S.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE
ORDER BY Metadata_Records DESC;
"
═══ SELECTED APPLICATION ACTIVITY SUMMARY ═══
Activity_Type                          Metadata_Records  Linked_Objects  Earliest_UTC         Latest_UTC
─────────────────────────────────────  ────────────────  ──────────────  ───────────────────  ───────────────────
com.apple.health.view.foryou           20                20              2020-04-03 18:23:35  2020-04-07 00:29:45
com.apple.notes.activity.edit-note     16                16              2020-03-28 00:37:25  2020-03-28 00:51:10
com.imgur.feed                         12                12              2020-03-22 15:01:45  2020-04-16 12:07:30
com.apple.health.view.data             12                12              2020-04-04 15:47:40  2020-04-07 00:29:45
com.imgur.profile                      6                 6               2020-03-22 15:01:45  2020-03-24 01:03:30
com.imgur.post                         6                 6               2020-03-22 15:06:40  2020-03-22 15:14:10
com.apple.Safari.UserActivity.Bookmarks  6               6               2020-03-28 01:32:05  2020-03-28 01:40:05
NSUserActivityTypeBrowsingWeb          6                 6               2020-03-28 01:07:00  2020-03-28 01:39:10
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ ORIGINAL EVIDENCE FILES ═══"
find original -maxdepth 2 -type f -printf '%f (%s bytes)\n' | sort

echo
echo "═══ HASH MANIFEST FILES ═══"
find output/hashes -maxdepth 1 -type f -printf '%f (%s bytes)\n' | sort
═══ ORIGINAL EVIDENCE FILES ═══
13-4-1.tar (17223813120 bytes)
ios_13_4_1.zip (9629286492 bytes)

═══ HASH MANIFEST FILES ═══
Accounts3_Source_SHA256.txt (500 bytes)
Accounts3_Working_SHA256.txt (311 bytes)
Calendar_Source_SHA256.txt (260 bytes)
Calendar_Working_SHA256.txt (260 bytes)
CallHistory_Verification.txt (257 bytes)
MobileGestalt_SHA256.txt (239 bytes)
Safari_Source_SHA256.txt (239 bytes)
Safari_Working_SHA256.txt (239 bytes)
SMS_Attachment_SHA256.txt (2026 bytes)
SMS_Source_SHA256.txt (455 bytes)
SMS_Working_SHA256.txt (278 bytes)
TAR_SHA256_Recheck.txt (86 bytes)
TAR_SHA256.txt (86 bytes)
ZIP_SHA256.txt (90 bytes)

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ CURRENT ORIGINAL ARCHIVE HASHES ═══"
sha256sum original/13-4-1.tar original/ios_13_4_1.zip

echo
echo "═══ CURRENT COREDUET DATABASE HASH ═══"
sha256sum working/knowledgec/knowledgeC.db

echo
echo "═══ COREDUET DATABASE SIZE ═══"
ls -lh working/knowledgec/knowledgeC.db
═══ CURRENT ORIGINAL ARCHIVE HASHES ═══
991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb  original/13-4-1.tar
761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551  original/ios_13_4_1.zip

═══ CURRENT COREDUET DATABASE HASH ═══
dfdf31fc6f3399777ad51ae28a345aac3365c53d09be79253ba70aceb1e1cd4b  working/knowledgec/knowledgeC.db

═══ COREDUET DATABASE SIZE ═══
-rw-r--r-- 1 kali kali 14M Nov  4 18:26 working/knowledgec/knowledgeC.db

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET SQLITE INTEGRITY CHECK ═══"

sqlite3 working/knowledgec/knowledgeC.db "PRAGMA integrity_check;"

echo
echo "═══ COREDUET DATABASE TABLE COUNT ═══"

sqlite3 working/knowledgec/knowledgeC.db "
SELECT COUNT(*) AS Table_Count
FROM sqlite_master
WHERE type = 'table';
"
═══ COREDUET SQLITE INTEGRITY CHECK ═══
ok

═══ COREDUET DATABASE TABLE COUNT ═══
16

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

cat > output/timeline/Case2_Master_Timeline_UTC.csv <<'EOF'
Date or Period UTC,Artifact,Verified Observation,Interpretation Limit
2020-03-22 01:13:20 to 2020-04-16 12:09:55,CoreDuet Apple Mail,48 mailbox activity metadata records linked to 48 activity objects and 4 encoded identifiers,Mailbo
x names and message contents not established
2020-03-22 13:05:24 to 2020-04-16 15:54:58,Photos asset metadata,59 asset records with 58 still images and 1 video,Creation metadata does not independently prove
capture or viewing
2020-03-22 15:01:45 to 2020-04-16 12:07:30,CoreDuet Imgur,12 feed activity records across the observed period,Specific posts and content not established
2020-03-22 15:06:40 to 2020-03-22 15:14:10,CoreDuet Imgur,6 post activity records,Specific post content not established
2020-03-22 15:01:45 to 2020-03-24 01:03:30,CoreDuet Imgur,6 profile activity records,Profile identity and viewed content not established
2020-03-27 18:32:42 to 2020-04-13 16:28:24,CoreDuet Bluetooth,57 event objects classified with Apple audio device metadata,Device model and user identity not esta
blished
2020-03-27 18:32:42 to 2020-03-30 13:14:06,CoreDuet Bluetooth,Product ID value 8194 recorded,Exact device model not established
2020-03-31 18:47:44 to 2020-04-13 16:28:24,CoreDuet Bluetooth,Product ID value 8206 recorded,Exact device model not established
2020-03-28 00:37:25 to 2020-03-28 00:51:10,CoreDuet Notes,16 edit-note activity records,Note contents not established
2020-03-28 01:07:00 to 2020-03-28 01:39:10,CoreDuet Safari,6 web-browsing activity records,URLs and page contents not established
2020-03-28 01:32:05 to 2020-03-28 01:40:05,CoreDuet Safari,6 bookmark activity records,Bookmark titles and URLs not established
2020-04-01 17:46:20 to 2020-04-15 16:17:25,CoreDuet Apple Maps,100 activity records linked to 77 encoded item identifiers,Destinations routes and physical movemen
t not established
2020-04-03 18:23:35 to 2020-04-07 00:29:45,CoreDuet Health,20 For You view activity records,Health measurements and medical conclusions not established
2020-04-04 15:47:40 to 2020-04-07 00:29:45,CoreDuet Health,12 data view activity records,Specific health data values not established
2020-04-05 11:33:55 to 2020-04-16 15:56:10,CoreDuet Photos metadata,Photos-related dataset sample events linked to Photos streams,Model or prediction records do n
ot prove intentional user actions
2020-04-08 to 2020-04-16,CoreDuet Charging,16 battery-level metadata records on smartTopOffCheckpoint stream,Checkpoint values are system metadata and not a compl
ete battery history
2020-04-13 19:16:16 to 2020-04-16 17:07:12,CoreDuet Charging,pluggedIn metadata associated with batterytemperature stream,Values do not establish exact charging s
tart and stop times
EOF

echo "Master timeline created."
Master timeline created.

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

awk '
NR > 2 {
    total++
    if ($4 == 1) video++
    else if ($4 == 0) still++
    if ($12 == 1) favorites++
    if ($13 == 1) hidden++
    if ($2 ~ /\.PNG$/ && $3 == "public.jpeg") png_jpeg++
}
END {
    print "Total asset records:", total
    print "Still-image records:", still
    print "Video records:", video
    print "Favorite records:", favorites+0
    print "Hidden records:", hidden+0
    print "PNG filenames with public.jpeg type:", png_jpeg+0
}' output/media/Photos_Asset_Metadata_UTC.txt
Total asset records: 59
Still-image records: 58
Video records: 1
Favorite records: 0
Hidden records: 0
PNG filenames with public.jpeg type: 2

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ MASTER TIMELINE VALIDATION ═══"

echo "Total timeline entries:"
awk 'END {print NR-1}' output/timeline/Case2_Master_Timeline_UTC.csv

echo
echo "CSV column count:"
awk -F',' 'NR==1 {print NF}' output/timeline/Case2_Master_Timeline_UTC.csv

echo
echo "First five entries:"
head -n 6 output/timeline/Case2_Master_Timeline_UTC.csv

echo
echo "Last five entries:"
tail -n 5 output/timeline/Case2_Master_Timeline_UTC.csv
═══ MASTER TIMELINE VALIDATION ═══
Total timeline entries:
17

CSV column count:
4

First five entries:
Date_or_Period_UTC,Artifact,Verified_Observation,Interpretation_Limit
2020-03-22 01:13:20 to 2020-04-16 12:09:55,CoreDuet Apple Mail,48 mailbox activity metadata records linked to 48 activity objects and 4 encoded identifiers,Mailbo
x names and message contents not established
2020-03-22 13:05:24 to 2020-04-16 15:54:58,Photos asset metadata,59 asset records with 58 still images and 1 video,Creation metadata does not independently prove
capture or viewing
2020-03-22 15:01:45 to 2020-04-16 12:07:30,CoreDuet Imgur,12 feed activity records across the observed period,Specific posts and content not established
2020-03-22 15:06:40 to 2020-03-22 15:14:10,CoreDuet Imgur,6 post activity records,Specific post content not established
2020-03-22 15:01:45 to 2020-03-24 01:03:30,CoreDuet Imgur,6 profile activity records,Profile identity and viewed content not established

Last five entries:
2020-04-03 18:23:35 to 2020-04-07 00:29:45,CoreDuet Health,20 For You view activity records,Health measurements and medical conclusions not established
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET EVENT AND STRUCTURED METADATA SCHEMA ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(Z_4EVENT);

PRAGMA table_info(ZSTRUCTUREDMETADATA);
"
═══ COREDUET EVENT AND STRUCTURED METADATA SCHEMA ═══
cid  name               type     notnull  dflt_value  pk
---  -----------------  -------  -------  ----------  --
0    Z_4CUSTOMMETADATA  INTEGER  0                    1
1    Z_11EVENT          INTEGER  0                    2
cid  name                                                                  type     notnull  dflt_value  pk
---  ------------------------------------------------------------------------  -------  -------  ----------  --
0    Z_PK                                                                  INTEGER  0                    1
1    Z_ENT                                                                 INTEGER  0                    0
2    Z_OPT                                                                 INTEGER  0                    0
3    Z_CDPORTRAITMETADATAKEY__ALGORITHM                                    INTEGER  0                    0
4    Z_CDPORTRAITMETADATAKEY__ASSETVERSION                                 INTEGER  0                    0
5    Z_DKAPPINSTALLMETADATAKEY__ISINSTALL                                  INTEGER  0                    0
6    Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISELIGIBLEFORPREDICTION           INTEGER  0                    0
7    Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISPUBLICLYINDEXABLE               INTEGER  0                    0
8    Z_DKAPPLICATIONMETADATAKEY__PROCESSIDENTIFIER                         INTEGER  0                    0
9    Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON                               INTEGER  0                    0
10   Z_DKBLUETOOTHMETADATAKEY__DEVICETYPE                                  INTEGER  0                    0
11   Z_DKBULLETINBOARDMETADATAKEY__HASDATE                                 INTEGER  0                    0
12   Z_DKDEVICEBATTERYPERCENTAGEMETADATAKEY__FULLYCHARGED                  INTEGER  0                    0
13   Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE                         INTEGER  0                    0
14   Z_DKDIGITALHEALTHMETADATAKEY__USAGETYPE                               INTEGER  0                    0
15   Z_DKDISCOVERABILITYUSAGEMETADATAKEY__CONTENTTYPE                      INTEGER  0                    0
16   Z_DKDISCOVERABILITYUSAGEMETADATAKEY__INELIGIBLEREASON                 INTEGER  0                    0
17   Z_DKDISCOVERABILITYUSAGEMETADATAKEY__STATE                            INTEGER  0                    0
18   Z_DKGLANCELAUNCHMETADATA__DEVICEIDENTIFIER                            INTEGER  0                    0
19   Z_DKINTENTMETADATAKEY__DIRECTION                                      INTEGER  0                    0
20   Z_DKINTENTMETADATAKEY__DONATEDBYSIRI                                  INTEGER  0                    0
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET EVENT-METADATA LINK RECORD COUNTS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(*) AS Total_Link_Records,
    COUNT(DISTINCT Z_4CUSTOMMETADATA) AS Distinct_Metadata_IDs,
    COUNT(DISTINCT Z_11EVENT) AS Distinct_Event_IDs,
    SUM(CASE WHEN Z_4CUSTOMMETADATA IS NULL THEN 1 ELSE 0 END) AS Null_Metadata_References,
    SUM(CASE WHEN Z_11EVENT IS NULL THEN 1 ELSE 0 END) AS Null_Event_References
FROM Z_4EVENT;
"
═══ COREDUET EVENT-METADATA LINK RECORD COUNTS ═══
Total_Link_Records  Distinct_Metadata_IDs  Distinct_Event_IDs  Null_Metadata_References  Null_Event_References
──────────────────  ─────────────────────  ──────────────────  ────────────────────────  ─────────────────────
615                 128                    334                 0                         0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== COREDUET METADATA LINK FREQUENCY ====="

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    Z_4CUSTOMMETADATA AS Metadata_ID,
    COUNT(*) AS Linked_Event_Records,
    COUNT(DISTINCT Z_11EVENT) AS Distinct_Events
FROM Z_4EVENT
GROUP BY Z_4CUSTOMMETADATA
ORDER BY Linked_Event_Records DESC, Metadata_ID
LIMIT 20;
"
===== COREDUET METADATA LINK FREQUENCY =====
Metadata_ID  Linked_Event_Records  Distinct_Events
-----------  --------------------  ---------------
17           57                    57
150          45                    45
151          36                    36
38           31                    31
16           26                    26
14           24                    24
20           23                    23
41           21                    21
88           16                    16
93           16                    16
95           16                    16
92           12                    12
96           12                    12
135          12                    12
133          11                    11
139          10                    10
21           8                     8
23           8                     8
3            7                     7
140          7                     7

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== COREDUET HIGH-FREQUENCY METADATA RECORD DETAILS ====="

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    M.Z_PK AS Metadata_ID,
    L.Linked_Events,
    M.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    M.Z_DKAPPLICATIONACTIVITYMETADATAKEY__TITLE AS Activity_Title,
    M.Z_DKAPPLICATIONMETADATAKEY__PROCESSIDENTIFIER AS Process_Identifier,
    M.Z_DKAPPLICATIONACTIVITYMETADATAKEY__USERACTIVITYUUID AS Activity_UUID
FROM ZSTRUCTUREDMETADATA M
JOIN (
    SELECT
        Z_4CUSTOMMETADATA,
        COUNT(DISTINCT Z_11EVENT) AS Linked_Events
    FROM Z_4EVENT
    GROUP BY Z_4CUSTOMMETADATA
) L
ON M.Z_PK = L.Z_4CUSTOMMETADATA
ORDER BY L.Linked_Events DESC, M.Z_PK
LIMIT 20;
"
===== COREDUET HIGH-FREQUENCY METADATA RECORD DETAILS =====
Metadata_ID  Linked_Events  Activity_Type           Activity_Title  Process_Identifier  Activity_UUID
-----------  -------------  ----------------------  --------------  ------------------  ------------------------------------
151          36
38           31
92           12
3            7
136          5              com.apple.mail.mailbox                                      95EE02BB-7BD4-4890-80F4-B579FF0E7960
86           4
87           4              com.apple.mail.mailbox                                      95EE02BB-7BD4-4890-80F4-B579FF0E7960
91           4
101          4
102          4
104          4
```

```
    M.Z_DKAPPLICATIONACTIVITYMETADATAKEY__USERACTIVITYUUID AS Activity_UUID
FROM ZSTRUCTUREDMETADATA M
JOIN (
    SELECT
        Z_4CUSTOMMETADATA,
        COUNT(DISTINCT Z_11EVENT) AS Linked_Events
    FROM Z_4EVENT
    GROUP BY Z_4CUSTOMMETADATA
) L
ON M.Z_PK = L.Z_4CUSTOMMETADATA
ORDER BY L.Linked_Events DESC, M.Z_PK
LIMIT 20;
"
===== COREDUET HIGH-FREQUENCY METADATA RECORD DETAILS =====
Metadata_ID  Linked_Events  Activity_Type          Activity_Title  Process_Identifier  Activity_UUID
-----------  -------------  ---------------------  --------------  ------------------  ------------------------------------
151          36
38           31
92           12
3            7
136          5              com.apple.mail.mailbox                                     95EE02BB-7BD4-4890-80F4-B579FF0E7960
86           4
87           4              com.apple.mail.mailbox                                     95EE02BB-7BD4-4890-80F4-B579FF0E7960
91           4
101          4
102          4
104          4
106          4
108          4
113          4
114          4
115          4
116          4
134          4              com.apple.mail.mailbox                                     95EE02BB-7BD4-4890-80F4-B579FF0E7960
98           3
107          3

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET APPLE MAIL METADATA RECORDS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    Z_PK AS Metadata_ID,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE AS Activity_Type,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__TITLE AS Activity_Title,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__CONTENTDESCRIPTION AS Content_Description,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__ITEMIDENTIFIER AS Item_Identifier,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__USERACTIVITYUUID AS Activity_UUID,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISELIGIBLEFORPREDICTION AS Prediction_Eligible,
    Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISPUBLICLYINDEXABLE AS Publicly_Indexable
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.mail.mailbox'
ORDER BY Z_PK;
"
═══ COREDUET APPLE MAIL METADATA RECORDS ═══
```

Metadata_ID	Activity_Type	Activity_Title	Content_Description	Item_Identifier	Activity_UUID	Prediction_Eligible	Publicly_Indexable
87	com.apple.mail.mailbox			7Xyg9VDd17nXS4pqHXqfuw==	95EE02BB-7BD4-4890-80F4-B579FF0E7960	0	0
132	com.apple.mail.mailbox			7Xyg9VDd17nXS4pqHXqfuw==	95EE02BB-7BD4-4890-80F4-B579FF0E7960	0	0
134	com.apple.mail.mailbox			7Xyg9VDd17nXS4pqHXqfuw==	95EE02BB-7BD4-4890-80F4-B579FF0E7960	0	0
136	com.apple.mail.mailbox			7Xyg9VDd17nXS4pqHXqfuw==	95EE02BB-7BD4-4890-80F4-B579FF0E7960	0	0
191	com.apple.mail.mailbox			7Xyg9VDd17nXS4pqHXqfuw==	95EE02BB-7BD4-4890-80F4-B579FF0E7960	0	0
422	com.apple.mail.mailbox			7Xyg9VDd17nXS4pqHXqfuw==	C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D	0	0
441	com.apple.mail.mailbox			c3iZ/Ahb5EswfNhFKCspQQ==	C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D	0	0

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ APPLE MAIL METADATA EVENT LINKS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    M.Z_PK AS Metadata_ID,
    M.Z_DKAPPLICATIONACTIVITYMETADATAKEY__USERACTIVITYUUID AS Activity_UUID,
    COUNT(L.Z_11EVENT) AS Linked_Event_Records,
    COUNT(DISTINCT L.Z_11EVENT) AS Distinct_Event_IDs,
    GROUP_CONCAT(DISTINCT L.Z_11EVENT) AS Event_IDs
FROM ZSTRUCTUREDMETADATA M
LEFT JOIN Z_4EVENT L
    ON M.Z_PK = L.Z_4CUSTOMMETADATA
WHERE M.Z_DKAPPLICATIONACTIVITYMETADATAKEY__ACTIVITYTYPE = 'com.apple.mail.mailbox'
GROUP BY M.Z_PK
ORDER BY M.Z_PK;
"
═══ APPLE MAIL METADATA EVENT LINKS ═══
Metadata_ID  Activity_UUID                           Linked_Event_Records  Distinct_Event_IDs  Event_IDs
───────────  ──────────────────────────────────────  ────────────────────  ──────────────────  ──────────────────────────────
87           95EE02BB-7BD4-4890-80F4-B579FF0E7960    4                     4                   14263,17096,20531,21352
132          95EE02BB-7BD4-4890-80F4-B579FF0E7960    2                     2                   17553,17555
134          95EE02BB-7BD4-4890-80F4-B579FF0E7960    4                     4                   17557,17559,17571,17573
136          95EE02BB-7BD4-4890-80F4-B579FF0E7960    5                     5                   17561,17594,17596,17603,17604
191          95EE02BB-7BD4-4890-80F4-B579FF0E7960    0                     0
422          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
441          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
444          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
446          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
455          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
458          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
460          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
472          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
491          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
505          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
506          C6CE4DF2-DA62-4CC0-80C5-0574E54FD66D    0                     0
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET ZOBJECT SCHEMA ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(ZOBJECT);
"
═══ COREDUET ZOBJECT SCHEMA ═══
cid  name                      type     notnull  dflt_value  pk
───  ────────────────────────  ───────  ───────  ──────────  ──
0    Z_PK                      INTEGER  0                    1
1    Z_ENT                     INTEGER  0                    0
2    Z_OPT                     INTEGER  0                    0
3    ZUUIDHASH                 INTEGER  0                    0
4    ZEVENT                    INTEGER  0                    0
5    ZSOURCE                   INTEGER  0                    0
6    ZCATEGORYTYPE             INTEGER  0                    0
7    ZINTEGERVALUE             INTEGER  0                    0
8    ZCOMPATIBILITYVERSION     INTEGER  0                    0
9    ZENDDAYOFWEEK             INTEGER  0                    0
10   ZENDSECONDOFDAY           INTEGER  0                    0
11   ZHASCUSTOMMETADATA        INTEGER  0                    0
12   ZHASSTRUCTUREDMETADATA    INTEGER  0                    0
13   ZSECONDSFROMGMT           INTEGER  0                    0
14   ZSHOULDSYNC               INTEGER  0                    0
15   ZSTARTDAYOFWEEK           INTEGER  0                    0
16   ZSTARTSECONDOFDAY         INTEGER  0                    0
17   ZVALUECLASS               INTEGER  0                    0
18   ZVALUEINTEGER             INTEGER  0                    0
19   ZVALUETYPECODE            INTEGER  0                    0
20   ZSTRUCTUREDMETADATA       INTEGER  0                    0
21   ZVALUE                    INTEGER  0                    0
22   Z9_VALUE                  INTEGER  0                    0
23   ZIDENTIFIERTYPE           INTEGER  0                    0
24   ZQUANTITYTYPE             INTEGER  0                    0
25   ZOBJECT                   INTEGER  0                    0
26   Z9_OBJECT                 INTEGER  0                    0
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ APPLE MAIL LINKED ZOBJECT RECORDS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    O.Z_PK AS Object_ID,
    O.ZEVENT AS Event_Type,
    O.ZSOURCE AS Source_ID,
    O.ZCATEGORYTYPE AS Category_Type,
    O.ZSTARTDATE AS Start_Date_Raw,
    O.ZENDDATE AS End_Date_Raw,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZVERBPHRASE AS Verb_Phrase,
    O.ZVALUESTRING AS Value_String
FROM Z_4EVENT L
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE L.Z_4CUSTOMMETADATA IN (87, 132, 134, 136)
ORDER BY O.Z_PK;
"
═══ APPLE MAIL LINKED ZOBJECT RECORDS ═══
```

Object_ID	Event_Type	Source_ID	Category_Type	Start_Date_Raw	End_Date_Raw	Stream_Name	Verb_Phrase	Value_String
14263				608002048	608002048	/charging/smartTopOffCheckpoint		
17096				608046784	608353408	/charging/smartTopOffCheckpoint		
17553				608395520	608395520	/photos/engagement/2To3Seconds		PFLDatasetSample-v1.0
17555				608395530	608395530	/photos/deletes/all		PFLDatasetSample-v1.0
17557				608395540	608395540	/photos/engagement/0To1Seconds		PFLDatasetSample-v1.0
17559				608395545	608395545	/photos/engagement/0To1Seconds		PFLDatasetSample-v1.0
17561				608395545	608395545	/photos/engagement/1To2Seconds		PFLDatasetSample-v1.0
17571				608395565	608395565	/photos/engagement/0To1Seconds		PFLDatasetSample-v1.0
17573				608395570	608395570	/photos/deletes/all		PFLDatasetSample-v1.0
17594				608395660	608395660	/photos/engagement/0To1Seconds		PFLDatasetSample-v1.0
17596				608395665	608395665	/photos/engagement/0To1Seconds		PFLDatasetSample-v1.0
17603				608395690	608395690	/photos/deletes/all		PFLDatasetSample-v1.0
17604				608395690	608395690	/photos/engagement/2To3Seconds		PFLDatasetSample-v1.0

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET ENTITY NAMES AND IDS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT Z_ENT, Z_NAME, Z_SUPER, Z_MAX
FROM Z_PRIMARYKEY
ORDER BY Z_ENT;
"
═══ COREDUET ENTITY NAMES AND IDS ═══
Z_ENT  Z_NAME                        Z_SUPER  Z_MAX
─────  ────────────────────────────  ───────  ─────
1      AdditionChangeSet             0        0
2      ContextualChangeRegistration  0        2
3      ContextualKeyPath             0        0
4      CustomMetadata                0        184
5      DeletionChangeSet             0        0
6      Histogram                     0        21
7      HistogramValue                0        217
8      KeyValue                      0        1794
9      Object                        0        22243
10     Category                      9        0
11     Event                         9        0
12     Identifier                    9        0
13     Quantity                      9        0
14     Relation                      9        0
15     Source                        0        930
16     StructuredMetadata            0        7746
17     SyncPeer                      0        1181

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ CORRECT CUSTOM METADATA TO EVENT RELATIONSHIP ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Custom_Metadata_ID,
    C.ZNAME AS Metadata_Name,
    C.ZSTRINGVALUE AS String_Value,
    C.ZINTEGERVALUE AS Integer_Value,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZSTARTDATE AS Start_Date_Raw,
    O.ZENDDATE AS End_Date_Raw
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
ORDER BY C.Z_PK, O.Z_PK
LIMIT 60;
"
═══ CORRECT CUSTOM METADATA TO EVENT RELATIONSHIP ═══
Custom_Metadata_ID  Metadata_Name                                 String_Value  Integer_Value  Event_Object_ID  Stream_Name            Start_Date_Raw  End_Da
te_Raw
──────────────────  ────────────────────────────────────────────  ────────────  ─────────────  ───────────────  ─────────────────────  ──────────────  ──────
──────
1                   _DKShareSheetSuggestionMetadataKey-attachments                           720              /sharesheet/feedback   606575613       606575
614
2                   _DKShareSheetSuggestionMetadataKey-attachments                           927              /sharesheet/feedback   606582201       606582
206
2                   _DKShareSheetSuggestionMetadataKey-attachments                           931              /sharesheet/feedback   606582396       606582
399
2                   _DKShareSheetSuggestionMetadataKey-attachments                           935              /sharesheet/feedback   606582490       606582
493
2                   _DKShareSheetSuggestionMetadataKey-attachments                           938              /sharesheet/feedback   606582552       606582
555
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ CUSTOM METADATA EVENT RELATIONSHIP SUMMARY ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.ZNAME AS Metadata_Name,
    COUNT(*) AS Relationship_Count,
    COUNT(DISTINCT C.Z_PK) AS Distinct_Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Distinct_Event_Objects,
    COUNT(DISTINCT O.ZSTREAMNAME) AS Distinct_Streams
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
GROUP BY C.ZNAME
ORDER BY Relationship_Count DESC, Metadata_Name;
"
═══ CUSTOM METADATA EVENT RELATIONSHIP SUMMARY ═══
```

Metadata_Name	Relationship_Count	Distinct_Metadata_Records	Distinct_Event_Objects	Distinct_Streams
_DKPhotosMetadataKey.identifier	101	40	101	8
pluggedIn	81	2	81	1
_DKBluetoothMetadataKey-isAppleAudioDevice	57	1	57	1
_DKBluetoothMetadataKey-productID	57	2	57	1
_DKShareSheetSuggestionMetadataKey-attachments	35	14	35	1
_DKShareSheetSuggestionMetadataKey-locationUUIDs	24	1	24	1
compatVersion	23	1	23	2
sentimentScore	21	1	21	1
UUID	16	8	16	1
aggDBitmap	16	4	16	1
batteryLevel	16	10	16	1
checkpoint	16	4	16	1
decisionMakerID	16	2	16	1
decisionTime	16	5	16	1
durationPrediction	16	5	16	1

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ PHOTOS METADATA EVENT STREAM DISTRIBUTION ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    O.ZSTREAMNAME AS Stream_Name,
    COUNT(*) AS Relationship_Count,
    COUNT(DISTINCT C.Z_PK) AS Metadata_Records,
    COUNT(DISTINCT O.Z_PK) AS Event_Objects,
    MIN(O.ZSTARTDATE) AS Earliest_Raw_Date,
    MAX(O.ZSTARTDATE) AS Latest_Raw_Date
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = '_DKPhotosMetadataKey.identifier'
GROUP BY O.ZSTREAMNAME
ORDER BY Relationship_Count DESC;
"
═══ PHOTOS METADATA EVENT STREAM DISTRIBUTION ═══
Stream_Name                  Relationship_Count  Metadata_Records  Event_Objects  Earliest_Raw_Date  Latest_Raw_Date
---------------------------  ------------------  ----------------  -------------  -----------------  ---------------
/photos/engagement/0To1Seconds  58                  33                58             607779235          608397170
/photos/engagement/1To2Seconds  18                  11                18             608350755          608397165
/photos/engagement/2To3Seconds  12                  11                12             607779355          608403240
/photos/deletes/all          5                   5                 5              608395530          608395745
/photos/share/all            3                   3                 3              608397205          608745370
/photos/share/airdrop        3                   3                 3              608397205          608745370
/photos/edit/all             1                   1                 1              608396670          608396670
/photos/deletes/recent       1                   1                 1              608395745          608395745

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ PHOTOS IDENTIFIER EVENT DETAILS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZSTRINGVALUE AS Photos_Identifier,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZSTARTDATE AS Start_Date_Raw,
    O.ZENDDATE AS End_Date_Raw
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = '_DKPhotosMetadataKey.identifier'
ORDER BY O.ZSTARTDATE, O.Z_PK
LIMIT 40;
"
═══ PHOTOS IDENTIFIER EVENT DETAILS ═══
```

Metadata_ID	Photos_Identifier	Event_Object_ID	Stream_Name	Start_Date_Raw	End_Date_Raw
47	921D69FD-ABAB-4932-ABB8-0D9EA250ABD6	12217	/photos/engagement/0To1Seconds	607779235	607779235
48	D58B51B1-8B77-4C7E-B671-FD114186834B	12219	/photos/engagement/0To1Seconds	607779250	607779250
49	37669042-037D-4CBB-9C68-A3581ABA5802	12220	/photos/engagement/0To1Seconds	607779250	607779250
50	DDCED3AD-B567-4800-82B7-DF2DD5F7F8E0	12221	/photos/engagement/0To1Seconds	607779250	607779250
51	8C8A1F6D-1E8A-4C2B-8162-89CE3D819179	12222	/photos/engagement/0To1Seconds	607779255	607779255
52	5FB39B5F-F70D-419D-8FB7-3FA468666B96	12223	/photos/engagement/0To1Seconds	607779265	607779265
53	8F23BDAD-9286-4FCB-AEB2-1CDE18A9FF21	12224	/photos/engagement/0To1Seconds	607779275	607779275
54	1A305CC8-95FE-41F9-BDFD-FEF6B85431E2	12225	/photos/engagement/0To1Seconds	607779295	607779295
55	83B59B2F-BB80-4B1A-8A45-A09A042D7298	12226	/photos/engagement/0To1Seconds	607779305	607779305
56	7E073B5C-871A-47DF-91F6-DAA5482B7B72	12227	/photos/engagement/0To1Seconds	607779310	607779310
57	F24831C4-D627-474C-B87C-403E40F4210C	12228	/photos/engagement/0To1Seconds	607779315	607779315
58	9D06954C-AF54-4711-9B2F-84CB05648D26	12229	/photos/engagement/0To1Seconds	607779315	607779315
59	F4E94CE7-8216-4B3A-99F1-9BF2B5E52A81	12230	/photos/engagement/0To1Seconds	607779320	607779320

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ PHOTOS EVENT TIMESTAMP CONVERSION ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    MIN(O.ZSTARTDATE) AS Earliest_Raw,
    datetime(MIN(O.ZSTARTDATE) + 978307200, 'unixepoch') AS Earliest_UTC,
    MAX(O.ZSTARTDATE) AS Latest_Raw,
    datetime(MAX(O.ZSTARTDATE) + 978307200, 'unixepoch') AS Latest_UTC
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = '_DKPhotosMetadataKey.identifier';
"
═══ PHOTOS EVENT TIMESTAMP CONVERSION ═══
Earliest_Raw  Earliest_UTC         Latest_Raw  Latest_UTC
────────────  ───────────────────  ──────────  ───────────────────
607779235     2020-04-05 11:33:55  608745370   2020-04-16 15:56:10

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
echo "═══ CHARGING PLUGGED-IN EVENT DETAILS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZNAME AS Metadata_Name,
    C.ZINTEGERVALUE AS Integer_Value,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    O.ZSTARTDATE AS Start_Date_Raw,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC,
    O.ZENDDATE AS End_Date_Raw
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = 'pluggedIn'
ORDER BY O.ZSTARTDATE, O.Z_PK;
"
```

```
═══ CHARGING PLUGGED-IN EVENT DETAILS ═══
Metadata_ID  Metadata_Name  Integer_Value  Event_Object_ID  Stream_Name                Start_Date_Raw  Start_UTC            End_Date_Raw
───────────  ─────────────  ─────────────  ───────────────  ─────────────────────────  ──────────────  ───────────────────  ────────────
150          pluggedIn      1              18910            /dasd/batterytemperature   608498176       2020-04-13 19:16:16  608498176
150          pluggedIn      1              18959            /dasd/batterytemperature   608498368       2020-04-13 19:19:28  608498368
150          pluggedIn      1              18970            /dasd/batterytemperature   608498560       2020-04-13 19:22:40  608498560
150          pluggedIn      1              18976            /dasd/batterytemperature   608498624       2020-04-13 19:23:44  608498624
150          pluggedIn      1              18978            /dasd/batterytemperature   608498688       2020-04-13 19:24:48  608498688
150          pluggedIn      1              19135            /dasd/batterytemperature   608499072       2020-04-13 19:31:12  608499072
150          pluggedIn      1              19187            /dasd/batterytemperature   608499136       2020-04-13 19:32:16  608499136
150          pluggedIn      1              19218            /dasd/batterytemperature   608499456       2020-04-13 19:37:36  608499456
150          pluggedIn      1              19230            /dasd/batterytemperature   608499776       2020-04-13 19:42:56  608499776
150          pluggedIn      1              19265            /dasd/batterytemperature   608500544       2020-04-13 19:55:44  608500544
150          pluggedIn      1              19351            /dasd/batterytemperature   608503744       2020-04-13 20:49:04  608503744
150          pluggedIn      1              19352            /dasd/batterytemperature   608503808       2020-04-13 20:50:08  608503808
150          pluggedIn      1              19365            /dasd/batterytemperature   608504320       2020-04-13 20:58:40  608504320
150          pluggedIn      1              19374            /dasd/batterytemperature   608504768       2020-04-13 21:06:08  608504768
150          pluggedIn      1              19411            /dasd/batterytemperature   608506624       2020-04-13 21:37:04  608506624
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ BATTERY LEVEL EVENT DETAILS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZINTEGERVALUE AS Battery_Level_Value,
    C.ZDOUBLEVALUE AS Battery_Level_Double,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = 'batteryLevel'
ORDER BY O.ZSTARTDATE, O.Z_PK;
"
═══ BATTERY LEVEL EVENT DETAILS ═══
```

Metadata_ID	Battery_Level_Value	Battery_Level_Double	Event_Object_ID	Stream_Name	Start_UTC
89		79.0	14263	/charging/smartTopOffCheckpoint	2020-04-08 01:27:28
99		82.0	14329	/charging/smartTopOffCheckpoint	2020-04-08 01:27:28
107		99.0	14356	/charging/smartTopOffCheckpoint	2020-04-08 05:33:52
116		100.0	14471	/charging/smartTopOffCheckpoint	2020-04-08 07:10:56
124		75.0	17096	/charging/smartTopOffCheckpoint	2020-04-08 13:53:04
126		83.0	17110	/charging/smartTopOffCheckpoint	2020-04-12 03:03:28
107		99.0	17131	/charging/smartTopOffCheckpoint	2020-04-12 07:12:00
116		100.0	17182	/charging/smartTopOffCheckpoint	2020-04-12 08:49:04
156		80.0	20531	/charging/smartTopOffCheckpoint	2020-04-15 01:38:08
161		86.0	20603	/charging/smartTopOffCheckpoint	2020-04-15 01:38:08
107		99.0	20629	/charging/smartTopOffCheckpoint	2020-04-15 06:47:28
116		100.0	20657	/charging/smartTopOffCheckpoint	2020-04-15 08:22:24
163		77.0	21352	/charging/smartTopOffCheckpoint	2020-04-15 10:53:52
99		82.0	21411	/charging/smartTopOffCheckpoint	2020-04-16 01:44:32
172		98.0	21438	/charging/smartTopOffCheckpoint	2020-04-16 06:06:56

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ TIME TILL TOP-OFF EVENT DETAILS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZNAME AS Metadata_Name,
    C.ZINTEGERVALUE AS Integer_Value,
    C.ZDOUBLEVALUE AS Double_Value,
    C.ZSTRINGVALUE AS String_Value,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = 'timeTillTopOff'
ORDER BY O.ZSTARTDATE, O.Z_PK;
"
═══ TIME TILL TOP-OFF EVENT DETAILS ═══
```

Metadata_ID	Metadata_Name	Integer_Value	Double_Value	String_Value	Event_Object_ID	Stream_Name	Start_UTC
90	timeTillTopOff		2760.0		14263	/charging/smartTopOffCheckpoint	2020-04-08 01:27:28
103	timeTillTopOff		6951.96		14329	/charging/smartTopOffCheckpoint	2020-04-08 01:27:28
105	timeTillTopOff		1236.22		14356	/charging/smartTopOffCheckpoint	2020-04-08 05:33:52
114	timeTillTopOff		900.0		14471	/charging/smartTopOffCheckpoint	2020-04-08 07:10:56
123	timeTillTopOff		7924.4		17096	/charging/smartTopOffCheckpoint	2020-04-08 13:53:04
125	timeTillTopOff		6615.74		17110	/charging/smartTopOffCheckpoint	2020-04-12 03:03:28
105	timeTillTopOff		1236.22		17131	/charging/smartTopOffCheckpoint	2020-04-12 07:12:00
114	timeTillTopOff		900.0		17182	/charging/smartTopOffCheckpoint	2020-04-12 08:49:04
155	timeTillTopOff		7624.4		20531	/charging/smartTopOffCheckpoint	2020-04-15 01:38:08
159	timeTillTopOff		5442.02		20603	/charging/smartTopOffCheckpoint	2020-04-15 01:38:08
162	timeTillTopOff		1224.43		20629	/charging/smartTopOffCheckpoint	2020-04-15 06:47:28
114	timeTillTopOff		900.0		20657	/charging/smartTopOffCheckpoint	2020-04-15 08:22:24

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ CHARGING CHECKPOINT EVENT DETAILS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZNAME AS Metadata_Name,
    C.ZINTEGERVALUE AS Integer_Value,
    C.ZDOUBLEVALUE AS Double_Value,
    C.ZSTRINGVALUE AS String_Value,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME = 'checkpoint'
ORDER BY O.ZSTARTDATE, O.Z_PK;
"
═══ CHARGING CHECKPOINT EVENT DETAILS ═══
Metadata_ID  Metadata_Name  Integer_Value  Double_Value  String_Value  Event_Object_ID  Stream_Name                          Start_UTC
───────────  ─────────────  ─────────────  ────────────  ────────────  ───────────────  ───────────────────────────────────  ───────────────────
91           checkpoint     0                                          14263            /charging/smartTopOffCheckpoint      2020-04-08 01:27:28
104          checkpoint     2                                          14329            /charging/smartTopOffCheckpoint      2020-04-08 01:27:28
108          checkpoint     5                                          14356            /charging/smartTopOffCheckpoint      2020-04-08 05:33:52
115          checkpoint     7                                          14471            /charging/smartTopOffCheckpoint      2020-04-08 07:10:56
91           checkpoint     0                                          17096            /charging/smartTopOffCheckpoint      2020-04-08 13:53:04
104          checkpoint     2                                          17110            /charging/smartTopOffCheckpoint      2020-04-12 03:03:28
108          checkpoint     5                                          17131            /charging/smartTopOffCheckpoint      2020-04-12 07:12:00
115          checkpoint     7                                          17182            /charging/smartTopOffCheckpoint      2020-04-12 08:49:04
91           checkpoint     0                                          20531            /charging/smartTopOffCheckpoint      2020-04-15 01:38:08
104          checkpoint     2                                          20603            /charging/smartTopOffCheckpoint      2020-04-15 01:38:08
108          checkpoint     5                                          20629            /charging/smartTopOffCheckpoint      2020-04-15 06:47:28
115          checkpoint     7                                          20657            /charging/smartTopOffCheckpoint      2020-04-15 08:22:24
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ BLUETOOTH METADATA EVENT DETAILS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    C.Z_PK AS Metadata_ID,
    C.ZNAME AS Metadata_Name,
    C.ZINTEGERVALUE AS Integer_Value,
    C.ZSTRINGVALUE AS String_Value,
    O.Z_PK AS Event_Object_ID,
    O.ZSTREAMNAME AS Stream_Name,
    datetime(O.ZSTARTDATE + 978307200, 'unixepoch') AS Start_UTC
FROM Z_4EVENT L
JOIN ZCUSTOMMETADATA C
    ON C.Z_PK = L.Z_4CUSTOMMETADATA
JOIN ZOBJECT O
    ON O.Z_PK = L.Z_11EVENT
WHERE C.ZNAME IN (
    '_DKBluetoothMetadataKey-productID',
    '_DKBluetoothMetadataKey-isAppleAudioDevice'
)
ORDER BY O.ZSTARTDATE, O.Z_PK;
"
═══ BLUETOOTH METADATA EVENT DETAILS ═══
```

Metadata_ID	Metadata_Name	Integer_Value	String_Value	Event_Object_ID	Stream_Name	Start_UTC
17	_DKBluetoothMetadataKey-isAppleAudioDevice	1		5241	/bluetooth/isConnected	2020-03-27 18:32:42
16	_DKBluetoothMetadataKey-productID	8194		5241	/bluetooth/isConnected	2020-03-27 18:32:42
17	_DKBluetoothMetadataKey-isAppleAudioDevice	1		5243	/bluetooth/isConnected	2020-03-27 18:32:52
16	_DKBluetoothMetadataKey-productID	8194		5243	/bluetooth/isConnected	2020-03-27 18:32:52
17	_DKBluetoothMetadataKey-isAppleAudioDevice	1		5244	/bluetooth/isConnected	2020-03-27 18:33:03
16	_DKBluetoothMetadataKey-productID	8194		5244	/bluetooth/isConnected	2020-03-27 18:33:03
17	_DKBluetoothMetadataKey-isAppleAudioDevice	1		5325	/bluetooth/isConnected	2020-03-27 18:33:05
16	_DKBluetoothMetadataKey-productID	8194		5325	/bluetooth/isConnected	2020-03-27 18:33:05
17	_DKBluetoothMetadataKey-isAppleAudioDevice	1		5430	/bluetooth/isConnected	2020-03-27 18:54:16
16	_DKBluetoothMetadataKey-productID	8194		5430	/bluetooth/isConnected	2020-03-27 18:54:16

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

CORE="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge"

echo "═══ COREDUET SOURCE FILES ═══"
sudo find "$CORE" -maxdepth 1 -type f \
  -printf '%f (%s bytes)\n' | sort

echo
echo "═══ DATABASE FILES ═══"
sudo find "$CORE" -maxdepth 1 -type f \
  \( -name 'knowledgeC.db' -o -name 'knowledgeC.db-wal' -o -name 'knowledgeC.db-shm' \) \
  -printf '%f (%s bytes)\n' | sort
═══ COREDUET SOURCE FILES ═══
knowledgeC.db (14213120 bytes)
knowledgeC.db-shm (32768 bytes)
knowledgeC.db-wal (3456712 bytes)

═══ DATABASE FILES ═══
knowledgeC.db (14213120 bytes)
knowledgeC.db-shm (32768 bytes)
knowledgeC.db-wal (3456712 bytes)

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

CORE="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge"

echo "═══ ORIGINAL SOURCE SHA-256 ═══"
sudo sha256sum \
"$CORE/knowledgeC.db" \
"$CORE/knowledgeC.db-shm" \
"$CORE/knowledgeC.db-wal"

echo
echo "═══ WORKING COPY SHA-256 ═══"
sha256sum \
working/knowledgec/knowledgeC.db \
working/knowledgec/knowledgeC.db-shm \
working/knowledgec/knowledgeC.db-wal
 ═══ ORIGINAL SOURCE SHA-256 ═══
ed7e2974440106f3fa1efd3b3ba8fd2290d45e09f54564f6dce0a9fcf4dd5be2  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db
c53c682332253b82be213fdfa4e2199d1efea9d0deefe73e0db5290f4a5efb38  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db-shm
a1b018f06164163cc6dbd14c353733b6d3e7dc40aca1d4f909e3ec2a5159c670  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db-wal

═══ WORKING COPY SHA-256 ═══
ed7e2974440106f3fa1efd3b3ba8fd2290d45e09f54564f6dce0a9fcf4dd5be2  working/knowledgec/knowledgeC.db
c53c682332253b82be213fdfa4e2199d1efea9d0deefe73e0db5290f4a5efb38  working/knowledgec/knowledgeC.db-shm
a1b018f06164163cc6dbd14c353733b6d3e7dc40aca1d4f909e3ec2a5159c670  working/knowledgec/knowledgeC.db-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET DATABASE INTEGRITY ═══"
sqlite3 working/knowledgec/knowledgeC.db \
"PRAGMA integrity_check;"

echo
echo "═══ COREDUET DATABASE TABLES ═══"
sqlite3 working/knowledgec/knowledgeC.db "
SELECT name
FROM sqlite_master
WHERE type='table'
ORDER BY name;
"
═══ COREDUET DATABASE INTEGRITY ═══
ok

═══ COREDUET DATABASE TABLES ═══
ZADDITIONCHANGESET
ZCONTEXTUALCHANGEREGISTRATION
ZCONTEXTUALKEYPATH
ZCUSTOMMETADATA
ZDELETIONCHANGESET
ZHISTOGRAM
ZHISTOGRAMVALUE
ZKEYVALUE
ZOBJECT
ZSOURCE
ZSTRUCTUREDMETADATA
ZSYNCPEER
Z_4EVENT
Z_METADATA
Z_MODELCACHE
Z_PRIMARYKEY

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ ZOBJECT TABLE STRUCTURE ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(ZOBJECT);
"

echo
echo "═══ ZSOURCE TABLE STRUCTURE ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(ZSOURCE);
"

echo
echo "═══ EVENT RELATIONSHIP STRUCTURE ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(Z_4EVENT);
"
═══ ZOBJECT TABLE STRUCTURE ═══
cid  name                      type     notnull  dflt_value  pk
───  ────────────────────────  ───────  ───────  ──────────  ──
0    Z_PK                      INTEGER  0                    1
1    Z_ENT                     INTEGER  0                    0
2    Z_OPT                     INTEGER  0                    0
3    ZUUIDHASH                 INTEGER  0                    0
4    ZEVENT                    INTEGER  0                    0
5    ZSOURCE                   INTEGER  0                    0
6    ZCATEGORYTYPE             INTEGER  0                    0
7    ZINTEGERVALUE             INTEGER  0                    0
8    ZCOMPATIBILITYVERSION     INTEGER  0                    0
9    ZENDDAYOFWEEK             INTEGER  0                    0
10   ZENDSECONDOFDAY           INTEGER  0                    0
11   ZHASCUSTOMMETADATA        INTEGER  0                    0
12   ZHASSTRUCTUREDMETADATA    INTEGER  0                    0
13   ZSECONDSFROMGMT           INTEGER  0                    0
14   ZSHOULDSYNC               INTEGER  0                    0
15   ZSTARTDAYOFWEEK           INTEGER  0                    0
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET RECORD COUNTS ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT 'ZOBJECT' AS Table_Name, COUNT(*) AS Records FROM ZOBJECT
UNION ALL
SELECT 'ZSOURCE', COUNT(*) FROM ZSOURCE
UNION ALL
SELECT 'ZSTRUCTUREDMETADATA', COUNT(*) FROM ZSTRUCTUREDMETADATA
UNION ALL
SELECT 'ZCUSTOMMETADATA', COUNT(*) FROM ZCUSTOMMETADATA
UNION ALL
SELECT 'Z_4EVENT', COUNT(*) FROM Z_4EVENT;
"
═══ COREDUET RECORD COUNTS ═══
Table_Name            Records
────────────────────  ───────
ZOBJECT               21652
ZSOURCE               769
ZSTRUCTUREDMETADATA   1738
ZCUSTOMMETADATA       135
Z_4EVENT              615

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET EVENT FIELD DISTRIBUTION ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    ZSTREAMNAME,
    ZVALUETYPECODE,
    ZVALUECLASS,
    COUNT(*) AS Records
FROM ZOBJECT
GROUP BY ZSTREAMNAME, ZVALUETYPECODE, ZVALUECLASS
ORDER BY Records DESC
LIMIT 30;
"
═══ COREDUET EVENT FIELD DISTRIBUTION ═══
```

ZSTREAMNAME	ZVALUETYPECODE	ZVALUECLASS	Records
/app/inFocus	6584185901589580638	1	4910
/device/batteryPercentage	-5678941118864335806	3	3395
/activity/level	5907650307545087287	2	1857
/display/isBacklit	-2475731913145812025	2	1746
/notification/usage	2196297603508581658	1	1392
/defaultPaired/nearby	-2475731913145812025	2	1300
/app/usage	6584185901589580638	1	769
/inferred/motion	271174270552846259	2	620
/knowledge-sync-deletion-bookmark/898F53A0-1714-5175-8C2D-9CD767341EB5	-635635896434291546	1	587
/activity/level/feedback	-635635896434291546	1	558
/knowledge-sync-addition-window/898F53A0-1714-5175-8C2D-9CD767341EB5	-635635896434291546	1	534
/user/isFirstBacklightOnAfterWakeup	-2475731913145812025	2	412
/event/tombstone	5520820237227302457	1	369
/knowledge-sync-deletion-bookmark/59B290D2-C297-5561-8F8A-FD7A69566885	-635635896434291546	1	365
/app/activity	6584185901589580638	1	325
/device/isLockedImputed	-2475731913145812025	2	322
/device/isLocked	-2475731913145812025	2	316
/media/nowPlaying	6584185901589580638	1	281
/keybag/isLocked	-2475731913145812025	2	226
/app/intents	6584185901589580638	1	182

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== COREDUET SOURCE DISTRIBUTION ====="
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    Z_PK AS Source_ID,
    ZBUNDLEID AS Bundle_ID,
    COUNT(*) AS Source_Record
FROM ZSOURCE
GROUP BY Z_PK, ZBUNDLEID
ORDER BY Source_Record DESC
LIMIT 25;
"
===== COREDUET SOURCE DISTRIBUTION =====
Source_ID  Bundle_ID                    Source_Record
---------  ---------------------------  -------------
1          com.apple.MobileSMS          1
2          com.apple.mobilemail         1
6          com.apple.mobilemail         1
7          com.apple.mobilemail         1
8          com.apple.mobilemail         1
11         com.hammerandchisel.discord  1
12         com.apple.mobileslideshow    1
13         com.apple.mobileslideshow    1
14         com.apple.mobileslideshow    1
15         com.apple.mobileslideshow    1
16         com.apple.mobileslideshow    1
17         com.apple.mobileslideshow    1
22         com.apple.mobileslideshow    1
23         com.apple.mobilemail         1
27         com.apple.mobileslideshow    1
28         com.apple.mobileslideshow    1
29         com.hammerandchisel.discord  1
30         com.hammerandchisel.discord  1
31         com.facebook.Messenger       1
32         com.facebook.Messenger       1
33         com.facebook.Messenger       1
```

```
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET EVENTS BY SOURCE APPLICATION ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    s.ZBUNDLEID AS Bundle_ID,
    COUNT(o.Z_PK) AS Event_Records,
    MIN(datetime(o.ZSTARTDATE + 978307200,'unixepoch')) AS First_Event_UTC,
    MAX(datetime(o.ZSTARTDATE + 978307200,'unixepoch')) AS Last_Event_UTC
FROM ZOBJECT o
LEFT JOIN ZSOURCE s ON o.ZSOURCE = s.Z_PK
GROUP BY s.ZBUNDLEID
ORDER BY Event_Records DESC
LIMIT 25;
"
═══ COREDUET EVENTS BY SOURCE APPLICATION ═══
Bundle_ID                          Event_Records  First_Event_UTC      Last_Event_UTC
─────────────────────────────────  ─────────────  ───────────────────  ───────────────────
                                   20905          0000-12-30 00:00:00  2020-04-16 17:19:47
com.apple.Maps                     140            2020-04-01 17:45:24  2020-04-15 16:17:25
com.apple.mobileslideshow          90             2020-03-22 13:13:24  2020-04-16 16:03:25
com.apple.mobilemail               48             2020-03-22 01:13:20  2020-04-16 12:09:55
com.apple.MobileSMS                46             2020-03-22 01:10:59  2020-04-16 12:59:29
com.apple.Health                   40             2020-04-03 18:23:35  2020-04-07 00:29:45
com.apple.mobilesafari             39             2020-03-28 00:58:40  2020-03-28 01:44:45
com.apple.InCallService            37             2020-03-23 20:02:52  2020-04-14 15:56:12
co.babypenguin.imo                 34             2020-03-22 15:16:19  2020-04-16 12:06:49
imgurmobile                        32             2020-03-22 15:01:45  2020-04-16 12:07:30
net.whatsapp.WhatsApp              20             2020-03-26 18:42:58  2020-04-16 12:05:44
com.apple.mobilenotes              19             2020-03-28 00:37:11  2020-03-28 00:51:10
```

```
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    Z_PK,
    ZSTREAMNAME,
    ZSTARTDATE,
    ZENDDATE,
    ZSOURCE
FROM ZOBJECT
ORDER BY ZSTARTDATE ASC
LIMIT 10;
"
```

```
==== COREDUET TIMESTAMP QUALITY CHECK ====
```

Total_Records	Null_Start_Dates	Minimum_Raw_Start	Maximum_Raw_Start	Negative_Start_Dates
21652	0	-63114076800	608750387	3

```
==== EARLIEST RAW TIMESTAMP SAMPLES ====
```

Z_PK	ZSTREAMNAME	ZSTARTDATE	ZENDDATE	ZSOURCE
457	/knowledge-sync-deletion-bookmark/59B290D2-C297-5561-8F8A-FD7A69566885	-63114076800	606535373.237509	
1944	/knowledge-sync-deletion-bookmark/898F53A0-1714-5175-8C2D-9CD767341EB5	-63114076800	606691606.690638	
2288	/knowledge-sync-deletion-bookmark/59B290D2-C297-5561-8F8A-FD7A69566885	-63114076800	606709529.078239	
1943	/knowledge-sync-addition-window/898F53A0-1714-5175-8C2D-9CD767341EB5	604099606.81132	606691606.690362	
2	/device/batteryPercentage	606519536	606519536	
6	/device/batteryPercentage	606519536	606519612	
7	/display/isBacklit	606519536	606519628	
8	/activity/level	606519536	606519628	
3	/device/isPluggedIn	606519537	606519540	
1	/activity/level/feedback	606519539	606519539	

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET TIMESTAMP QUALITY CHECK ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(*) AS Total_Records,
    SUM(CASE WHEN ZSTARTDATE IS NULL THEN 1 ELSE 0 END) AS Null_Start_Dates,
    MIN(ZSTARTDATE) AS Minimum_Raw_Start,
    MAX(ZSTARTDATE) AS Maximum_Raw_Start,
    SUM(CASE WHEN ZSTARTDATE < 0 THEN 1 ELSE 0 END) AS Negative_Start_Dates
FROM ZOBJECT;
"

echo
echo "═══ EARLIEST RAW TIMESTAMP SAMPLES ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    Z_PK,
    ZSTREAMNAME,
    ZSTARTDATE,
    ZENDDATE,
    ZSOURCE
FROM ZOBJECT
ORDER BY ZSTARTDATE ASC
LIMIT 10;
"
═══ COREDUET TIMESTAMP QUALITY CHECK ═══
Total_Records  Null_Start_Dates  Minimum_Raw_Start  Maximum_Raw_Start  Negative_Start_Dates
─────────────  ─────────────────  ─────────────────  ─────────────────  ────────────────────
21652          0                  -63114076800       608750387          3
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ VALID COREDUET ACTIVITY DATE RANGE ═══"
sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(*) AS Valid_Timestamp_Records,
    datetime(MIN(ZSTARTDATE)+978307200,'unixepoch') AS Earliest_UTC,
    datetime(MAX(ZSTARTDATE)+978307200,'unixepoch') AS Latest_UTC
FROM ZOBJECT
WHERE ZSTARTDATE > 0
AND ZSTARTDATE < 700000000;
"
═══ VALID COREDUET ACTIVITY DATE RANGE ═══
Valid_Timestamp_Records  Earliest_UTC         Latest_UTC
─────────────────────────  ──────────────────  ──────────────────
21649                    2020-02-22 21:26:46  2020-04-16 17:19:47

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET EVENT METADATA RELATIONSHIP ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
PRAGMA table_info(Z_4EVENT);
"
═══ COREDUET EVENT METADATA RELATIONSHIP ═══
cid  name                type     notnull  dflt_value  pk
───  ──────────────────  ───────  ───────  ──────────  ──
0    Z_4CUSTOMMETADATA   INTEGER  0                    1
1    Z_11EVENT           INTEGER  0                    2

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET METADATA RECORD COUNTS AND RELATIONSHIPS ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT 'Structured metadata' AS Item, COUNT(*) AS Records
FROM ZSTRUCTUREDMETADATA
UNION ALL
SELECT 'Custom metadata', COUNT(*)
FROM ZCUSTOMMETADATA
UNION ALL
SELECT 'Event-metadata links', COUNT(*)
FROM Z_4EVENT
UNION ALL
SELECT 'Links to existing events', COUNT(*)
FROM Z_4EVENT e
JOIN ZOBJECT o ON o.Z_PK = e.Z_11EVENT
UNION ALL
SELECT 'Links to existing custom metadata', COUNT(*)
FROM Z_4EVENT e
JOIN ZCUSTOMMETADATA c ON c.Z_PK = e.Z_4CUSTOMMETADATA
UNION ALL
SELECT 'Events referencing structured metadata', COUNT(*)
FROM ZOBJECT o
JOIN ZSTRUCTUREDMETADATA s ON s.Z_PK = o.ZSTRUCTUREDMETADATA;
"
═══ COREDUET METADATA RECORD COUNTS AND RELATIONSHIPS ═══
Item                                    Records
──────────────────────────────────────  ──────────
Structured metadata                     1738
Custom metadata                         135
```

```
FROM ZSTRUCTUREDMETADATA
UNION ALL
SELECT 'Custom metadata', COUNT(*)
FROM ZCUSTOMMETADATA
UNION ALL
SELECT 'Event-metadata links', COUNT(*)
FROM Z_4EVENT
UNION ALL
SELECT 'Links to existing events', COUNT(*)
FROM Z_4EVENT e
JOIN ZOBJECT o ON o.Z_PK = e.Z_11EVENT
UNION ALL
SELECT 'Links to existing custom metadata', COUNT(*)
FROM Z_4EVENT e
JOIN ZCUSTOMMETADATA c ON c.Z_PK = e.Z_4CUSTOMMETADATA
UNION ALL
SELECT 'Events referencing structured metadata', COUNT(*)
FROM ZOBJECT o
JOIN ZSTRUCTUREDMETADATA s ON s.Z_PK = o.ZSTRUCTUREDMETADATA;
"
==== COREDUET METADATA RECORD COUNTS AND RELATIONSHIPS ====
Item                                    Records
--------------------------------------  -------
Structured metadata                     1738
Custom metadata                         135
Event-metadata links                    615
Links to existing events                615
Links to existing custom metadata       615
Events referencing structured metadata  7600

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== COREDUET SELECTED ACTIVITY STREAM SUMMARY ====="

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    ZSTREAMNAME AS Activity_Stream,
    COUNT(*) AS Records,
    COUNT(ZVALUESTRING) AS ValueString_Records,
    COUNT(ZSTRING) AS String_Records,
    COUNT(ZVALUEINTEGER) AS Integer_Records,
    COUNT(ZVALUEDOUBLE) AS Double_Records,
    COUNT(DISTINCT ZSOURCE) AS Distinct_Sources,
    datetime(MIN(ZSTARTDATE)+978307200,'unixepoch') AS Earliest_UTC,
    datetime(MAX(ZSTARTDATE)+978307200,'unixepoch') AS Latest_UTC
FROM ZOBJECT
WHERE ZSTARTDATE > 0
  AND ZSTARTDATE < 700000000
  AND (
      ZSTREAMNAME LIKE '/app/%'
      OR ZSTREAMNAME LIKE '/notification/%'
      OR ZSTREAMNAME LIKE '/inferred/motion%'
      OR ZSTREAMNAME LIKE '/photos/engagement%'
      OR ZSTREAMNAME LIKE '/media/nowPlaying%'
      OR ZSTREAMNAME LIKE '/device/isLocked%'
  )
GROUP BY ZSTREAMNAME
ORDER BY Records DESC;
"
===== COREDUET SELECTED ACTIVITY STREAM SUMMARY =====
Activity_Stream                Records  ValueString_Records  String_Records  Integer_Records  Double_Records  Distinct_Sources  Earliest_UTC         Latest_UTC
_____
/app/inFocus                   4910     4910                 0               4910            4910            0                 2020-03-21 21:42:46  2020-04-16 1
7:19:47
/notification/usage            1392     1392                 0               1392            1392            0                 2020-03-22 01:05:38  2020-04-16 1
```

```
      ZSTREAMNAME LIKE '/app/%'
      OR ZSTREAMNAME LIKE '/notification/%'
      OR ZSTREAMNAME LIKE '/inferred/motion%'
      OR ZSTREAMNAME LIKE '/photos/engagement%'
      OR ZSTREAMNAME LIKE '/media/nowPlaying%'
      OR ZSTREAMNAME LIKE '/device/isLocked%'
  )
GROUP BY ZSTREAMNAME
ORDER BY Records DESC;
"
═══ COREDUET SELECTED ACTIVITY STREAM SUMMARY ═══
```

Activity_Stream	Records	ValueString_Records	String_Records	Integer_Records	Double_Records	Distinct_Sources	Earliest_UTC	Latest_UTC
/app/inFocus	4910	4910	0	4910	4910	0	2020-03-21 21:42:46	2020-04-16 17:19:47
/notification/usage	1392	1392	0	1392	1392	0	2020-03-22 01:05:38	2020-04-16 17:16:13
/app/usage	769	769	0	769	769	0	2020-03-21 21:42:46	2020-04-16 16:02:52
/inferred/motion	620	0	0	620	620	0	2020-03-21 21:43:30	2020-04-16 11:43:41
/app/activity	325	325	0	325	325	325	2020-03-22 01:13:20	2020-04-16 16:03:25
/device/isLockedImputed	322	0	0	322	322	0	2020-03-21 21:40:00	2020-04-16 13:57:44
/device/isLocked	316	0	0	316	316	0	2020-03-21 21:40:00	2020-04-16 13:57:44
/media/nowPlaying	281	281	0	281	281	0	2020-03-22 13:56:16	2020-04-15 16:22:54
/app/intents	182	169	0	182	182	182	2020-03-22 01:10:59	2020-04-16 12:59:29
/app/install	102	102	0	102	102	0	2020-03-21 21:44:00	2020-04-16 15:53:00
/photos/engagement/0To1Seconds	58	58	0	58	58	0	2020-04-05 11:33:55	2020-04-12 15:12:50
/photos/engagement/1To2Seconds	18	18	0	18	18	0	2020-04-12 02:19:15	2020-04-12 15:12:45

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET APP ACTIVITY AND INTENT SOURCES ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    o.ZSTREAMNAME AS Activity_Stream,
    s.ZBUNDLEID AS Application_Bundle_ID,
    COUNT(*) AS Records,
    datetime(MIN(o.ZSTARTDATE)+978307200,'unixepoch') AS First_Seen_UTC,
    datetime(MAX(o.ZSTARTDATE)+978307200,'unixepoch') AS Last_Seen_UTC
FROM ZOBJECT o
LEFT JOIN ZSOURCE s ON s.Z_PK = o.ZSOURCE
WHERE o.ZSTREAMNAME IN ('/app/activity','/app/intents')
  AND o.ZSTARTDATE > 0
  AND o.ZSTARTDATE < 700000000
GROUP BY o.ZSTREAMNAME, s.ZBUNDLEID
ORDER BY o.ZSTREAMNAME, Records DESC;
"
═══ COREDUET APP ACTIVITY AND INTENT SOURCES ═══
```

Activity_Stream	Application_Bundle_ID	Records	First_Seen_UTC	Last_Seen_UTC
/app/activity	com.apple.Maps	100	2020-04-01 17:46:20	2020-04-15 16:17:25
/app/activity	com.apple.mobileslideshow	51	2020-03-22 13:13:40	2020-04-16 16:03:25
/app/activity	com.apple.mobilemail	48	2020-03-22 01:13:20	2020-04-16 12:09:55
/app/activity	com.apple.Health	40	2020-04-03 18:23:35	2020-04-07 00:29:45
/app/activity	imgurmobile	25	2020-03-22 15:01:45	2020-04-16 12:07:30
/app/activity	com.apple.mobilesafari	25	2020-03-28 00:58:40	2020-03-28 01:44:45
/app/activity	com.apple.mobilenotes	16	2020-03-28 00:37:25	2020-03-28 00:51:10
/app/activity	net.kortina.labs.Venmo	5	2020-03-25 00:35:10	2020-03-25 00:45:00
/app/activity	com.apple.mobiletimer	5	2020-04-12 13:23:40	2020-04-16 12:11:25
/app/activity	com.enchantedcloud.photovault	3	2020-04-15 00:40:15	2020-04-15 00:42:20
/app/activity	com.atebits.Tweetie2	2	2020-03-24 01:38:00	2020-03-24 13:45:25
/app/activity	com.apple.mobilephone	2	2020-04-12 15:28:50	2020-04-13 11:53:05
/app/activity	com.apple.mobilecal	2	2020-04-12 13:34:35	2020-04-12 13:35:10
/app/activity	com.apple.weather	1	2020-03-28 01:25:50	2020-03-28 01:25:50
/app/intents	com.apple.MobileSMS	44	2020-03-22 01:10:59	2020-04-16 12:59:29

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ COREDUET STRUCTURED METADATA FIELD POPULATION ═══"

sqlite3 -header -column working/knowledgec/knowledgeC.db "
SELECT
    COUNT(*) AS Total_Metadata_Records,
    COUNT(Z_CDPORTRAITMETADATAKEY__ALGORITHM) AS Portrait_Algorithm,
    COUNT(Z_CDPORTRAITMETADATAKEY__ASSETVERSION) AS Portrait_Asset_Version,
    COUNT(Z_DKAPPINSTALLMETADATAKEY__ISINSTALL) AS App_Install_Flag,
    COUNT(Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISELIGIBLEFORPREDICTION) AS Prediction_Flag,
    COUNT(Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISPUBLICLYINDEXABLE) AS Public_Index_Flag,
    COUNT(Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON) AS Audio_Route_Reason,
    COUNT(Z_DKDEVICEBATTERYPERCENTAGEMETADATAKEY__FULLYCHARGED) AS Fully_Charged_Flag,
    COUNT(Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE) AS Adapter_Type
FROM ZSTRUCTUREDMETADATA;
"
═══ COREDUET STRUCTURED METADATA FIELD POPULATION ═══
Total_Metadata_Records  Portrait_Algorithm  Portrait_Asset_Version  App_Install_Flag  Prediction_Flag  Public_Index_Flag  Audio_Route_Reason  Fully_Charged_Flag
Adapter_Type
──────────────────────  ──────────────────  ──────────────────────  ────────────────  ───────────────  ─────────────────  ──────────────────  ──────────────────
──────────
1738                    22                  22                      45                334              325                22                  1
4

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ >....
UNION ALL

SELECT 'Prediction Eligible',
       CAST(Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISELIGIBLEFORPREDICTION AS TEXT),
       COUNT(*)
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISELIGIBLEFORPREDICTION IS NOT NULL
GROUP BY Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISELIGIBLEFORPREDICTION

UNION ALL

SELECT 'Publicly Indexable',
       CAST(Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISPUBLICLYINDEXABLE AS TEXT),
       COUNT(*)
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISPUBLICLYINDEXABLE IS NOT NULL
GROUP BY Z_DKAPPLICATIONACTIVITYMETADATAKEY__ISPUBLICLYINDEXABLE

UNION ALL

SELECT 'Audio Route Reason',
       CAST(Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON AS TEXT),
       COUNT(*)
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON IS NOT NULL
GROUP BY Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON

UNION ALL

SELECT 'Adapter Type',
       CAST(Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE AS TEXT),
       COUNT(*)
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE IS NOT NULL
GROUP BY Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE
```

```
       CAST(Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON AS TEXT),
       COUNT(*)
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON IS NOT NULL
GROUP BY Z_DKAUDIOMETADATAKEY__ROUTECHANGEREASON

UNION ALL

SELECT 'Adapter Type',
       CAST(Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE AS TEXT),
       COUNT(*)
FROM ZSTRUCTUREDMETADATA
WHERE Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE IS NOT NULL
GROUP BY Z_DKDEVICEISPLUGGEDINMETADATAKEY__ADAPTERTYPE

ORDER BY Metadata_Field, Stored_Value;
"
==== COREDUET STRUCTURED METADATA VALUE DISTRIBUTIONS ====
Metadata_Field       Stored_Value  Records
-------------------  ------------  -------
Adapter Type         -536854523    1
Adapter Type         -536854525    1
Adapter Type         -536854526    1
Adapter Type         -536854528    1
App Install Flag     0             2
App Install Flag     1             43
Audio Route Reason   0             1
Audio Route Reason   1             7
Audio Route Reason   2             2
Audio Route Reason   3             8
Audio Route Reason   4             3
Audio Route Reason   8             1
Prediction Eligible  0             194
Prediction Eligible  1             140
Publicly Indexable   0             316
Publicly Indexable   1             9

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
fired_ttl | INTEGER
disallow_propose_new_time | INTEGER
structured_data | BLOB
local_structured_data | BLOB
junk_status | INTEGER
conference_url | TEXT
contact_identifier | TEXT
recurrence_set | TEXT
flags | INTEGER
contact_name | TEXT

═══ CALENDAR ITEM TIMESTAMP SAMPLE ═══

start_date:
536457600.0
537667200.0
539222400.0
540259200.0
540432000.0

end_date:
536543999.0
537753599.0
539308799.0
540345599.0
540518399.0

creation_date:
608391316.844578

last_modified:
581035593.0
581035593.0
581035593.0
581035593.0
581035593.0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

python3 - <<'PY'
import sqlite3

db = "working/calendar/Calendar.sqlitedb"
con = sqlite3.connect(f"file:{db}?mode=ro", uri=True)
con.row_factory = sqlite3.Row

print("===== CALENDAR SOURCES =====")
for r in con.execute("""
    SELECT ROWID, title, type
    FROM Calendar
    ORDER BY ROWID
"""):
    print(dict(r))

print("\n===== TIMED CALENDAR ITEM =====")
r = con.execute("""
    SELECT ROWID, summary, start_date, end_date,
           creation_date, last_modified, all_day,
           calendar_id, status, has_recurrences
    FROM CalendarItem
    WHERE ROWID = 119
""").fetchone()

print(dict(r) if r else "Record 119 not found")

con.close()
PY
===== CALENDAR SOURCES =====
{'ROWID': 1, 'title': 'Default', 'type': None}
{'ROWID': 2, 'title': 'DEFAULT_CALENDAR_NAME', 'type': None}
{'ROWID': 3, 'title': 'DEFAULT_TASK_CALENDAR_NAME', 'type': None}
{'ROWID': 5, 'title': 'Facebook Birthdays', 'type': None}
{'ROWID': 6, 'title': 'Found in Mail', 'type': None}
{'ROWID': 7, 'title': 'Found in Natural Language', 'type': None}
```

```
    ORDER BY ROWID
"""):
    print(dict(r))

print("\n==== TIMED CALENDAR ITEM ====")
r = con.execute("""
    SELECT ROWID, summary, start_date, end_date,
           creation_date, last_modified, all_day,
           calendar_id, status, has_recurrences
    FROM CalendarItem
    WHERE ROWID = 119
""").fetchone()

print(dict(r) if r else "Record 119 not found")

con.close()
PY
==== CALENDAR SOURCES ====
{'ROWID': 1, 'title': 'Default', 'type': None}
{'ROWID': 2, 'title': 'DEFAULT_CALENDAR_NAME', 'type': None}
{'ROWID': 3, 'title': 'DEFAULT_TASK_CALENDAR_NAME', 'type': None}
{'ROWID': 5, 'title': 'Facebook Birthdays', 'type': None}
{'ROWID': 6, 'title': 'Found in Mail', 'type': None}
{'ROWID': 7, 'title': 'Found in Natural Language', 'type': None}
{'ROWID': 9, 'title': 'Birthdays', 'type': None}
{'ROWID': 10, 'title': 'thisisdfir@gmail.com', 'type': None}
{'ROWID': 11, 'title': 'thisisdfir@gmail.com', 'type': None}
{'ROWID': 12, 'title': 'Work', 'type': None}
{'ROWID': 13, 'title': 'Home', 'type': None}
{'ROWID': 14, 'title': 'US Holidays', 'type': None}
{'ROWID': 15, 'title': 'thisisdfir@gmail.com', 'type': None}
{'ROWID': 16, 'title': 'thisisdfir@gmail.com', 'type': None}

==== TIMED CALENDAR ITEM ====
{'ROWID': 119, 'summary': 'Pick up lunch', 'start_date': 608392800.0, 'end_date': 608396400.0, 'creation_date': 608391316.844578, 'last_modified': 608391316.84693
, 'all_day': 0, 'calendar_id': 13, 'status': 0, 'has_recurrences': 0}

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

ROOT="$HOME/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/PhotoData"

echo "===== SOURCE SHA256 ====="
sudo find "$ROOT" -maxdepth 1 -type f -name 'Photos.sqlite*' \
  -exec sha256sum {} \;

echo
echo "===== WORKING COPY SHA256 ====="
sha256sum working/photos/Photos.sqlite*
===== SOURCE SHA256 =====
3d0689786153247456ce2cd6ffd035de6249b228956ee0d73f6e37f79659c806  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media
/PhotoData/Photos.sqlite
374447748e8e2ad952f059f2c9e9fb350e5dcd4f019681516d5894730ba4bc8c  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media
/PhotoData/Photos.sqlite-wal
a29019f6d00d6f0539063e27c50561f57145b4c0bfa83872be2f429055353ab2  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media
/PhotoData/Photos.sqlite-shm

===== WORKING COPY SHA256 =====
3d0689786153247456ce2cd6ffd035de6249b228956ee0d73f6e37f79659c806  working/photos/Photos.sqlite
a29019f6d00d6f0539063e27c50561f57145b4c0bfa83872be2f429055353ab2  working/photos/Photos.sqlite-shm
374447748e8e2ad952f059f2c9e9fb350e5dcd4f019681516d5894730ba4bc8c  working/photos/Photos.sqlite-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ PHOTOS DATABASE INTEGRITY ═══"
sqlite3 working/photos/Photos.sqlite "PRAGMA integrity_check;"

echo
echo "═══ PHOTOS DATABASE TABLES ═══"
sqlite3 working/photos/Photos.sqlite "
SELECT name
FROM sqlite_master
WHERE type='table'
ORDER BY name;
"
═══ PHOTOS DATABASE INTEGRITY ═══
ok

═══ PHOTOS DATABASE TABLES ═══
ACHANGE
ATRANSACTION
ATRANSACTIONSTRING
ZADDITIONALASSETATTRIBUTES
ZADJUSTMENT
ZALBUMLIST
ZASSETANALYSISSTATE
ZASSETDESCRIPTION
ZCLOUDFEEDENTRY
ZCLOUDMASTER
ZCLOUDMASTERMEDIAMETADATA
ZCLOUDRESOURCE
ZCLOUDSHAREDALBUMINVITATIONRECORD
ZCLOUDSHAREDCOMMENT
ZCODEC
ZCOMPUTEDASSETATTRIBUTES
ZDEFERREDREBUILDFACE
ZDETECTEDFACE
ZDETECTEDFACEGROUP
ZDETECTEDFACEPRINT
```

```
═══ RELATED ASSET TABLE SCHEMAS ═══
── ZADDITIONALASSETATTRIBUTES ──
Z_PK (INTEGER)
Z_ENT (INTEGER)
Z_OPT (INTEGER)
ZALLOWEDFORANALYSIS (INTEGER)
ZCAMERACAPTUREDEVICE (INTEGER)
ZCLOUDAVALANCHEPICKTYPE (INTEGER)
ZCLOUDGROUPINGSTATE (INTEGER)
ZCLOUDKINDSUBTYPE (INTEGER)
ZCLOUDRECOVERYSTATE (INTEGER)
ZCLOUDSTATERECOVERYATTEMPTSCOUNT (INTEGER)
ZDESTINATIONASSETCOPYSTATE (INTEGER)
ZEMBEDDEDTHUMBNAILHEIGHT (INTEGER)
ZEMBEDDEDTHUMBNAILLENGTH (INTEGER)
ZEMBEDDEDTHUMBNAILOFFSET (INTEGER)
ZEMBEDDEDTHUMBNAILWIDTH (INTEGER)
ZIMPORTEDBY (INTEGER)
ZINFERREDTIMEZONEOFFSET (INTEGER)
ZLOCATIONHASH (INTEGER)
ZORIGINALFILESIZE (INTEGER)
ZORIGINALHEIGHT (INTEGER)
ZORIGINALORIENTATION (INTEGER)
ZORIGINALRESOURCECHOICE (INTEGER)
ZORIGINALWIDTH (INTEGER)
ZPENDINGPLAYCOUNT (INTEGER)
ZPENDINGSHARECOUNT (INTEGER)
ZPENDINGVIEWCOUNT (INTEGER)
ZPLAYCOUNT (INTEGER)
ZPTPTRASHEDSTATE (INTEGER)
ZREVERSELOCATIONDATAISVALID (INTEGER)
ZSCENEANALYSISVERSION (INTEGER)
ZSHARECOUNT (INTEGER)
ZSHARETYPE (INTEGER)
ZSHIFTEDLOCATIONISVALID (INTEGER)
ZTIMEZONEOFFSET (INTEGER)
ZUPLOADATTEMPTS (INTEGER)
ZVARIATIONSUGGESTIONSTATES (INTEGER)
ZVIDEOCPDISPLAYTIMESCALE (INTEGER)
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ MAIN PHOTO ASSET SCHEMA ═══"
sqlite3 working/photos/Photos.sqlite "PRAGMA table_info(ZGENERICASSET);"

echo
echo "═══ RELATED ASSET TABLE SCHEMAS ═══"
for T in ZADDITIONALASSETATTRIBUTES ZEXTENDEDATTRIBUTES ZINTERNALRESOURCE ZCLOUDRESOURCE ZGENERICALBUM; do
    echo "── $T ──"
    sqlite3 working/photos/Photos.sqlite "PRAGMA table_info($T);" |
    awk -F'|' '{print $2 " (" $3 ")"}'
done

echo
echo "═══ IMPORTANT TABLE COUNTS ═══"
for T in ZGENERICASSET ZGENERICALBUM ZADDITIONALASSETATTRIBUTES ZEXTENDEDATTRIBUTES ZINTERNALRESOURCE ZCLOUDRESOURCE ZDETECTEDFACE ZPERSON; do
    printf "%-35s " "$T"
    sqlite3 working/photos/Photos.sqlite "SELECT COUNT(*) FROM $T;"
done
═══ MAIN PHOTO ASSET SCHEMA ═══
0|Z_PK|INTEGER|0||1
1|Z_ENT|INTEGER|0||0
2|Z_OPT|INTEGER|0||0
3|ZAVALANCHEPICKTYPE|INTEGER|0||0
4|ZCLOUDDELETESTATE|INTEGER|0||0
5|ZCLOUDDOWNLOADREQUESTS|INTEGER|0||0
6|ZCLOUDHASCOMMENTSBYME|INTEGER|0||0
7|ZCLOUDHASCOMMENTSCONVERSATION|INTEGER|0||0
8|ZCLOUDHASUNSEENCOMMENTS|INTEGER|0||0
9|ZCLOUDISDELETABLE|INTEGER|0||0
10|ZCLOUDISMYASSET|INTEGER|0||0
11|ZCLOUDLOCALSTATE|INTEGER|0||0
12|ZCLOUDPLACEHOLDERKIND|INTEGER|0||0
13|ZCOMPLETE|INTEGER|0||0
14|ZCUSTOMRENDEREDVALUE|INTEGER|0||0
15|ZDEFERREDPROCESSINGNEEDED|INTEGER|0||0
16|ZDEPTHSTATES|INTEGER|0||0
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    Z_PK AS Asset_ID,
    ZFILENAME AS Filename,
    ZUNIFORMTYPEIDENTIFIER AS Media_Type,
    ZKIND AS Kind,
    ZWIDTH AS Width,
    ZHEIGHT AS Height,
    ROUND(ZDURATION, 2) AS Duration,
    datetime(ZDATECREATED + 978307200, 'unixepoch') AS Created_UTC,
    datetime(ZMODIFICATIONDATE + 978307200, 'unixepoch') AS Modified_UTC,
    ZFAVORITE AS Favorite,
    ZHIDDEN AS Hidden
FROM ZGENERICASSET
ORDER BY ZDATECREATED;
" | tee output/media/Photos_Asset_Metadata_UTC.txt
```

Asset_ID	Filename	Media_Type	Kind	Width	Height	Duration	Created_UTC	Modified_UTC	Favorite	Hidden
1	IMG_0001.JPG	public.jpeg	0	1125	1137	0.0	2020-03-22 13:05:24	2020-04-06 15:02:38	0	0
2	IMG_0002.JPG	public.jpeg	0	3550	3150	0.0	2020-03-22 13:05:39	2020-04-06 15:02:38	0	0
3	IMG_0003.JPG	public.jpeg	0	1668	860	0.0	2020-03-22 13:10:50	2020-04-13 19:32:52	0	0
5	IMG_0005.JPG	public.jpeg	0	1522	1775	0.0	2020-03-22 13:25:02	2020-04-06 15:02:38	0	0
6	IMG_0006.JPG	public.jpeg	0	1170	1175	0.0	2020-03-22 13:25:02	2020-04-06 15:02:38	0	0
7	IMG_0007.JPG	public.jpeg	0	1178	781	0.0	2020-03-22 13:25:02	2020-04-12 18:51:13	0	0
8	IMG_0008.JPG	public.jpeg	0	1125	1156	0.0	2020-03-22 13:25:02	2020-04-13 19:32:52	0	0
4	IMG_0004.JPG	public.jpeg	0	1117	886	0.0	2020-03-22 13:25:02	2020-04-06 15:02:38	0	0
9	IMG_0009.JPG	public.jpeg	0	828	637	0.0	2020-03-22 13:31:03	2020-04-06 15:02:38	0	0
10	IMG_0010.JPG	public.jpeg	0	620	553	0.0	2020-03-22 13:40:14	2020-04-12 18:51:12	0	0
11	IMG_0011.JPG	public.jpeg	0	450	465	0.0	2020-03-22 14:03:09	2020-04-06 15:02:38	0	0
12	IMG_0012.JPG	public.jpeg	0	828	763	0.0	2020-03-22 15:06:41	2020-04-12 18:51:13	0	0
13	IMG_0013.JPG	public.jpeg	0	638	884	0.0	2020-03-22 15:08:14	2020-04-06 15:02:38	0	0
14	IMG_0014.JPG	public.jpeg	0	1079	1121	0.0	2020-03-22 15:09:16	2020-04-12 18:51:13	0	0
15	IMG_0015.PNG	public.jpeg	0	640	611	0.0	2020-03-22 18:42:11	2020-04-13 19:32:52	0	0
16	IMG_0016.JPG	public.jpeg	0	640	640	0.0	2020-03-22 18:43:51	2020-04-06 15:02:38	0	0
17	IMG_0017.JPG	public.jpeg	0	640	285	0.0	2020-03-22 18:49:20	2020-04-06 15:02:38	0	0

```
    ZREMOTEAVAILABILITY AS Remote_Availability
FROM ZINTERNALRESOURCE
ORDER BY ZASSET, Z_PK
LIMIT 30;
"
```

==== INTERNAL RESOURCE COUNTS ====

Resource_Type	Record_Count	Total_Bytes
0	152	65289680
1	3	13667957
3	3	5613149

==== RESOURCE AVAILABILITY ====

Local_Availability	Remote_Availability	Record_Count
-1	1	94
1	1	64

==== RESOURCE FILE REFERENCES ====

Resource_ID	Asset_ID	Resource_Type	Data_Length	Width	Height	Local_Availability	Remote_Availability
1	1	0	241802	1125	1137	1	1
36	1	0	55938	414	418	-1	1
2	2	0	1187913	3550	3150	1	1
28	2	0	37845	440	390	-1	1
31	2	0	365929	1882	1670	-1	1
3	3	0	161764	1668	860	1	1
20	3	0	42331	579	299	-1	1
4	4	0	366221	1117	886	1	1
29	4	0	33880	467	370	-1	1
5	5	0	1762050	1522	1775	1	1
35	5	0	47772	384	448	-1	1
6	6	0	994503	1170	1175	1	1
26	6	0	71266	414	416	-1	1
7	7	0	563391	1178	781	1	1
23	7	0	48665	511	339	-1	1
8	8	0	676542	1125	1156	1	1
32	8	0	32584	410	421	-1	1
9	9	0	135392	828	637	1	1

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ >....
sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    ZRESOURCETYPE AS Resource_Type,
    COUNT(*) AS Record_Count,
    SUM(ZDATALENGTH) AS Total_Bytes
FROM ZINTERNALRESOURCE
GROUP BY ZRESOURCETYPE
ORDER BY ZRESOURCETYPE;
"

echo
echo "===== RESOURCE AVAILABILITY ====="
sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    ZLOCALAVAILABILITY AS Local_Availability,
    ZREMOTEAVAILABILITY AS Remote_Availability,
    COUNT(*) AS Record_Count
FROM ZINTERNALRESOURCE
GROUP BY ZLOCALAVAILABILITY, ZREMOTEAVAILABILITY
ORDER BY ZLOCALAVAILABILITY, ZREMOTEAVAILABILITY;
"

echo
echo "===== RESOURCE FILE REFERENCES ====="
sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    Z_PK AS Resource_ID,
    ZASSET AS Asset_ID,
    ZRESOURCETYPE AS Resource_Type,
    ZDATALENGTH AS Data_Length,
    ZUNORIENTEDWIDTH AS Width,
    ZUNORIENTEDHEIGHT AS Height,
    ZLOCALAVAILABILITY AS Local_Availability,
    ZREMOTEAVAILABILITY AS Remote_Availability
FROM ZINTERNALRESOURCE
ORDER BY ZASSET, Z_PK
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

BASE="$HOME/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1"

echo "═══ DCIM DIRECTORY ═══"
sudo find "$BASE/private/var/mobile/Media/DCIM" \
  -maxdepth 3 -type f 2>/dev/null | head -40

echo
echo "═══ PHOTODATA MEDIA DIRECTORIES ═══"
sudo find "$BASE/private/var/mobile/Media/PhotoData" \
  -maxdepth 3 -type d 2>/dev/null | head -60

echo
echo "═══ MEDIA FILE COUNTS ═══"
for D in "$BASE/private/var/mobile/Media/DCIM" \
         "$BASE/private/var/mobile/Media/PhotoData"; do
    echo "── $D ──"
    sudo find "$D" -type f 2>/dev/null | wc -l
done
═══ DCIM DIRECTORY ═══
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0010.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0033.MOV
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0032.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0010.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0057.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0051.PNG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0013.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0046.PNG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0027.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0001.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0039.PNG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/._IMG_0017.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0012.JPG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0044.PNG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0043.PNG
/home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0004.JPG
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    Z_PK AS Asset_ID,
    ZFILENAME AS Filename,
    ZDIRECTORY AS Directory,
    ZUNIFORMTYPEIDENTIFIER AS File_Type
FROM ZGENERICASSET
ORDER BY Z_PK;
"
```

Asset_ID	Filename	Directory	File_Type
1	IMG_0001.JPG	DCIM/100APPLE	public.jpeg
2	IMG_0002.JPG	DCIM/100APPLE	public.jpeg
3	IMG_0003.JPG	DCIM/100APPLE	public.jpeg
4	IMG_0004.JPG	DCIM/100APPLE	public.jpeg
5	IMG_0005.JPG	DCIM/100APPLE	public.jpeg
6	IMG_0006.JPG	DCIM/100APPLE	public.jpeg
7	IMG_0007.JPG	DCIM/100APPLE	public.jpeg
8	IMG_0008.JPG	DCIM/100APPLE	public.jpeg
9	IMG_0009.JPG	DCIM/100APPLE	public.jpeg
10	IMG_0010.JPG	DCIM/100APPLE	public.jpeg
11	IMG_0011.JPG	DCIM/100APPLE	public.jpeg
12	IMG_0012.JPG	DCIM/100APPLE	public.jpeg
13	IMG_0013.JPG	DCIM/100APPLE	public.jpeg
14	IMG_0014.JPG	DCIM/100APPLE	public.jpeg
15	IMG_0015.PNG	DCIM/100APPLE	public.jpeg
16	IMG_0016.JPG	DCIM/100APPLE	public.jpeg
17	IMG_0017.JPG	DCIM/100APPLE	public.jpeg
18	IMG_0018.JPG	DCIM/100APPLE	public.jpeg
19	IMG_0019.JPG	DCIM/100APPLE	public.jpeg
20	IMG_0020.PNG	DCIM/100APPLE	public.png
21	IMG_0021.JPG	DCIM/100APPLE	public.jpeg
22	IMG_0022.PNG	DCIM/100APPLE	public.jpeg
23	IMG_0023.JPG	DCIM/100APPLE	public.jpeg
24	IMG_0024.JPG	DCIM/100APPLE	public.jpeg

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DCIM="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE"

echo "═══ ACTUAL MEDIA FILES ═══"
sudo find "$DCIM" -maxdepth 1 -type f ! -name '._*' \
  -printf '%f\n' | sort

echo
echo "═══ FILE COUNTS ═══"
echo -n "All files: "
sudo find "$DCIM" -maxdepth 1 -type f | wc -l

echo -n "AppleDouble sidecars: "
sudo find "$DCIM" -maxdepth 1 -type f -name '._*' | wc -l

echo -n "Non-sidecar files: "
sudo find "$DCIM" -maxdepth 1 -type f ! -name '._*' | wc -l
═══ ACTUAL MEDIA FILES ═══
IMG_0001.JPG
IMG_0002.JPG
IMG_0003.JPG
IMG_0004.JPG
IMG_0005.JPG
IMG_0006.JPG
IMG_0007.JPG
IMG_0008.JPG
IMG_0009.JPG
IMG_0010.JPG
IMG_0011.JPG
IMG_0012.JPG
IMG_0013.JPG
IMG_0014.JPG
IMG_0015.PNG
IMG_0016.JPG
IMG_0017.JPG
IMG_0018.JPG
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    Z_PK AS Resource_ID,
    ZASSET AS Asset_ID,
    ZRESOURCETYPE AS Resource_Type,
    ZFILEID AS File_ID,
    ZDATALENGTH AS Data_Length,
    ZLOCALAVAILABILITY AS Local_Availability,
    ZREMOTEAVAILABILITY AS Remote_Availability
FROM ZINTERNALRESOURCE
WHERE ZASSET = 37
ORDER BY Z_PK;
"
Resource_ID  Asset_ID  Resource_Type  File_ID  Data_Length  Local_Availability  Remote_Availability
-----------  --------  -------------  -------  -----------  ------------------  -------------------
89           37        1              905720   9125655      1                   1
90           37        0              906147   74957        1                   1
91           37        1              -1       1010642      -1                  1
92           37        0              -1       22059        -1                  1
93           37        1              -1       3531660      -1                  1

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DCIM="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE"

echo "===== VIDEO FILE METADATA ====="
sudo stat "$DCIM/IMG_0033.MOV" "$DCIM/IMG_0037.MOV"

echo
echo "===== VIDEO FILE SHA-256 ====="
sudo sha256sum "$DCIM/IMG_0033.MOV" "$DCIM/IMG_0037.MOV"

echo
echo "===== VIDEO FILE TYPES ====="
sudo file "$DCIM/IMG_0033.MOV" "$DCIM/IMG_0037.MOV"
===== VIDEO FILE METADATA =====
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0033.MOV
  Size: 3370449         Blocks: 6584       IO Block: 4096   regular file
Device: 8,1     Inode: 4754241     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:45:24.928564207 -0500
Modify: 2020-04-11 13:56:11.000000000 -0400
Change: 2026-11-04 03:45:24.964562454 -0500
 Birth: 2026-11-04 03:45:24.928564207 -0500
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0037.MOV
  Size: 9125655         Blocks: 17824      IO Block: 4096   regular file
Device: 8,1     Inode: 4754249     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:45:25.004560506 -0500
Modify: 2020-04-11 21:57:11.000000000 -0400
Change: 2026-11-04 03:45:25.216550176 -0500
 Birth: 2026-11-04 03:45:25.004560506 -0500

===== VIDEO FILE SHA-256 =====
bb8f797b7754975f1b88c0b578b2a223d6171a102804f6971f7eedbeb21ab6bc  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0033.MOV
cb6f2a93460381150ba049a2a73b875e9f9e8b47eccf2b9fa6645dfa55a331c7  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0037.MOV

===== VIDEO FILE TYPES =====
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ which exiftool

DCIM="$HOME/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE"

sudo exiftool "$DCIM/IMG_0033.MOV" "$DCIM/IMG_0037.MOV"
/usr/bin/exiftool
======== /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0033.MOV
ExifTool Version Number         : 13.25
File Name                       : IMG_0033.MOV
Directory                       : /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE
File Size                       : 3.4 MB
File Modification Date/Time     : 2020:04:11 13:56:11-04:00
File Access Date/Time           : 2026:11:04 18:13:45-05:00
File Inode Change Date/Time     : 2026:11:04 03:45:24-05:00
File Permissions                : -rw-r--r--
File Type                       : MOV
File Type Extension             : mov
MIME Type                       : video/quicktime
Major Brand                     : Apple QuickTime (.MOV/QT)
Minor Version                   : 0.0.0
Compatible Brands               : qt
Media Data Size                 : 3366439
Media Data Offset               : 36
Movie Header Version            : 0
Create Date                     : 2020:04:11 17:56:09
Modify Date                     : 2020:04:11 17:56:11
Time Scale                      : 600
Duration                        : 2.70 s
Preferred Rate                  : 1
Preferred Volume                : 100.00%
Preview Time                    : 0 s
Preview Duration                : 0 s
Poster Time                     : 0 s
Selection Time                  : 0 s
Selection Duration              : 0 s
Current Time                    : 0 s
Next Track ID                   : 5
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== PHOTOS ALBUM TABLES ====="
sqlite3 working/photos/Photos.sqlite "
SELECT name
FROM sqlite_master
WHERE type='table'
AND (name LIKE '%ALBUM%' OR name LIKE '%ASSET%')
ORDER BY name;
"

echo
echo "===== ALBUM TABLE STRUCTURE ====="
sqlite3 -header -column working/photos/Photos.sqlite "
PRAGMA table_info(ZGENERICALBUM);
"

echo
echo "===== ALBUM RECORD COUNTS ====="
sqlite3 -header -column working/photos/Photos.sqlite "
SELECT COUNT(*) AS Total_Albums FROM ZGENERICALBUM;
"
===== PHOTOS ALBUM TABLES =====
ZADDITIONALASSETATTRIBUTES
ZALBUMLIST
ZASSETANALYSISSTATE
ZASSETDESCRIPTION
ZCLOUDSHAREDALBUMINVITATIONRECORD
ZCOMPUTEDASSETATTRIBUTES
ZGENERICALBUM
ZGENERICASSET
ZMEDIAANALYSISASSETATTRIBUTES
Z_25ALBUMLISTS
Z_26ASSETS
Z_34MEMORIESBEINGCURATEDASSETS
Z_34MEMORIESBEINGEXTENDEDCURATEDASSETS
Z_34MEMORIESBEINGMOVIECURATEDASSETS
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    Z_PK AS Album_ID,
    ZTITLE AS Album_Title,
    ZKIND AS Album_Kind,
    ZCACHEDCOUNT AS Cached_Asset_Count,
    ZCACHEDPHOTOSCOUNT AS Cached_Photo_Count,
    ZCACHEDVIDEOSCOUNT AS Cached_Video_Count
FROM ZGENERICALBUM
ORDER BY Z_PK;
"
```

Album_ID	Album_Title	Album_Kind	Cached_Asset_Count	Cached_Photo_Count	Cached_Video_Count
1		1609	0	0	0
2		1628	0	0	0
3	progress-ota-restore	3572	0	0	0
4	progress-sync	3571	0	0	0
5		1607	0	0	0
6		1614	0	0	0
7		1622	0	0	0
8		4002	0	0	0
9		4003	0	0	0
10		1617	0	0	0
11		1608	0	0	0
12		1611	0	0	0
13		4006	0	0	0
14		1613	0	0	0
15		1600	0	0	0
16		1610	0	0	0
17		1606	0	0	0
18		1619	0	0	0
19		3999	0	0	0
20		1602	0	0	0
21		1623	0	0	0
22		1621	0	0	0

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ ALBUM-ASSET RELATIONSHIP SCHEMA ═══"
sqlite3 -header -column working/photos/Photos.sqlite "
PRAGMA table_info(Z_26ASSETS);
"

echo
echo "═══ ALBUM-ASSET RELATIONSHIP RECORDS ═══"
sqlite3 -header -column working/photos/Photos.sqlite "
SELECT * FROM Z_26ASSETS;
"
═══ ALBUM-ASSET RELATIONSHIP SCHEMA ═══
cid  name            type     notnull  dflt_value  pk
───  ──────────────  ───────  ───────  ──────────  ──
0    Z_26ALBUMS      INTEGER  0                    1
1    Z_34ASSETS      INTEGER  0                    2
2    Z_FOK_34ASSETS  INTEGER  0                    0

═══ ALBUM-ASSET RELATIONSHIP RECORDS ═══
Z_26ALBUMS  Z_34ASSETS  Z_FOK_34ASSETS
──────────  ──────────  ──────────────
39          12          2048
39          13          3072
39          14          4096
40          28          2048
40          29          3072

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -header -column working/photos/Photos.sqlite "
SELECT
    a.Z_PK AS Asset_ID,
    a.ZFILENAME AS Filename,
    a.ZUNIFORMTYPEIDENTIFIER AS File_Type,
    datetime(a.ZDATECREATED + 978307200, 'unixepoch') AS Created_UTC,
    CASE
        WHEN al.Z_26ALBUMS = 39 THEN 'Imgur'
        WHEN al.Z_26ALBUMS = 40 THEN 'WhatsApp'
    END AS Album
FROM Z_26ASSETS al
JOIN ZGENERICASSET a ON a.Z_PK = al.Z_34ASSETS
ORDER BY al.Z_26ALBUMS, a.Z_PK;
"
Asset_ID  Filename      File_Type    Created_UTC          Album
--------  ------------  -----------  -------------------  --------
12        IMG_0012.JPG  public.jpeg  2020-03-22 15:06:41  Imgur
13        IMG_0013.JPG  public.jpeg  2020-03-22 15:08:14  Imgur
14        IMG_0014.JPG  public.jpeg  2020-03-22 15:09:16  Imgur
28        IMG_0028.JPG  public.jpeg  2020-03-27 00:38:00  WhatsApp
29        IMG_0029.JPG  public.jpeg  2020-03-27 00:38:00  WhatsApp

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

SAFARI="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Safari"

echo "===== COPYING SAFARI DATABASE FILES ====="
sudo cp "$SAFARI/History.db" working/safari/
sudo cp "$SAFARI/History.db-shm" working/safari/
sudo cp "$SAFARI/History.db-wal" working/safari/

sudo chown -R kali:kali working/safari

echo
echo "===== SOURCE HASHES ====="
sudo bash -c "cd '$SAFARI' && sha256sum History.db History.db-shm History.db-wal" \
  | tee output/hashes/Safari_Source_SHA256.txt

echo
echo "===== WORKING COPY HASHES ====="
(cd working/safari && sha256sum History.db History.db-shm History.db-wal) \
  | tee output/hashes/Safari_Working_SHA256.txt

echo
echo "===== HASH COMPARISON ====="
diff -u output/hashes/Safari_Source_SHA256.txt \
  output/hashes/Safari_Working_SHA256.txt

echo
echo "===== WORKING COPY FILES ====="
ls -lh working/safari/
===== COPYING SAFARI DATABASE FILES =====

===== SOURCE HASHES =====
b33a703a085bc84e49a955d453b36a0be12082721adc9a1efa9ad8923738b811  History.db
1023506566fe45893aee01e2c51f27946f09c1af833f6ad7c9bd8b88c4976cb0  History.db-shm
82873188823dc8c426830af2ab09e809f0954b6fd89fb58a8ebcd11db0ea6de4  History.db-wal

===== WORKING COPY HASHES =====
```

```
echo "===== SOURCE HASHES ====="
sudo bash -c "cd '$SAFARI' && sha256sum History.db History.db-shm History.db-wal" \
  | tee output/hashes/Safari_Source_SHA256.txt

echo
echo "===== WORKING COPY HASHES ====="
(cd working/safari && sha256sum History.db History.db-shm History.db-wal) \
  | tee output/hashes/Safari_Working_SHA256.txt

echo
echo "===== HASH COMPARISON ====="
diff -u output/hashes/Safari_Source_SHA256.txt \
  output/hashes/Safari_Working_SHA256.txt

echo
echo "===== WORKING COPY FILES ====="
ls -lh working/safari/
===== COPYING SAFARI DATABASE FILES =====

===== SOURCE HASHES =====
b33a703a085bc84e49a955d453b36a0be12082721adc9a1efa9ad8923738b811  History.db
1023506566fe45893aee01e2c51f27946f09c1af833f6ad7c9bd8b88c4976cb0  History.db-shm
82873188823dc8c426830af2ab09e809f0954b6fd89fb58a8ebcd11db0ea6de4  History.db-wal

===== WORKING COPY HASHES =====
b33a703a085bc84e49a955d453b36a0be12082721adc9a1efa9ad8923738b811  History.db
1023506566fe45893aee01e2c51f27946f09c1af833f6ad7c9bd8b88c4976cb0  History.db-shm
82873188823dc8c426830af2ab09e809f0954b6fd89fb58a8ebcd11db0ea6de4  History.db-wal

===== HASH COMPARISON =====

===== WORKING COPY FILES =====
total 164K
-rw-r--r-- 1 kali kali 96K Nov  4 16:05 History.db
-rw-r--r-- 1 kali kali 32K Nov  4 16:05 History.db-shm
-rw-r--r-- 1 kali kali 33K Nov  4 16:05 History.db-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/safari/History.db <<'SQL'
.headers on
.mode column

.print '===== DATABASE INTEGRITY ====='
PRAGMA integrity_check;

.print '===== TABLES AND VIEWS ====='
SELECT name, type
FROM sqlite_master
WHERE type IN ('table', 'view')
ORDER BY name;

.print '===== DATABASE SCHEMA ====='
.schema
SQL
===== DATABASE INTEGRITY =====
integrity_check
---------------
ok
===== TABLES AND VIEWS =====
name                    type
----------------------  -----
history_client_versions  table
history_event_listeners  table
history_events           table
history_items            table
history_items_to_tags    table
history_tags             table
history_tombstones       table
history_visits           table
metadata                 table
sqlite_sequence          table
===== DATABASE SCHEMA =====
CREATE TABLE history_items (id INTEGER PRIMARY KEY AUTOINCREMENT,url TEXT NOT NULL UNIQUE,domain_expansion TEXT NULL,visit_count INTEGER NOT NULL,daily_visit_coun
```

```
history_events           table
history_items            table
history_items_to_tags    table
history_tags             table
history_tombstones       table
history_visits           table
metadata                 table
sqlite_sequence          table
═══ DATABASE SCHEMA ═══
CREATE TABLE history_items (id INTEGER PRIMARY KEY AUTOINCREMENT,url TEXT NOT NULL UNIQUE,domain_expansion TEXT NULL,visit_count INTEGER NOT NULL,daily_visit_coun
ts BLOB NOT NULL,weekly_visit_counts BLOB NULL,autocomplete_triggers BLOB NULL,should_recompute_derived_visit_counts INTEGER NOT NULL,visit_count_score INTEGER NO
T NULL);
CREATE TABLE sqlite_sequence(name,seq);
CREATE TABLE history_visits (id INTEGER PRIMARY KEY AUTOINCREMENT,history_item INTEGER NOT NULL REFERENCES history_items(id) ON DELETE CASCADE,visit_time REAL NOT
 NULL,title TEXT NULL,load_successful BOOLEAN NOT NULL DEFAULT 1,http_non_get BOOLEAN NOT NULL DEFAULT 0,synthesized BOOLEAN NOT NULL DEFAULT 0,redirect_source IN
TEGER NULL UNIQUE REFERENCES history_visits(id) ON DELETE CASCADE,redirect_destination INTEGER NULL UNIQUE REFERENCES history_visits(id) ON DELETE CASCADE,origin
INTEGER NOT NULL DEFAULT 0,generation INTEGER NOT NULL DEFAULT 0,attributes INTEGER NOT NULL DEFAULT 0,score INTEGER NOT NULL DEFAULT 0);
CREATE TABLE history_tombstones (id INTEGER PRIMARY KEY AUTOINCREMENT,start_time REAL NOT NULL,end_time REAL NOT NULL,url TEXT,generation INTEGER NOT NULL DEFAULT
 0);
CREATE TABLE metadata (key TEXT NOT NULL UNIQUE, value);
CREATE TABLE history_client_versions (client_version INTEGER PRIMARY KEY,last_seen REAL NOT NULL);
CREATE TABLE history_event_listeners (listener_name TEXT PRIMARY KEY NOT NULL UNIQUE,last_seen REAL NOT NULL);
CREATE TABLE history_events (id INTEGER PRIMARY KEY AUTOINCREMENT,event_type TEXT NOT NULL,event_time REAL NOT NULL,pending_listeners TEXT NOT NULL,value BLOB);
CREATE TABLE history_tags (id INTEGER PRIMARY KEY,type INTEGER NOT NULL,level INTEGER NOT NULL,identifier TEXT NOT NULL,title TEXT NOT NULL,modification_timestamp
 REAL NOT NULL,item_count INTEGER NOT NULL DEFAULT 0);
CREATE TABLE history_items_to_tags (history_item INTEGER NOT NULL,tag_id INTEGER NOT NULL,timestamp REAL NOT NULL,FOREIGN KEY(tag_id) REFERENCES history_tags(id)
ON DELETE CASCADE,FOREIGN KEY(history_item) REFERENCES history_items(id) ON DELETE CASCADE,UNIQUE(history_item, tag_id) ON CONFLICT REPLACE);
CREATE TRIGGER increment_count_on_insert AFTER INSERT ON history_items_to_tags BEGIN  UPDATE history_tags SET item_count = item_count + 1 WHERE id = NEW.tag_id;EN
D;
CREATE TRIGGER decrement_count_on_delete BEFORE DELETE ON history_items_to_tags BEGIN  UPDATE history_tags SET item_count = item_count - 1 WHERE id = OLD.tag_id;E
ND;
CREATE INDEX history_items__domain_expansion ON history_items (domain_expansion);
CREATE INDEX history_visits__last_visit ON history_visits (history_item, visit_time DESC, synthesized ASC);
CREATE INDEX history_visits__origin ON history_visits (origin, generation);
CREATE INDEX history_tombstones__generation ON history_tombstones (generation);
CREATE INDEX history_tombstones__end_time ON history_tombstones (end_time);

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/safari/History.db <<'SQL'
.headers on
.mode column

.print '===== SAFARI TABLE RECORD COUNTS ====='
SELECT 'history_items' AS table_name, COUNT(*) AS records FROM history_items
UNION ALL
SELECT 'history_visits', COUNT(*) FROM history_visits
UNION ALL
SELECT 'history_tombstones', COUNT(*) FROM history_tombstones
UNION ALL
SELECT 'history_tags', COUNT(*) FROM history_tags
UNION ALL
SELECT 'history_events', COUNT(*) FROM history_events
UNION ALL
SELECT 'metadata', COUNT(*) FROM metadata;

.print '===== HISTORY ITEM AND VISIT RELATIONSHIP ====='
SELECT
    COUNT(DISTINCT history_item) AS items_with_visits,
    COUNT(*) AS total_visits
FROM history_visits;
SQL
===== SAFARI TABLE RECORD COUNTS =====
table_name          records
------------------  -------
history_items       10
history_visits      16
history_tombstones  0
history_tags        1
history_events      0
metadata            10
===== HISTORY ITEM AND VISIT RELATIONSHIP =====
items_with_visits  total_visits
-----------------  ------------
```

```
.print '===== HISTORY ITEM AND VISIT RELATIONSHIP ====='
SELECT
    COUNT(DISTINCT history_item) AS items_with_visits,
    COUNT(*) AS total_visits
FROM history_visits;
SQL
===== SAFARI TABLE RECORD COUNTS =====
table_name           records
-------------------  -------
history_items        10
history_visits       16
history_tombstones   0
history_tags         1
history_events       0
metadata             10
===== HISTORY ITEM AND VISIT RELATIONSHIP =====
items_with_visits  total_visits
-----------------  ------------
10                 16

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/safari/History.db <<'SQL'
.headers on
.mode column
.width 6 8 22 35 12

.print '===== SAFARI VISIT TIMELINE (UTC) ====='
SELECT
    v.id AS visit_id,
    v.history_item AS item_id,
    datetime(v.visit_time + 978307200, 'unixepoch') AS visit_time_utc,
    CASE
        WHEN instr(substr(i.url, instr(i.url, '://') + 3), '/') > 0
        THEN substr(
            substr(i.url, instr(i.url, '://') + 3),
            1,
            instr(substr(i.url, instr(i.url, '://') + 3), '/') - 1
        )
        ELSE substr(i.url, instr(i.url, '://') + 3)
    END AS domain,
    v.load_successful
FROM history_visits v
JOIN history_items i ON i.id = v.history_item
ORDER BY v.visit_time ASC;

.print '===== SAFARI TIMESTAMP RANGE ====='
SELECT
    datetime(MIN(visit_time) + 978307200, 'unixepoch') AS first_visit_utc,
    datetime(MAX(visit_time) + 978307200, 'unixepoch') AS last_visit_utc,
    COUNT(*) AS total_visits
FROM history_visits;
SQL
===== SAFARI VISIT TIMELINE (UTC) =====
visit_  item_id   visit_time_utc          domain                                load_success
------  --------  ----------------------  ------------------------------------  ------------
2       1         2020-03-28 00:58:36     www.google.com                        1
```

```
JOIN history_items i ON i.id = v.history_item
ORDER BY v.visit_time ASC;

.print '===== SAFARI TIMESTAMP RANGE ====='
SELECT
    datetime(MIN(visit_time) + 978307200, 'unixepoch') AS first_visit_utc,
    datetime(MAX(visit_time) + 978307200, 'unixepoch') AS last_visit_utc,
    COUNT(*) AS total_visits
FROM history_visits;
SQL
==== SAFARI VISIT TIMELINE (UTC) ====
visit_  item_id   visit_time_utc          domain                  load_success
------  --------  ----------------------  ----------------------  ------------
2       1         2020-03-28 00:58:36     www.google.com          1
1       1         2020-03-28 00:58:37     www.google.com          1
3       2         2020-03-28 00:59:03     www.google.com          1
4       2         2020-03-28 00:59:31     www.google.com          1
5       2         2020-03-28 00:59:32     www.google.com          1
6       1         2020-03-28 01:00:17     www.google.com          1
7       3         2020-03-28 01:02:05     www.google.com          1
9       4         2020-03-28 01:02:44     www.google.com          1
8       4         2020-03-28 01:02:44     www.google.com          1
11      5         2020-03-28 01:03:51     www.nhl.com             1
10      5         2020-03-28 01:03:53     www.nhl.com             1
12      6         2020-03-28 01:06:32     www.apple.com           1
13      7         2020-03-28 01:07:41     www.apple.com           1
14      8         2020-03-28 01:38:48     www.cultofmac.com       1
15      9         2020-03-28 01:43:49     dfir.pubpub.org         1
16      10        2020-03-28 01:44:37     dfir.pubpub.org         1
===== SAFARI TIMESTAMP RANGE =====
first_  last_vis  total_visits
------  --------  ------------
2020-0  2020-03-  16
3-28 0  28 01:44
0:58:3  :37
6

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

python3 - <<'PY'
import sqlite3
from urllib.parse import urlsplit

db = "working/safari/History.db"
con = sqlite3.connect(f"file:{db}?mode=ro", uri=True)

query = """
SELECT v.id, v.history_item, v.visit_time, v.title, i.url
FROM history_visits v
JOIN history_items i ON i.id = v.history_item
ORDER BY v.visit_time ASC
"""

print("===== SAFARI PAGE TITLES AND URL PATHS =====")
print(f"{'Visit':<7} {'Item':<6} {'UTC':<20} {'Domain':<28} {'Path':<35} Title")

for visit_id, item_id, visit_time, title, url in con.execute(query):
    parts = urlsplit(url)
    utc = __import__("datetime").datetime.fromtimestamp(
        visit_time + 978307200,
        __import__("datetime").timezone.utc
    ).strftime("%Y-%m-%d %H:%M:%S")

    print(
        f"{visit_id:<7} {item_id:<6} {utc:<20} "
        f"{parts.hostname or '':<28} {(parts.path or '/')[:34]:<35} "
        f"{(title or '[No title]')[:70]}"
    )

con.close()
PY
===== SAFARI PAGE TITLES AND URL PATHS =====
Visit   Item   UTC                  Domain                       Path                                Title
2       1      2020-03-28 00:58:36  www.google.com               /search                             when does mlb start 2020 - Google Search
```

```
print("==== SAFARI PAGE TITLES AND URL PATHS ====")
print(f"{'Visit':<7} {'Item':<6} {'UTC':<20} {'Domain':<28} {'Path':<35} Title")

for visit_id, item_id, visit_time, title, url in con.execute(query):
    parts = urlsplit(url)
    utc = __import__("datetime").datetime.fromtimestamp(
        visit_time + 978307200,
        __import__("datetime").timezone.utc
    ).strftime("%Y-%m-%d %H:%M:%S")

    print(
        f"{visit_id:<7} {item_id:<6} {utc:<20} "
        f"{parts.hostname or '':<28} {(parts.path or '/')[:34]:<35} "
        f"{(title or '[No title]')[:70]}"
    )

con.close()
PY
==== SAFARI PAGE TITLES AND URL PATHS ====
Visit   Item   UTC                  Domain                       Path                                 Title
2       1      2020-03-28 00:58:36  www.google.com               /search                              when does mlb start 2020 - Google Search
1       1      2020-03-28 00:58:37  www.google.com               /search                              when does mlb start 2020 - Google Search
3       2      2020-03-28 00:59:03  www.google.com               /amp/s/www.mlb.com/amp/news/mlb-20   MLB 2020 season delayed
4       2      2020-03-28 00:59:31  www.google.com               /amp/s/www.mlb.com/amp/news/mlb-20   MLB 2020 season delayed
5       2      2020-03-28 00:59:32  www.google.com               /amp/s/www.mlb.com/amp/news/mlb-20   MLB 2020 season delayed
6       1      2020-03-28 01:00:17  www.google.com               /search                              when does mlb start 2020 - Google Search
7       3      2020-03-28 01:02:05  www.google.com               /search                              when does mlb start 2020 - Google Search
9       4      2020-03-28 01:02:44  www.google.com               /search                              Is the NHL going to resume? - Google Search
8       4      2020-03-28 01:02:44  www.google.com               /search                              Is the NHL going to resume? - Google Search
11      5      2020-03-28 01:03:51  www.nhl.com                  /news/commissioner-gary-bettman-nh   NHL expects to resume season after pause for coronavirus
10      5      2020-03-28 01:03:53  www.nhl.com                  /news/commissioner-gary-bettman-nh   NHL expects to resume season after pause for coronavirus
12      6      2020-03-28 01:06:32  www.apple.com                /                                    Apple
13      7      2020-03-28 01:07:41  www.apple.com                /ipad-pro/                           iPad Pro - Apple
14      8      2020-03-28 01:38:48  www.cultofmac.com            /                                    Cult of Mac | Tech and culture through an Apple lens
15      9      2020-03-28 01:43:49  dfir.pubpub.org              /                                    DFIR Review
16      10     2020-03-28 01:44:37  dfir.pubpub.org              /pub/e5xlbw88                        iOS Mobile Installation Logs · DFIR Review

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ >....
sqlite3 -readonly working/safari/History.db <<'SQL'
.headers on
.mode column

.print '===== SAFARI METADATA STRUCTURE ====='
SELECT
    key,
    typeof(value) AS value_type,
    CASE
        WHEN typeof(value) IN ('text', 'blob')
        THEN length(value)
        ELSE NULL
    END AS value_length
FROM metadata
ORDER BY key;

.print '===== SAFARI CLIENT VERSIONS ====='
SELECT
    client_version,
    datetime(last_seen + 978307200, 'unixepoch') AS last_seen_utc
FROM history_client_versions
ORDER BY last_seen;

.print '===== SAFARI EVENT LISTENERS ====='
SELECT listener_name,
       datetime(last_seen + 978307200, 'unixepoch') AS last_seen_utc
FROM history_event_listeners
ORDER BY listener_name;

.print '===== SAFARI TAG RECORDS ====='
SELECT id, type, level, item_count
FROM history_tags
ORDER BY id;

.print '===== DELETION RECORD COUNT ====='
SELECT COUNT(*) AS tombstone_records
```

```
       datetime(last_seen + 978307200, 'unixepoch') AS last_seen_utc
FROM history_event_listeners
ORDER BY listener_name;

.print '===== SAFARI TAG RECORDS ====='
SELECT id, type, level, item_count
FROM history_tags
ORDER BY id;

.print '===== DELETION RECORD COUNT ====='
SELECT COUNT(*) AS tombstone_records
FROM history_tombstones;
SQL
===== SAFARI METADATA STRUCTURE =====
key                                     value_type  value_length
--------------------------------------  ----------  ------------
cached_sync_circle_size                 integer
current_generation                      integer
fetch_throttler_data                    blob        53
last_maintenance_date                   integer
last_synced_generation                  integer
push_notifications_initialized          integer
push_throttler_data                     blob        53
server_change_token                     blob        302
sync_circle_size_retrieval_throttler_data  blob        53
sync_with_manatee_container             integer
===== SAFARI CLIENT VERSIONS =====
===== SAFARI EVENT LISTENERS =====
===== SAFARI TAG RECORDS =====
id  type  level  item_count
--  ----  -----  ----------
1   1     200    1
===== DELETION RECORD COUNT =====
tombstone_records
-----------------
0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

CALENDAR="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Calendar"

mkdir -p working/calendar output/hashes

sudo cp "$CALENDAR/Calendar.sqlitedb" working/calendar/
sudo cp "$CALENDAR/Calendar.sqlitedb-shm" working/calendar/
sudo cp "$CALENDAR/Calendar.sqlitedb-wal" working/calendar/

sudo chown -R kali:kali working/calendar

echo "===== SOURCE HASHES ====="
sudo bash -c "cd '$CALENDAR' && sha256sum Calendar.sqlitedb Calendar.sqlitedb-shm Calendar.sqlitedb-wal" \
  | tee output/hashes/Calendar_Source_SHA256.txt

echo
echo "===== WORKING COPY HASHES ====="
(cd working/calendar && sha256sum Calendar.sqlitedb Calendar.sqlitedb-shm Calendar.sqlitedb-wal) \
  | tee output/hashes/Calendar_Working_SHA256.txt

echo
echo "===== HASH COMPARISON ====="
diff -u output/hashes/Calendar_Source_SHA256.txt \
  output/hashes/Calendar_Working_SHA256.txt
===== SOURCE HASHES =====
df625ce21ee114775855debb8a6e17eea17a95a068beb3de714b175c4e7448be  Calendar.sqlitedb
a4818b000143cad123f87ade4d125eaccdf94b61f5e315ca8ba3145f9c432639  Calendar.sqlitedb-shm
0c6ec7c6e12875b22cad8a90b0976fddf41cf3309be3149748c67ae8ad26b9ec  Calendar.sqlitedb-wal

===== WORKING COPY HASHES =====
df625ce21ee114775855debb8a6e17eea17a95a068beb3de714b175c4e7448be  Calendar.sqlitedb
a4818b000143cad123f87ade4d125eaccdf94b61f5e315ca8ba3145f9c432639  Calendar.sqlitedb-shm
0c6ec7c6e12875b22cad8a90b0976fddf41cf3309be3149748c67ae8ad26b9ec  Calendar.sqlitedb-wal

===== HASH COMPARISON =====
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/calendar/Calendar.sqlitedb <<'SQL'
.headers on
.mode column

.print '===== CALENDAR DATABASE INTEGRITY ====='
PRAGMA integrity_check;

.print '===== CALENDAR TABLES ====='
SELECT name, type
FROM sqlite_master
WHERE type IN ('table', 'view')
ORDER BY name;

.print '===== CALENDAR DATABASE SCHEMA ====='
.schema
SQL
===== CALENDAR DATABASE INTEGRITY =====
integrity_check
---------------
ok
===== CALENDAR TABLES =====
name                      type
------------------------  -----
Alarm                     table
AlarmChanges              table
Attachment                table
AttachmentChanges         table
Calendar                  table
CalendarChanges           table
CalendarItem              table
CalendarItemChanges       table
Category                  table
CategoryLink              table
ClientCursor              table
ClientCursorConsumed      table
```

```
integrity_check
---------------
ok
===== CALENDAR TABLES =====
name                              type
--------------------------------  -----
Alarm                             table
AlarmChanges                      table
Attachment                        table
AttachmentChanges                 table
Calendar                          table
CalendarChanges                   table
CalendarItem                      table
CalendarItemChanges               table
Category                          table
CategoryLink                      table
ClientCursor                      table
ClientCursorConsumed              table
ClientSequence                    table
Contact                           table
ContactChanges                    table
EventAction                       table
EventActionChanges                table
ExceptionDate                     table
ExceptionDateChanges              table
Identity                          table
Location                          table
Notification                      table
NotificationChanges               table
OccurrenceCache                   table
OccurrenceCacheDays               table
Participant                       table
ParticipantChanges                table
Recurrence                        table
RecurrenceChanges                 table
ResourceChange                    table
ScheduledTaskCache                table
Sharee                            table
ShareeChanges                     table
```

```
SuggestedEventInfo         table
_SqliteDatabaseProperties  table
sqlite_sequence            table
═══ CALENDAR DATABASE SCHEMA ═══
CREATE TABLE _SqliteDatabaseProperties (key TEXT, value TEXT, UNIQUE(key));
CREATE TABLE Store (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT, default_alarm_offset INTEGER, type INTEGER, constraint_path TEXT, disabled INTEGER, extern
al_id TEXT, persistent_id TEXT, flags INTEGER, creator_bundle_id TEXT, creator_code_signing_identity TEXT, only_creator_can_modify INTEGER, external_mod_tag TEXT,
 preferred_event_private_value INTEGER, strictest_event_private_value INTEGER, last_sync_error INTEGER, last_sync_start REAL, last_sync_end REAL, delegated_accoun
t_owner_store_id TEXT, delegated_account_default_calendar_for_new_events_id INTEGER, shows_notifications INTEGER, flags2 INTEGER);
CREATE TABLE sqlite_sequence(name,seq);
CREATE TABLE StoreChanges (record INTEGER, type INTEGER, sequence_number INTEGER);
CREATE TRIGGER delete_store_members AFTER DELETE ON Store
BEGIN
DELETE FROM Calendar WHERE store_id = OLD.ROWID;
END;
CREATE TRIGGER delete_store_changes AFTER DELETE ON Store
BEGIN
DELETE FROM CalendarChanges WHERE store_id = OLD.ROWID;DELETE FROM CalendarItemChanges WHERE store_id = OLD.ROWID;DELETE FROM AlarmChanges WHERE store_id = OLD.RO
WID;DELETE FROM RecurrenceChanges WHERE store_id = OLD.ROWID;DELETE FROM ParticipantChanges WHERE store_id = OLD.ROWID;
END;
CREATE INDEX StoreExternalId on Store(external_id);
CREATE TABLE Calendar (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, store_id INTEGER, title TEXT, flags INTEGER, color TEXT, symbolic_color_name TEXT, color_is_displa
y INTEGER, type TEXT, supported_entity_types INTEGER, external_id TEXT, external_mod_tag TEXT, external_id_tag TEXT, external_rep BLOB, display_order INTEGER, UUI
D TEXT, shared_owner_name TEXT, shared_owner_address TEXT, sharing_status INTEGER, sharing_invitation_response INTEGER, published_URL TEXT, is_published INTEGER,
invitation_status INTEGER, sync_token TEXT, self_identity_id INTEGER, self_identity_email TEXT, self_identity_phone_number TEXT, owner_identity_id INTEGER, owner_
identity_email TEXT, owner_identity_phone_number TEXT, notes TEXT, bulk_requests BLOB, subcal_account_id TEXT, push_key TEXT, digest BLOB);
CREATE TABLE CalendarChanges (record INTEGER, type INTEGER, sequence_number INTEGER, store_id INTEGER, flags INTEGER, external_id TEXT, external_id_tag TEXT, UUID
 TEXT, old_flags INTEGER);
CREATE INDEX CalendarExternalId on Calendar(external_id);
CREATE INDEX CalendarStoreId on Calendar(store_id);
CREATE INDEX CalendarUUID on Calendar(UUID);
CREATE INDEX CalendarSelfIdentityId on Calendar(self_identity_id);
CREATE TRIGGER delete_calendar_members AFTER DELETE ON Calendar
BEGIN
DELETE FROM OccurrenceCacheDays where calendar_id = OLD.ROWID;DELETE FROM OccurrenceCache where calendar_id = OLD.ROWID;DELETE FROM CalendarItem WHERE calendar_id
 = OLD.ROWID;DELETE FROM OccurrenceCacheDays WHERE count = 0;DELETE FROM Notification WHERE calendar_id = OLD.ROWID; DELETE FROM sharee where owner_id = OLD.ROWID
;DELETE FROM Alarm WHERE calendar_owner_id = OLD.ROWID;
END;
CREATE TABLE Recurrence (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, frequency INTEGER, interval INTEGER, week_start INTEGER, count INTEGER, cached_end_date REAL, ca
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

python3 - <<'PY'
import sqlite3

db = "working/calendar/Calendar.sqlitedb"
con = sqlite3.connect(f"file:{db}?mode=ro", uri=True)

tables = [
    "CalendarItem", "Calendar", "Store", "Alarm",
    "Participant", "Recurrence", "Attachment",
    "Location", "Notification", "ResourceChange"
]

print("===== CALENDAR TABLE RECORD COUNTS =====")
for table in tables:
    try:
        count = con.execute(f'SELECT COUNT(*) FROM "{table}"').fetchone()[0]
        print(f"{table}: {count}")
    except sqlite3.Error as e:
        print(f"{table}: unavailable ({e})")

con.close()
PY
===== CALENDAR TABLE RECORD COUNTS =====
CalendarItem: 119
Calendar: 14
Store: 5
Alarm: 0
Participant: 0
Recurrence: 26
Attachment: 0
Location: 1
Notification: 0
ResourceChange: 0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

python3 - <<'PY'
import sqlite3

db = "working/calendar/Calendar.sqlitedb"
con = sqlite3.connect(f"file:{db}?mode=ro", uri=True)

print("===== CALENDAR ITEM COLUMN STRUCTURE =====")
for row in con.execute("PRAGMA table_info(CalendarItem)"):
    print(f"{row[1]} | {row[2]}")

print("\n===== CALENDAR ITEM TIMESTAMP SAMPLE =====")
columns = {r[1] for r in con.execute("PRAGMA table_info(CalendarItem)")}

for field in ["start_date", "end_date", "creation_date", "last_modified"]:
    if field in columns:
        print(f"\n{field}:")
        for row in con.execute(
            f'SELECT "{field}" FROM CalendarItem '
            f'WHERE "{field}" IS NOT NULL LIMIT 5'
        ):
            print(row[0])

con.close()
PY
===== CALENDAR ITEM COLUMN STRUCTURE =====
ROWID | INTEGER
summary | TEXT
location_id | INTEGER
client_location_id | INTEGER
description | TEXT
start_date | REAL
start_tz | TEXT
end_date | REAL
end_tz | TEXT
all_day | INTEGER
```

```
all_day | INTEGER
calendar_id | INTEGER
orig_item_id | INTEGER
orig_date | REAL
organizer_id | INTEGER
self_attendee_id | INTEGER
status | INTEGER
invitation_status | INTEGER
availability | INTEGER
privacy_level | INTEGER
url | TEXT
last_modified | REAL
sequence_num | INTEGER
birthday_id | INTEGER
modified_properties | INTEGER
external_tracking_status | INTEGER
external_id | TEXT
external_mod_tag | TEXT
unique_identifier | TEXT
external_schedule_id | TEXT
external_rep | BLOB
response_comment | TEXT
last_synced_response_comment | TEXT
hidden | INTEGER
has_recurrences | INTEGER
has_attendees | INTEGER
UUID | TEXT
entity_type | INTEGER
priority | INTEGER
due_date | REAL
due_tz | TEXT
due_all_day | INTEGER
completion_date | REAL
creation_date | REAL
action | TEXT
app_link | BLOB
display_order | INTEGER
created_by_id | INTEGER
modified_by_id | INTEGER
```

```
Session  Actions  Edit  View  Help
display_order | INTEGER
created_by_id | INTEGER
modified_by_id | INTEGER
shared_item_created_date | REAL
shared_item_created_tz | TEXT
shared_item_modified_date | REAL
shared_item_modified_tz | TEXT
invitation_changed_properties | INTEGER
default_alarm_removed | INTEGER
phantom_master | INTEGER
participation_status_modified_date | REAL
calendar_scale | TEXT
travel_time | INTEGER
travel_advisory_behavior | INTEGER
start_location_id | INTEGER
suggested_event_info_id | INTEGER
first_alert_date | REAL
proposed_start_date | REAL
can_forward | INTEGER
location_prediction_state | INTEGER
fired_ttl | INTEGER
disallow_propose_new_time | INTEGER
structured_data | BLOB
local_structured_data | BLOB
junk_status | INTEGER
conference_url | TEXT
contact_identifier | TEXT
recurrence_set | TEXT
flags | INTEGER
contact_name | TEXT

==== CALENDAR ITEM TIMESTAMP SAMPLE ====

start_date:
536457600.0
537667200.0
539222400.0
540259200.0
540432000.0
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

SELECT * FROM ABMultiValueLabel;
SELECT * FROM ABMultiValueEntryKey;
SQL
value
--------------
_$!<Mobile>!$_
TextNow

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

PRAGMA table_info(ABMultiValueLabel);
PRAGMA table_info(ABMultiValueEntryKey);
PRAGMA table_info(ABMultiValue);
SQL
cid  name   type  notnull  dflt_value  pk
---  -----  ----  -------  ----------  --
0    value  TEXT  0                    0
cid  name   type  notnull  dflt_value  pk
---  -----  ----  -------  ----------  --
0    value  TEXT  0                    0
cid  name        type     notnull  dflt_value            pk
---  ----------  -------  -------  --------------------  --
0    UID         INTEGER  0                              1
1    record_id   INTEGER  0                              0
2    property    INTEGER  0                              0
3    identifier  INTEGER  0                              0
4    label       INTEGER  0                              0
5    value       TEXT     0                              0
6    guid        TEXT     1        ab_generate_guid()    0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Account_ID,
    CASE
        WHEN AccountIdentifier IS NULL THEN 'NULL'
        WHEN length(AccountIdentifier) = 0 THEN 'Empty'
        ELSE 'Present'
    END AS Identifier_Status,
    Flags,
    DefaultSourceID
FROM ABAccount
ORDER BY ROWID;
SQL
Account_ID  Identifier_Status  Flags  DefaultSourceID
──────────  ─────────────────  ─────  ───────────────
1           Present            0      0
2           Present            0      0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

SOURCE="$PWD/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CallHistoryDB"

echo "═══ CALL HISTORY EVIDENCE FILES ═══"

sudo find "$SOURCE" -maxdepth 1 -type f \
-printf '%f | %s bytes\n' | sort
═══ CALL HISTORY EVIDENCE FILES ═══
[sudo] password for kali:
CallHistory.storedata | 73728 bytes
CallHistory.storedata-shm | 32768 bytes
CallHistory.storedata-wal | 3588552 bytes
CallHistoryTemp.storedata | 73728 bytes
CallHistoryTemp.storedata-shm | 32768 bytes
CallHistoryTemp.storedata-wal | 0 bytes
com.apple.callhistory.databaseInfo.plist | 298 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

mkdir -p working/callhistory

SOURCE="$PWD/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CallHistoryDB"

sudo find "$SOURCE" -maxdepth 1 -type f \
-exec cp -p {} working/callhistory/ \;

sudo chown -R kali:kali working/callhistory

echo "═══ WORKING COPIES ═══"
ls -lh working/callhistory/
═══ WORKING COPIES ═══
total 3.7M
-rw-r--r-- 1 kali kali  72K Mar 21  2020 CallHistory.storedata
-rw-r--r-- 1 kali kali  32K Apr 16  2020 CallHistory.storedata-shm
-rw-r--r-- 1 kali kali 3.5M Apr 16  2020 CallHistory.storedata-wal
-rw-r--r-- 1 kali kali  72K Mar 21  2020 CallHistoryTemp.storedata
-rw-r--r-- 1 kali kali  32K Apr 16  2020 CallHistoryTemp.storedata-shm
-rw-r--r-- 1 kali kali    0 Mar 21  2020 CallHistoryTemp.storedata-wal
-rw-r--r-- 1 kali kali  298 Apr 13  2020 com.apple.callhistory.databaseInfo.plist

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

SOURCE="$PWD/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CallHistoryDB"

mkdir -p output/hashes

for f in "$SOURCE"/*; do
    [ -f "$f" ] || continue
    name=$(basename "$f")
    original=$(sudo sha256sum "$f" | awk '{print $1}')
    copy=$(sha256sum "working/callhistory/$name" | awk '{print $1}')

    printf "%s | %s\n" "$name" "$([ "$original" = "$copy" ] && echo MATCH || echo MISMATCH)"
done | tee output/hashes/CallHistory_Verification.txt
CallHistory.storedata | MATCH
CallHistory.storedata-shm | MATCH
CallHistory.storedata-wal | MATCH
CallHistoryTemp.storedata | MATCH
CallHistoryTemp.storedata-shm | MATCH
CallHistoryTemp.storedata-wal | MATCH
com.apple.callhistory.databaseInfo.plist | MATCH

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

PRAGMA database_list;

SELECT name, type
FROM sqlite_master
WHERE type IN ('table', 'view')
ORDER BY type, name;
SQL
seq  name  file
---  ----  ----------------------------------------------------------
0    main  /home/kali/SBT-DF205-Case2/working/callhistory/CallHistory.s
           toredata
name                          type
----------------------------  -----
ZCALLDBPROPERTIES             table
ZCALLRECORD                   table
ZHANDLE                       table
Z_2REMOTEPARTICIPANTHANDLES   table
Z_METADATA                    table
Z_MODELCACHE                  table
Z_PRIMARYKEY                  table

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

PRAGMA table_info(ZCALLRECORD);
PRAGMA table_info(ZHANDLE);
PRAGMA table_info(Z_2REMOTEPARTICIPANTHANDLES);
SQL
cid  name                            type       notnull  dflt_value  pk
---  ------------------------------  ---------  -------  ----------  --
0    Z_PK                            INTEGER    0                    1
1    Z_ENT                           INTEGER    0                    0
2    Z_OPT                           INTEGER    0                    0
3    ZANSWERED                       INTEGER    0                    0
4    ZCALL_CATEGORY                  INTEGER    0                    0
5    ZCALLTYPE                       INTEGER    0                    0
6    ZDISCONNECTED_CAUSE             INTEGER    0                    0
7    ZFACE_TIME_DATA                 INTEGER    0                    0
8    ZFILTERED_OUT_REASON            INTEGER    0                    0
9    ZHANDLE_TYPE                    INTEGER    0                    0
10   ZNUMBER_AVAILABILITY            INTEGER    0                    0
11   ZORIGINATED                     INTEGER    0                    0
12   ZREAD                           INTEGER    0                    0
13   ZVERIFICATIONSTATUS             INTEGER    0                    0
14   ZDATE                           TIMESTAMP  0                    0
15   ZDURATION                       FLOAT      0                    0
16   ZISO_COUNTRY_CODE               VARCHAR    0                    0
17   ZLOCATION                       VARCHAR    0                    0
18   ZNAME                           VARCHAR    0                    0
19   ZSERVICE_PROVIDER               VARCHAR    0                    0
20   ZUNIQUE_ID                      VARCHAR    0                    0
21   ZLOCALPARTICIPANTUUID           BLOB       0                    0
22   ZOUTGOINGLOCALPARTICIPANTUUID   BLOB       0                    0
23   ZADDRESS                        BLOB       0                    0
cid  name              type     notnull  dflt_value  pk
```

```
0    Z_PK                          INTEGER    0                  1
1    Z_ENT                         INTEGER    0                  0
2    Z_OPT                         INTEGER    0                  0
3    ZANSWERED                     INTEGER    0                  0
4    ZCALL_CATEGORY                INTEGER    0                  0
5    ZCALLTYPE                     INTEGER    0                  0
6    ZDISCONNECTED_CAUSE           INTEGER    0                  0
7    ZFACE_TIME_DATA               INTEGER    0                  0
8    ZFILTERED_OUT_REASON          INTEGER    0                  0
9    ZHANDLE_TYPE                  INTEGER    0                  0
10   ZNUMBER_AVAILABILITY          INTEGER    0                  0
11   ZORIGINATED                   INTEGER    0                  0
12   ZREAD                         INTEGER    0                  0
13   ZVERIFICATIONSTATUS           INTEGER    0                  0
14   ZDATE                         TIMESTAMP  0                  0
15   ZDURATION                     FLOAT      0                  0
16   ZISO_COUNTRY_CODE             VARCHAR    0                  0
17   ZLOCATION                     VARCHAR    0                  0
18   ZNAME                         VARCHAR    0                  0
19   ZSERVICE_PROVIDER             VARCHAR    0                  0
20   ZUNIQUE_ID                    VARCHAR    0                  0
21   ZLOCALPARTICIPANTUUID         BLOB       0                  0
22   ZOUTGOINGLOCALPARTICIPANTUUID BLOB       0                  0
23   ZADDRESS                      BLOB       0                  0
cid  name              type     notnull  dflt_value  pk
---  ----------------  -------  -------  ----------  --
0    Z_PK              INTEGER  0                    1
1    Z_ENT             INTEGER  0                    0
2    Z_OPT             INTEGER  0                    0
3    ZTYPE             INTEGER  0                    0
4    ZNORMALIZEDVALUE  VARCHAR  0                    0
5    ZVALUE            BLOB     0                    0
cid  name                          type     notnull  dflt_value  pk
---  ----------------------------  -------  -------  ----------  --
0    Z_2REMOTEPARTICIPANTCALLS     INTEGER  0                    1
1    Z_3REMOTEPARTICIPANTHANDLES   INTEGER  0                    2

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

SELECT 'ZCALLRECORD' AS Table_Name, COUNT(*) AS Records FROM ZCALLRECORD
UNION ALL
SELECT 'ZHANDLE', COUNT(*) FROM ZHANDLE
UNION ALL
SELECT 'Z_2REMOTEPARTICIPANTHANDLES',
       COUNT(*) FROM Z_2REMOTEPARTICIPANTHANDLES;
SQL
Table_Name                    Records
----------------------------  -------
ZCALLRECORD                   37
ZHANDLE                       37
Z_2REMOTEPARTICIPANTHANDLES   37

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

SELECT
    Z_PK AS Call_ID,
    datetime(ZDATE + 978307200, 'unixepoch') AS Date_UTC,
    ROUND(ZDURATION, 2) AS Duration_Seconds,
    ZORIGINATED AS Originated,
    ZANSWERED AS Answered,
    ZCALLTYPE AS Call_Type,
    ZCALL_CATEGORY AS Call_Category
FROM ZCALLRECORD
ORDER BY ZDATE;
SQL
```

Call_ID	Date_UTC	Duration_Seconds	Originated	Answered	Call_Type	Call_Category
12	2020-03-23 20:02:52	0.0	0	0	1	1
16	2020-03-24 17:37:18	0.0	0	0	1	1
26	2020-03-26 17:51:45	0.0	0	0	1	1
27	2020-03-27 16:25:36	0.0	0	0	1	1
28	2020-03-27 19:55:03	0.0	0	0	1	1
29	2020-04-01 20:06:38	0.0	0	0	1	1
30	2020-04-03 16:10:54	0.0	0	0	1	1
31	2020-04-05 20:42:18	23.09	1	0	1	1
32	2020-04-06 20:34:33	0.0	0	0	1	1
33	2020-04-06 21:48:08	0.0	0	0	1	1
34	2020-04-06 22:43:00	0.0	0	0	1	1
35	2020-04-07 18:10:42	0.0	0	0	1	1
36	2020-04-07 19:04:54	0.0	0	0	1	1
37	2020-04-07 20:10:27	0.0	0	0	1	1
38	2020-04-08 19:32:06	0.0	0	0	1	1
39	2020-04-08 19:53:39	0.0	0	0	1	1
40	2020-04-08 20:21:59	0.0	0	0	1	1
41	2020-04-08 22:12:22	0.0	0	0	1	1

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

SELECT 'CALLTYPE' AS Field, ZCALLTYPE AS Value,
       COUNT(*) AS Records
FROM ZCALLRECORD
GROUP BY ZCALLTYPE

UNION ALL

SELECT 'CALL_CATEGORY', ZCALL_CATEGORY, COUNT(*)
FROM ZCALLRECORD
GROUP BY ZCALL_CATEGORY

UNION ALL

SELECT 'ORIGINATED', ZORIGINATED, COUNT(*)
FROM ZCALLRECORD
GROUP BY ZORIGINATED

UNION ALL

SELECT 'ANSWERED', ZANSWERED, COUNT(*)
FROM ZCALLRECORD
GROUP BY ZANSWERED
ORDER BY Field, Value;
SQL
Field          Value  Records
-------------  -----  -------
ANSWERED       0      36
ANSWERED       1      1
CALLTYPE       1      36
CALLTYPE       8      1
CALL_CATEGORY  1      36
```

```
SELECT 'ANSWERED', ZANSWERED, COUNT(*)
FROM ZCALLRECORD
GROUP BY ZANSWERED
ORDER BY Field, Value;
SQL
Field           Value  Records
--------------  -----  -------
ANSWERED        0      36
ANSWERED        1      1
CALLTYPE        1      36
CALLTYPE        8      1
CALL_CATEGORY   1      36
CALL_CATEGORY   2      1
ORIGINATED      0      33
ORIGINATED      1      4

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

SELECT
    Z_PK AS Handle_ID,
    ZTYPE AS Handle_Type,
    CASE
        WHEN ZNORMALIZEDVALUE IS NULL THEN 'NULL'
        WHEN length(ZNORMALIZEDVALUE) = 0 THEN 'Empty'
        ELSE 'Present'
    END AS Normalized_Value_Status
FROM ZHANDLE
ORDER BY Z_PK;

SELECT
    Z_2REMOTEPARTICIPANTCALLS AS Call_ID,
    Z_3REMOTEPARTICIPANTHANDLES AS Handle_ID
FROM Z_2REMOTEPARTICIPANTHANDLES
ORDER BY Call_ID;
SQL
Handle_ID  Handle_Type  Normalized_Value_Status
---------  -----------  -----------------------
12         2            Present
16         2            Present
26         2            Present
27         2            Present
28         2            Present
29         2            Present
30         2            Present
31         2            Present
32         2            Present
33         2            Present
34         2            Present
35         2            Present
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistory.storedata <<'SQL'
.headers on
.mode column

SELECT * FROM ZCALLDBPROPERTIES;

SELECT * FROM Z_PRIMARYKEY;

SELECT * FROM Z_METADATA;
SQL
Z_PK  Z_ENT  Z_OPT  ZTIMER_ALL  ZTIMER_INCOMING  ZTIMER_LAST  ZTIMER_LIFETIME  ZTIMER_OUTGOING
----  -----  -----  ----------  ---------------  -----------  ---------------  ---------------
1     1      4      0.0         12.0             0.0          99.0             87.0
Z_ENT  Z_NAME           Z_SUPER  Z_MAX
-----  ---------------  -------  -----
1      CallDBProperties  0        1
2      CallRecord        0        65
3      Handle            0        65
Z_VERSION  Z_UUID                                Z_PLIST
---------  ------------------------------------  ----------
1          08FD4B50-4329-4517-8C95-1A9BD073665C  bplist00◆
```

```
_

NSStoreModelVersionIdentifiers[NSStoreType_

_NSAutoVacuumLevel_

NSStoreModelVersionHashesDigest_

NSStoreModelVersionHashes_

NSPersistenceFrameworkVersion_
 NSStoreModelVersionHashesVersion◆       R20VSQLiteQ2_
XwcqeQTKuuYDFau4o0mDgJIoBhhalQvYGDkkJXlqDsZQDDvoeUTJbx3kR1kn
51EaMWQY7DBwy4pxNgeiVMv/WGw==◆




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CallDBPropertiesO
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```
_NSAutoVacuumLevel_

NSStoreModelVersionHashesDigest_

NSStoreModelVersionHashes_

NSPersistenceFrameworkVersion_
 NSStoreModelVersionHashesVersion◆       R20VSQLiteQ2_
XwcqeQTKuuYDFau4o0mDgJIoBhhalQvYGDkkJXlqDsZQDDvoeUTJbx3kR1kn
51EaMWQY7DBwy4pxNgeiVMv/WGw==◆




ZCallRecordVHandle_

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┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/callhistory/CallHistoryTemp.storedata <<'SQL'
.headers on
.mode column

PRAGMA integrity_check;

SELECT name, type
FROM sqlite_master
WHERE type IN ('table', 'view')
ORDER BY name;
SQL
integrity_check
---------------
ok
name                          type
----------------------------  -----
ZCALLDBPROPERTIES             table
ZCALLRECORD                   table
ZHANDLE                       table
Z_2REMOTEPARTICIPANTHANDLES   table
Z_METADATA                    table
Z_MODELCACHE                  table
Z_PRIMARYKEY                  table

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

for DB in CallHistory.storedata CallHistoryTemp.storedata; do
    echo "═══ $DB ═══"
    sqlite3 -readonly "working/callhistory/$DB" "
    SELECT 'ZCALLRECORD' AS table_name, COUNT(*) AS records FROM ZCALLRECORD
    UNION ALL
    SELECT 'ZHANDLE', COUNT(*) FROM ZHANDLE
    UNION ALL
    SELECT 'Z_2REMOTEPARTICIPANTHANDLES',
           COUNT(*) FROM Z_2REMOTEPARTICIPANTHANDLES;
    "
done
═══ CallHistory.storedata ═══
ZCALLRECORD|37
ZHANDLE|37
Z_2REMOTEPARTICIPANTHANDLES|37
═══ CallHistoryTemp.storedata ═══
ZCALLRECORD|0
ZHANDLE|0
Z_2REMOTEPARTICIPANTHANDLES|0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

python3 - <<'PY'
import plistlib
from pathlib import Path

p = Path("working/callhistory/com.apple.callhistory.databaseInfo.plist")

with p.open("rb") as f:
    data = plistlib.load(f)

print("===== DATABASE INFO PLIST =====")
print("Plist type:", type(data).__name__)

if isinstance(data, dict):
    for key, value in data.items():
        if isinstance(value, bytes):
            print(f"{key}: <binary data, {len(value)} bytes>")
        else:
            print(f"{key}: {value}")
else:
    print(data)

print("\n===== FILE DETAILS =====")
print("Size:", p.stat().st_size, "bytes")
PY

sha256sum working/callhistory/com.apple.callhistory.databaseInfo.plist
===== DATABASE INFO PLIST =====
Plist type: dict
DatabaseVersionPerm: 20
DatabaseVersionTemp: 20

===== FILE DETAILS =====
Size: 298 bytes
705e14a760dbe754a3f14b00b9223e21addb970e44a66c8bb6be644e0519476e  working/callhistory/com.apple.callhistory.databaseInfo.plist
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Message_ID,
    is_from_me AS From_Me,
    service AS Service,
    CASE
        WHEN text IS NULL THEN 'NULL'
        WHEN length(text) = 0 THEN 'EMPTY'
        ELSE 'PRESENT'
    END AS Text_Status,
    length(text) AS Text_Length,
    CASE
        WHEN attributedBody IS NULL THEN 'No'
        ELSE 'Yes'
    END AS Attributed_Body_Present,
    CASE
        WHEN attributedBody IS NULL THEN 0
        ELSE length(attributedBody)
    END AS Attributed_Body_Length,
    cache_has_attachments AS Attachment_Flag
FROM message
WHERE ROWID IN (29,30,31,32,36,37)
ORDER BY ROWID;
SQL
Message_ID  From_Me  Service   Text_Status  Text_Length  Attributed_Body_Present  Attributed_Body_Length  Attachment_Flag
----------  -------  --------  -----------  -----------  -----------------------  ----------------------  ---------------
29          0        iMessage  NULL                      No                       0                       0
30          0        iMessage  NULL                      No                       0                       0
31          0        iMessage  NULL                      No                       0                       0
32          0        iMessage  NULL                      No                       0                       0
36          1        iMessage  PRESENT      1            Yes                      313                     1
37          0        iMessage  PRESENT      1            Yes                      367                     1
```

```
sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Message_ID,
    is_from_me AS From_Me,
    service AS Service,
    CASE
        WHEN text IS NULL THEN 'NULL'
        WHEN length(text) = 0 THEN 'EMPTY'
        ELSE 'PRESENT'
    END AS Text_Status,
    length(text) AS Text_Length,
    CASE
        WHEN attributedBody IS NULL THEN 'No'
        ELSE 'Yes'
    END AS Attributed_Body_Present,
    CASE
        WHEN attributedBody IS NULL THEN 0
        ELSE length(attributedBody)
    END AS Attributed_Body_Length,
    cache_has_attachments AS Attachment_Flag
FROM message
WHERE ROWID IN (29,30,31,32,36,37)
ORDER BY ROWID;
SQL
```

Message_ID	From_Me	Service	Text_Status	Text_Length	Attributed_Body_Present	Attributed_Body_Length	Attachment_Flag
29	0	iMessage	NULL		No	0	0
30	0	iMessage	NULL		No	0	0
31	0	iMessage	NULL		No	0	0
32	0	iMessage	NULL		No	0	0
36	1	iMessage	PRESENT	1	Yes	313	1
37	0	iMessage	PRESENT	1	Yes	367	1

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

PRAGMA table_info(message);
SQL
cid  name                                type     notnull  dflt_value  pk
---  ----------------------------------  -------  -------  ----------  --
0    ROWID                               INTEGER  0                    1
1    guid                                TEXT     1                    0
2    text                                TEXT     0                    0
3    replace                             INTEGER  0        0           0
4    service_center                      TEXT     0                    0
5    handle_id                           INTEGER  0        0           0
6    subject                             TEXT     0                    0
7    country                             TEXT     0                    0
8    attributedBody                      BLOB     0                    0
9    version                             INTEGER  0        0           0
10   type                                INTEGER  0        0           0
11   service                             TEXT     0                    0
12   account                             TEXT     0                    0
13   account_guid                        TEXT     0                    0
14   error                               INTEGER  0        0           0
15   date                                INTEGER  0                    0
16   date_read                           INTEGER  0                    0
17   date_delivered                      INTEGER  0                    0
18   is_delivered                        INTEGER  0        0           0
19   is_finished                         INTEGER  0        0           0
20   is_emote                            INTEGER  0        0           0
21   is_from_me                          INTEGER  0        0           0
22   is_empty                            INTEGER  0        0           0
23   is_delayed                          INTEGER  0        0           0
24   is_auto_reply                       INTEGER  0        0           0
25   is_prepared                         INTEGER  0        0           0
26   is_read                             INTEGER  0        0           0
```

26	is_read	INTEGER	0	0	0
27	is_system_message	INTEGER	0	0	0
28	is_sent	INTEGER	0	0	0
29	has_dd_results	INTEGER	0	0	0
30	is_service_message	INTEGER	0	0	0
31	is_forward	INTEGER	0	0	0
32	was_downgraded	INTEGER	0	0	0
33	is_archive	INTEGER	0	0	0
34	cache_has_attachments	INTEGER	0	0	0
35	cache_roomnames	TEXT	0		0
36	was_data_detected	INTEGER	0	0	0
37	was_deduplicated	INTEGER	0	0	0
38	is_audio_message	INTEGER	0	0	0
39	is_played	INTEGER	0	0	0
40	date_played	INTEGER	0		0
41	item_type	INTEGER	0	0	0
42	other_handle	INTEGER	0	0	0
43	group_title	TEXT	0		0
44	group_action_type	INTEGER	0	0	0
45	share_status	INTEGER	0	0	0
46	share_direction	INTEGER	0	0	0
47	is_expirable	INTEGER	0	0	0
48	expire_state	INTEGER	0	0	0
49	message_action_type	INTEGER	0	0	0
50	message_source	INTEGER	0	0	0
51	associated_message_guid	TEXT	0		0
52	associated_message_type	INTEGER	0	0	0
53	balloon_bundle_id	TEXT	0		0
54	payload_data	BLOB	0		0
55	expressive_send_style_id	TEXT	0		0
56	associated_message_range_location	INTEGER	0	0	0
57	associated_message_range_length	INTEGER	0	0	0
58	time_expressive_send_played	INTEGER	0		0
59	message_summary_info	BLOB	0		0
60	ck_sync_state	INTEGER	0	0	0
61	ck_record_id	TEXT	0		0
62	ck_record_change_tag	TEXT	0		0
63	destination_caller_id	TEXT	0		0
64	sr_ck_sync_state	INTEGER	0	0	0

```
35  cache_roomnames                  TEXT     0                 0
36  was_data_detected                INTEGER  0        0        0
37  was_deduplicated                 INTEGER  0        0        0
38  is_audio_message                 INTEGER  0        0        0
39  is_played                        INTEGER  0        0        0
40  date_played                      INTEGER  0                 0
41  item_type                        INTEGER  0        0        0
42  other_handle                     INTEGER  0        0        0
43  group_title                      TEXT     0                 0
44  group_action_type                INTEGER  0        0        0
45  share_status                     INTEGER  0        0        0
46  share_direction                  INTEGER  0        0        0
47  is_expirable                     INTEGER  0        0        0
48  expire_state                     INTEGER  0        0        0
49  message_action_type              INTEGER  0        0        0
50  message_source                   INTEGER  0        0        0
51  associated_message_guid          TEXT     0                 0
52  associated_message_type          INTEGER  0        0        0
53  balloon_bundle_id                TEXT     0                 0
54  payload_data                     BLOB     0                 0
55  expressive_send_style_id         TEXT     0                 0
56  associated_message_range_location INTEGER 0        0        0
57  associated_message_range_length  INTEGER  0        0        0
58  time_expressive_send_played      INTEGER  0                 0
59  message_summary_info             BLOB     0                 0
60  ck_sync_state                    INTEGER  0        0        0
61  ck_record_id                     TEXT     0                 0
62  ck_record_change_tag             TEXT     0                 0
63  destination_caller_id            TEXT     0                 0
64  sr_ck_sync_state                 INTEGER  0        0        0
65  sr_ck_record_id                  TEXT     0                 0
66  sr_ck_record_change_tag          TEXT     0                 0
67  is_corrupt                       INTEGER  0        0        0
68  reply_to_guid                    TEXT     0                 0
69  sort_id                          INTEGER  0                 0
70  is_spam                          INTEGER  0        0        0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Message_ID,
    is_empty AS Is_Empty,
    is_system_message AS System_Message,
    is_service_message AS Service_Message,
    is_audio_message AS Audio_Message,
    item_type AS Item_Type,
    type AS Message_Type,
    length(attributedBody) AS Attributed_Body_Bytes,
    length(payload_data) AS Payload_Bytes,
    length(message_summary_info) AS Summary_Info_Bytes,
    associated_message_type AS Associated_Type,
    CASE
        WHEN associated_message_guid IS NULL THEN 'No'
        ELSE 'Yes'
    END AS Associated_GUID_Present,
    CASE
        WHEN balloon_bundle_id IS NULL THEN 'No'
        ELSE 'Yes'
    END AS Balloon_Bundle_Present
FROM message
WHERE ROWID IN (29,30,31,32,36,37)
ORDER BY ROWID;
SQL
Message_ID  Is_Empty  System_Message  Service_Message  Audio_Message  Item_Type  Message_Type  Attributed_Body_Bytes  Payload_Bytes  Summary_Info_Bytes  Associate
d_Type  Associated_GUID_Present  Balloon_Bundle_Present
----------  --------  -------------  ---------------  -------------  ---------  ------------  ---------------------  -------------  ------------------  ---------
------  -----------------------  ----------------------
29          0         0              0                0              4          0                                                                        0
        No                         No
30          0         0              0                0              4          0                                                                        0
```

```
    is_service_message AS Service_Message,
    is_audio_message AS Audio_Message,
    item_type AS Item_Type,
    type AS Message_Type,
    length(attributedBody) AS Attributed_Body_Bytes,
    length(payload_data) AS Payload_Bytes,
    length(message_summary_info) AS Summary_Info_Bytes,
    associated_message_type AS Associated_Type,
    CASE
        WHEN associated_message_guid IS NULL THEN 'No'
        ELSE 'Yes'
    END AS Associated_GUID_Present,
    CASE
        WHEN balloon_bundle_id IS NULL THEN 'No'
        ELSE 'Yes'
    END AS Balloon_Bundle_Present
FROM message
WHERE ROWID IN (29,30,31,32,36,37)
ORDER BY ROWID;
SQL
```

Message_ID	Is_Empty	System_Message	Service_Message	Audio_Message	Item_Type	Message_Type	Attributed_Body_Bytes	Payload_Bytes	Summary_Info_Bytes	Associated_Type	Associated_GUID_Present	Balloon_Bundle_Present
29	0	0	0	0	4	0				0	No	No
30	0	0	0	0	4	0				0	No	No
31	0	0	0	0	4	0				0	No	No
32	0	0	0	0	4	0				0	No	No
36	0	0	0	0	0	0	313		51	0	No	No
37	0	0	0	0	0	0	367		51	0	No	No

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook \
-maxdepth 1 -type f \
-printf '%f | %s bytes\n' | sort
[sudo] password for kali:
AddressBookImages.sqlitedb | 8192 bytes
AddressBookImages.sqlitedb-shm | 32768 bytes
AddressBookImages.sqlitedb-wal | 1774688 bytes
AddressBook.sqlitedb | 303104 bytes
AddressBook.sqlitedb-shm | 32768 bytes
AddressBook.sqlitedb-wal | 1746912 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

SOURCE="$PWD/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook"

echo "===== COPYING CONTACTS DATABASES ====="

sudo find "$SOURCE" -maxdepth 1 -type f \
\( -name 'AddressBook.sqlitedb*' -o -name 'AddressBookImages.sqlitedb*' \) \
-exec cp -p {} working/contacts/ \;

sudo chown -R kali:kali working/contacts

echo
echo "===== WORKING COPIES ====="
ls -lh working/contacts/

echo
echo "===== SOURCE SHA-256 ====="
sudo find "$SOURCE" -maxdepth 1 -type f \
\( -name 'AddressBook.sqlitedb*' -o -name 'AddressBookImages.sqlitedb*' \) \
-exec sha256sum {} \;

echo
echo "===== WORKING COPY SHA-256 ====="
sha256sum working/contacts/*
===== COPYING CONTACTS DATABASES =====

===== WORKING COPIES =====
total 3.8M
-rw-r--r-- 1 kali kali 8.0K Mar 21  2020 AddressBookImages.sqlitedb
-rw-r--r-- 1 kali kali  32K Apr 16  2020 AddressBookImages.sqlitedb-shm
-rw-r--r-- 1 kali kali 1.7M Mar 27  2020 AddressBookImages.sqlitedb-wal
-rw-r--r-- 1 kali kali 296K Apr 15  2020 AddressBook.sqlitedb
-rw-r--r-- 1 kali kali  32K Apr 16  2020 AddressBook.sqlitedb-shm
-rw-r--r-- 1 kali kali 1.7M Apr 16  2020 AddressBook.sqlitedb-wal

===== SOURCE SHA-256 =====
```

```
echo
echo "===== WORKING COPY SHA-256 ====="
sha256sum working/contacts/*
===== COPYING CONTACTS DATABASES =====

===== WORKING COPIES =====
total 3.8M
-rw-r--r-- 1 kali kali 8.0K Mar 21  2020 AddressBookImages.sqlitedb
-rw-r--r-- 1 kali kali  32K Apr 16  2020 AddressBookImages.sqlitedb-shm
-rw-r--r-- 1 kali kali 1.7M Mar 27  2020 AddressBookImages.sqlitedb-wal
-rw-r--r-- 1 kali kali 296K Apr 15  2020 AddressBook.sqlitedb
-rw-r--r-- 1 kali kali  32K Apr 16  2020 AddressBook.sqlitedb-shm
-rw-r--r-- 1 kali kali 1.7M Apr 16  2020 AddressBook.sqlitedb-wal

===== SOURCE SHA-256 =====
70a07bde7ddc9d267c878aa37aa5d491ab4a343742090e37eea64e6b46ccce09  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook/AddressBook.sqlitedb-wal
3aa702c1594bee56a3ea936176b8098b410190e35c2870b8c76cc73c05317d2e  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook/AddressBookImages.sqlitedb
b192223794126273b36253caae0c0777c698163410a2801b74a972a9ceab4c7d  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook/AddressBookImages.sqlitedb-wal
f403f296338035658d0e93b9071f13d2990b7c954137cdea1325990bfd9c3186  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook/AddressBook.sqlitedb-shm
daaa4a8b67b9e18b6656dde286ba03c1faa08cebde78ca789019d3d90a2c0fa9  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook/AddressBookImages.sqlitedb-shm
0dc61afb1f37c1f196369968bc682c53024dabf1a1e657792e11148dfc1d1010  /home/kali/SBT-DF205-Case2/working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/AddressBook/AddressBook.sqlitedb

===== WORKING COPY SHA-256 =====
3aa702c1594bee56a3ea936176b8098b410190e35c2870b8c76cc73c05317d2e  working/contacts/AddressBookImages.sqlitedb
daaa4a8b67b9e18b6656dde286ba03c1faa08cebde78ca789019d3d90a2c0fa9  working/contacts/AddressBookImages.sqlitedb-shm
b192223794126273b36253caae0c0777c698163410a2801b74a972a9ceab4c7d  working/contacts/AddressBookImages.sqlitedb-wal
0dc61afb1f37c1f196369968bc682c53024dabf1a1e657792e11148dfc1d1010  working/contacts/AddressBook.sqlitedb
f403f296338035658d0e93b9071f13d2990b7c954137cdea1325990bfd9c3186  working/contacts/AddressBook.sqlitedb-shm
70a07bde7ddc9d267c878aa37aa5d491ab4a343742090e37eea64e6b46ccce09  working/contacts/AddressBook.sqlitedb-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "===== CONTACTS DATABASE INTEGRITY ====="
sqlite3 -readonly working/contacts/AddressBook.sqlitedb \
"PRAGMA integrity_check;"

echo
echo "===== CONTACTS IMAGES DATABASE INTEGRITY ====="
sqlite3 -readonly working/contacts/AddressBookImages.sqlitedb \
"PRAGMA integrity_check;"
===== CONTACTS DATABASE INTEGRITY =====
Error: in prepare, unknown tokenizer: ab_cf_tokenizer

===== CONTACTS IMAGES DATABASE INTEGRITY =====
ok

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ CONTACTS DATABASE SCHEMA ═══"
sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

SELECT name, type
FROM sqlite_master
WHERE type IN ('table', 'view')
ORDER BY type, name;
SQL

echo
echo "═══ CONTACTS IMAGES DATABASE SCHEMA ═══"
sqlite3 -readonly working/contacts/AddressBookImages.sqlitedb <<'SQL'
.headers on
.mode column

SELECT name, type
FROM sqlite_master
WHERE type IN ('table', 'view')
ORDER BY type, name;
SQL
═══ CONTACTS DATABASE SCHEMA ═══
name                              type
--------------------------------  -----
ABAccount                         table
ABGroup                           table
ABGroupChanges                    table
ABGroupMembers                    table
ABMultiValue                      table
ABMultiValueEntry                 table
ABMultiValueEntryKey              table
ABMultiValueLabel                 table
ABPerson                          table
ABPersonChanges                   table
```

SQL

==== CONTACTS DATABASE SCHEMA ====

name	type
ABAccount	table
ABGroup	table
ABGroupChanges	table
ABGroupMembers	table
ABMultiValue	table
ABMultiValueEntry	table
ABMultiValueEntryKey	table
ABMultiValueLabel	table
ABPerson	table
ABPersonChanges	table
ABPersonFullTextSearch	table
ABPersonFullTextSearch_content	table
ABPersonFullTextSearch_docsize	table
ABPersonFullTextSearch_segdir	table
ABPersonFullTextSearch_segments	table
ABPersonFullTextSearch_stat	table
ABPersonLink	table
ABPersonMultiValueDeletes	table
ABPersonSearchKey	table
ABPhoneLastFour	table
ABStore	table
ClientCursor	table
ClientCursorConsumed	table
ClientSequence	table
FirstSortSectionCount	table
FirstSortSectionCountTotal	table
LastSortSectionCount	table
LastSortSectionCountTotal	table
_SqliteDatabaseProperties	table
sqlite_sequence	table

==== CONTACTS IMAGES DATABASE SCHEMA ====

name	type
ABFullSizeImage	table

```
FirstSortSectionCountTotal      table
LastSortSectionCount            table
LastSortSectionCountTotal       table
_SqliteDatabaseProperties       table
sqlite_sequence                 table

═══ CONTACTS IMAGES DATABASE SCHEMA ═══
name                            type
──────────────────────────────  ──────
ABFullSizeImage                 table
ABThumbnailImage                table
_SqliteDatabaseProperties       table

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

PRAGMA table_info(ABPerson);
PRAGMA table_info(ABMultiValue);
PRAGMA table_info(ABAccount);
SQL
cid  name                     type     notnull  dflt_value          pk
---  -------------------------  -------  -------  ------------------  --
0    ROWID                    INTEGER  0                            1
1    First                    TEXT     0                            0
2    Last                     TEXT     0                            0
3    Middle                   TEXT     0                            0
4    FirstPhonetic            TEXT     0                            0
5    MiddlePhonetic           TEXT     0                            0
6    LastPhonetic             TEXT     0                            0
7    Organization             TEXT     0                            0
8    Department               TEXT     0                            0
9    Note                     TEXT     0                            0
10   Kind                     INTEGER  0                            0
11   Birthday                 TEXT     0                            0
12   JobTitle                 TEXT     0                            0
13   Nickname                 TEXT     0                            0
14   Prefix                   TEXT     0                            0
15   Suffix                   TEXT     0                            0
16   FirstSort                TEXT     0                            0
17   LastSort                 TEXT     0                            0
18   CreationDate             INTEGER  0                            0
19   ModificationDate         INTEGER  0                            0
20   CompositeNameFallback    TEXT     0                            0
21   ExternalIdentifier       TEXT     0                            0
22   ExternalModificationTag  TEXT     0                            0
23   ExternalUUID             TEXT     0                            0
24   StoreID                  INTEGER  0                            0
```

```
Session  Actions  Edit  View  Help
10   Kind                        INTEGER  0                          0
11   Birthday                    TEXT     0                          0
12   JobTitle                    TEXT     0                          0
13   Nickname                    TEXT     0                          0
14   Prefix                      TEXT     0                          0
15   Suffix                      TEXT     0                          0
16   FirstSort                   TEXT     0                          0
17   LastSort                    TEXT     0                          0
18   CreationDate                INTEGER  0                          0
19   ModificationDate            INTEGER  0                          0
20   CompositeNameFallback       TEXT     0                          0
21   ExternalIdentifier          TEXT     0                          0
22   ExternalModificationTag     TEXT     0                          0
23   ExternalUUID                TEXT     0                          0
24   StoreID                     INTEGER  0                          0
25   DisplayName                 TEXT     0                          0
26   ExternalRepresentation      BLOB     0                          0
27   FirstSortSection            TEXT     0                          0
28   LastSortSection             TEXT     0                          0
29   FirstSortLanguageIndex      INTEGER  0        2147483647        0
30   LastSortLanguageIndex       INTEGER  0        2147483647        0
31   PersonLink                  INTEGER  0        -1                0
32   ImageURI                    TEXT     0                          0
33   IsPreferredName             INTEGER  0        1                 0
34   guid                        TEXT     1        ab_generate_guid()  0
35   PhonemeData                 TEXT     0                          0
36   AlternateBirthday           TEXT     0                          0
37   MapsData                    TEXT     0                          0
38   FirstPronunciation          TEXT     0                          0
39   MiddlePronunciation         TEXT     0                          0
40   LastPronunciation           TEXT     0                          0
41   OrganizationPhonetic        TEXT     0                          0
42   OrganizationPronunciation   TEXT     0                          0
43   PreviousFamilyName          TEXT     0                          0
44   PreferredLikenessSource     TEXT     0                          0
45   PreferredPersonaIdentifier  TEXT     0                          0
46   PreferredChannel            TEXT     0                          0
47   DowntimeWhitelist           TEXT     0                          0
48   ImageType                   TEXT     0                          0
```

```
30   LastSortLanguageIndex       INTEGER  0        2147483647           0
31   PersonLink                  INTEGER  0        -1                   0
32   ImageURI                    TEXT     0                             0
33   IsPreferredName             INTEGER  0        1                    0
34   guid                        TEXT     1        ab_generate_guid()   0
35   PhonemeData                 TEXT     0                             0
36   AlternateBirthday           TEXT     0                             0
37   MapsData                    TEXT     0                             0
38   FirstPronunciation          TEXT     0                             0
39   MiddlePronunciation         TEXT     0                             0
40   LastPronunciation           TEXT     0                             0
41   OrganizationPhonetic        TEXT     0                             0
42   OrganizationPronunciation   TEXT     0                             0
43   PreviousFamilyName          TEXT     0                             0
44   PreferredLikenessSource     TEXT     0                             0
45   PreferredPersonaIdentifier  TEXT     0                             0
46   PreferredChannel            TEXT     0                             0
47   DowntimeWhitelist           TEXT     0                             0
48   ImageType                   TEXT     0                             0
49   ImageHash                   BLOB     0                             0
cid  name         type     notnull  dflt_value          pk
---  -----------  -------  -------  ------------------  --
0    UID          INTEGER  0                            1
1    record_id    INTEGER  0                            0
2    property     INTEGER  0                            0
3    identifier   INTEGER  0                            0
4    label        INTEGER  0                            0
5    value        TEXT     0                            0
6    guid         TEXT     1        ab_generate_guid()  0
cid  name                type     notnull  dflt_value          pk
---  ------------------  -------  -------  ------------------  --
0    ROWID               INTEGER  0                            1
1    AccountIdentifier   TEXT     0                            0
2    Flags               INTEGER  0                            0
3    DefaultSourceID     INTEGER  0                            0
4    guid                TEXT     1        ab_generate_guid()  0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

SELECT 'ABPerson' AS Table_Name, COUNT(*) AS Records FROM ABPerson
UNION ALL
SELECT 'ABMultiValue', COUNT(*) FROM ABMultiValue
UNION ALL
SELECT 'ABAccount', COUNT(*) FROM ABAccount
UNION ALL
SELECT 'ABGroup', COUNT(*) FROM ABGroup;
SQL
Table_Name    Records
------------  -------
ABPerson      2
ABMultiValue  3
ABAccount     2
ABGroup       0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Contact_ID,
    CASE WHEN First IS NOT NULL THEN 'Present' ELSE 'NULL' END AS First_Name,
    CASE WHEN Last IS NOT NULL THEN 'Present' ELSE 'NULL' END AS Last_Name,
    CASE WHEN Organization IS NOT NULL THEN 'Present' ELSE 'NULL' END AS Organization,
    Kind,
    CreationDate,
    ModificationDate
FROM ABPerson
ORDER BY ROWID;
SQL
Contact_ID  First_Name  Last_Name  Organization  Kind  CreationDate  ModificationDate
──────────  ──────────  ─────────  ────────────  ────  ────────────  ────────────────
1           Present     Present    NULL          0     606532293     607027441
2           Present     Present    NULL          0     607027393     607027406

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/contacts/AddressBook.sqlitedb <<'SQL'
.headers on
.mode column

SELECT
    UID,
    record_id,
    property,
    identifier,
    label,
    CASE
        WHEN value IS NULL THEN 'NULL'
        WHEN length(value) = 0 THEN 'Empty'
        ELSE 'Present'
    END AS Value_Status
FROM ABMultiValue
ORDER BY record_id, UID;
SQL
UID  record_id  property  identifier  label  Value_Status
---  ---------  --------  ----------  -----  ------------
4    1          3         0           1      Present
5    1          3         1           2      Present
6    1          4         0                  Present
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
PRAGMA integrity_check;

SELECT name, sql
FROM sqlite_master
WHERE type='table'
ORDER BY name;
SQL
ok
_SqliteDatabaseProperties|CREATE TABLE _SqliteDatabaseProperties (key TEXT, value TEXT, UNIQUE(key))
attachment|CREATE TABLE attachment (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NOT NULL, created_date INTEGER DEFAULT 0, start_date INTEGER DEFAULT
 0, filename TEXT, uti TEXT, mime_type TEXT, transfer_state INTEGER DEFAULT 0, is_outgoing INTEGER DEFAULT 0, user_info BLOB, transfer_name TEXT, total_bytes INTE
GER DEFAULT 0, is_sticker INTEGER DEFAULT 0, sticker_user_info BLOB, attribution_info BLOB, hide_attachment INTEGER DEFAULT 0, ck_sync_state INTEGER DEFAULT 0, ck
_server_change_token_blob BLOB, ck_record_id TEXT, original_guid TEXT UNIQUE NOT NULL, sr_ck_sync_state INTEGER DEFAULT 0, sr_ck_server_change_token_blob BLOB, sr
_ck_record_id TEXT)
chat|CREATE TABLE chat (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NOT NULL, style INTEGER, state INTEGER, account_id TEXT, properties BLOB, chat_i
dentifier TEXT, service_name TEXT, room_name TEXT, account_login TEXT, is_archived INTEGER DEFAULT 0, last_addressed_handle TEXT, display_name TEXT, group_id TEXT
, is_filtered INTEGER, successful_query INTEGER, engram_id TEXT, server_change_token TEXT, ck_sync_state INTEGER DEFAULT 0, original_group_id TEXT, last_read_mess
age_timestamp INTEGER DEFAULT 0, sr_server_change_token TEXT, sr_ck_sync_state INTEGER DEFAULT 0, cloudkit_record_id TEXT, sr_cloudkit_record_id TEXT, last_addres
sed_sim_id TEXT, is_blackholed INTEGER DEFAULT 0)
chat_handle_join|CREATE TABLE chat_handle_join (chat_id INTEGER REFERENCES chat (ROWID) ON DELETE CASCADE, handle_id INTEGER REFERENCES handle (ROWID) ON DELETE C
ASCADE, UNIQUE(chat_id, handle_id))
chat_message_join|CREATE TABLE chat_message_join (chat_id INTEGER REFERENCES chat (ROWID) ON DELETE CASCADE, message_id INTEGER REFERENCES message (ROWID) ON DELE
TE CASCADE, message_date INTEGER DEFAULT 0, PRIMARY KEY (chat_id, message_id))
deleted_messages|CREATE TABLE deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid TEXT NOT NULL)
handle|CREATE TABLE handle (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, id TEXT NOT NULL, country TEXT, service TEXT NOT NULL, uncanonicalized_id TEXT, person
_centric_id TEXT, UNIQUE (id, service) )
kvtable|CREATE TABLE kvtable (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, key TEXT UNIQUE NOT NULL, value BLOB NOT NULL)
message|CREATE TABLE message (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NOT NULL, text TEXT, replace INTEGER DEFAULT 0, service_center TEXT, handl
e_id INTEGER DEFAULT 0, subject TEXT, country TEXT, attributedBody BLOB, version INTEGER DEFAULT 0, type INTEGER DEFAULT 0, service TEXT, account TEXT, account_gu
id TEXT, error INTEGER DEFAULT 0, date INTEGER, date_read INTEGER, date_delivered INTEGER, is_delivered INTEGER DEFAULT 0, is_finished INTEGER DEFAULT 0, is_emote
 INTEGER DEFAULT 0, is_from_me INTEGER DEFAULT 0, is_empty INTEGER DEFAULT 0, is_delayed INTEGER DEFAULT 0, is_auto_reply INTEGER DEFAULT 0, is_prepared INTEGER D
EFAULT 0, is_read INTEGER DEFAULT 0, is_system_message INTEGER DEFAULT 0, is_sent INTEGER DEFAULT 0, has_dd_results INTEGER DEFAULT 0, is_service_message INTEGER
 DEFAULT 0, is_forward INTEGER DEFAULT 0, was_downgraded INTEGER DEFAULT 0, is_archive INTEGER DEFAULT 0, cache_has_attachments INTEGER DEFAULT 0, cache_roomnames
 TEXT, was_data_detected INTEGER DEFAULT 0, was_deduplicated INTEGER DEFAULT 0, is_audio_message INTEGER DEFAULT 0, is_played INTEGER DEFAULT 0, date_played INTEGE
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT 'Messages' AS Evidence_Item, COUNT(*) AS Total FROM message
UNION ALL
SELECT 'Chats', COUNT(*) FROM chat
UNION ALL
SELECT 'Handles', COUNT(*) FROM handle
UNION ALL
SELECT 'Attachments', COUNT(*) FROM attachment
UNION ALL
SELECT 'Deleted Message References', COUNT(*) FROM deleted_messages;
SQL
Evidence_Item                Total
---------------------------  -----
Messages                     48
Chats                        17
Handles                      17
Attachments                  7
Deleted Message References   0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Message_ID,
    datetime(date / 1000000000 + 978307200, 'unixepoch') AS UTC_Time,
    is from me AS From_Me,
    service AS Service,
    handle id AS Handle_ID,
    cache has attachments AS Has_Attachment
FROM message
ORDER BY date ASC;
SQL
```

Message_ID	UTC_Time	From_Me	Service	Handle_ID	Has_Attachment
1	2020-03-22 01:10:59	0	SMS	1	0
2	2020-03-22 15:16:16	0	SMS	2	0
3	2020-03-22 19:21:40	0	SMS	3	0
4	2020-03-23 01:06:38	0	SMS	4	0
5	2020-03-23 01:38:03	0	SMS	5	0
6	2020-03-23 17:41:36	0	SMS	6	0
7	2020-03-24 01:21:34	0	SMS	7	0
8	2020-03-24 01:27:40	0	SMS	8	0
9	2020-03-24 01:52:23	0	SMS	9	0
10	2020-03-25 00:47:38	0	SMS	10	0
11	2020-03-25 00:49:24	0	SMS	11	0
12	2020-03-25 17:54:08	0	SMS	6	0
13	2020-03-26 01:11:36	0	SMS	12	0
14	2020-03-26 20:55:46	0	SMS	6	0
15	2020-03-27 18:41:13	0	iMessage	13	1
16	2020-03-28 18:18:43	0	SMS	6	0
17	2020-03-30 08:15:36	0	SMS	6	0
18	2020-03-31 11:45:04	0	SMS	6	0
19	2020-04-01 17:39:22	0	SMS	6	0

```
13          2020-03-26 01:11:36  0        SMS       12         0
14          2020-03-26 20:55:46  0        SMS       6          0
15          2020-03-27 18:41:13  0        iMessage  13         1
16          2020-03-28 18:18:43  0        SMS       6          0
17          2020-03-30 08:15:36  0        SMS       6          0
18          2020-03-31 11:45:04  0        SMS       6          0
19          2020-04-01 17:39:22  0        SMS       6          0
20          2020-04-02 20:05:48  0        SMS       6          0
21          2020-04-04 18:44:25  0        SMS       6          0
22          2020-04-05 00:50:39  0        SMS       14         0
23          2020-04-06 01:22:44  0        SMS       15         0
24          2020-04-06 13:38:30  0        SMS       6          0
25          2020-04-07 01:05:15  0        SMS       1          0
26          2020-04-07 13:59:59  0        SMS       6          0
27          2020-04-08 18:31:44  0        SMS       6          0
28          2020-04-09 19:03:12  0        SMS       6          0
29          2020-04-11 18:36:12  0        iMessage  0          0
30          2020-04-11 18:36:24  0        iMessage  13         0
31          2020-04-11 18:40:51  0        iMessage  0          0
32          2020-04-11 18:44:14  0        iMessage  13         0
33          2020-04-12 02:08:31  0        SMS       16         0
34          2020-04-12 12:54:21  1        iMessage  13         0
35          2020-04-12 12:55:23  0        iMessage  13         0
36          2020-04-12 12:56:45  1        iMessage  13         1
37          2020-04-12 12:58:01  0        iMessage  13         1
38          2020-04-12 13:16:19  1        iMessage  13         1
39          2020-04-12 13:21:15  0        iMessage  13         1
40          2020-04-12 16:14:26  0        SMS       17         0
41          2020-04-12 16:15:00  1        iMessage  13         0
42          2020-04-12 16:16:15  0        SMS       17         0
43          2020-04-15 16:20:41  1        SMS       17         0
44          2020-04-15 17:19:51  0        SMS       6          0
45          2020-04-16 11:45:50  1        iMessage  13         1
46          2020-04-16 12:17:58  1        iMessage  13         0
47          2020-04-16 12:56:49  0        iMessage  13         0
48          2020-04-16 12:59:29  0        iMessage  13         0

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Handle_ID,
    service AS Service,
    country AS Country,
    CASE
        WHEN length(id) > 6
        THEN substr(id, 1, 3) || ' ... ' || substr(id, -3)
        ELSE '[short identifier]'
    END AS Masked_Identifier
FROM handle
ORDER BY ROWID;
SQL
```

Handle_ID	Service	Country	Masked_Identifier
1	SMS	us	[short identifier]
2	SMS	us	+18 ... 597
3	SMS	us	+13 ... 906
4	SMS	us	[short identifier]
5	SMS	us	+17 ... 496
6	SMS	us	[short identifier]
7	SMS	us	+12 ... 620
8	SMS	us	+13 ... 923
9	SMS	us	[short identifier]
10	SMS	us	+12 ... 110
11	SMS	us	[short identifier]
12	SMS	us	+13 ... 930
13	iMessage	us	jos ... com
14	SMS	us	+18 ... 436
15	SMS	us	[short identifier]
16	SMS	us	[short identifier]
17	SMS	us	+19 ... 479

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Attachment_ID,
    mime_type AS MIME_Type,
    uti AS UTI,
    total_bytes AS Size_Bytes,
    transfer_state AS Transfer_State,
    CASE
        WHEN filename IS NOT NULL
        THEN '[filename present]'
        ELSE '[no filename]'
    END AS Filename_Status
FROM attachment
ORDER BY ROWID;
SQL
Attachment_ID  MIME_Type        UTI                                                  Size_Bytes  Transfer_State  Filename_Status
1                               dyn.age81a5dzq7y066dbtf0g82peqf4hk2pdrb00n5xy        770534      5               [filename present]
2              image/jpeg       public.jpeg                                          676542      0               [filename present]
3              image/jpeg       public.jpeg                                          676542      5               [filename present]
4              image/jpeg       public.jpeg                                          368536      5               [filename present]
5              application/pdf  com.adobe.pdf                                        846528      5               [filename present]
6              application/pdf  com.adobe.pdf                                        7073725     5               [filename present]
7                               dyn.age81a5dzq7y066dbtf0g82peqf4hk2pdrb00n5xy        1598        5               [filename present]

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    maj.message_id AS Message_ID,
    maj.attachment_id AS Attachment_ID,
    a.mime_type AS MIME_Type,
    a.total_bytes AS Size_Bytes,
    a.transfer_state AS Transfer_State
FROM message_attachment_join AS maj
JOIN attachment AS a
    ON a.ROWID = maj.attachment_id
ORDER BY maj.message_id, maj.attachment_id;
SQL
Message_ID  Attachment_ID  MIME_Type         Size_Bytes  Transfer_State
──────────  ─────────────  ────────────────  ──────────  ──────────────
15          1                                770534      5
36          3              image/jpeg        676542      5
37          4              image/jpeg        368536      5
38          5              application/pdf   846528      5
39          6              application/pdf   7073725     5
45          7                                1598        5

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    m.ROWID AS Message_ID,
    datetime(m.date / 1000000000 + 978307200, 'unixepoch') AS UTC_Time,
    m.is_from_me AS From_Me,
    m.service AS Service,
    m.handle_id AS Handle_ID,
    CASE
        WHEN m.text IS NOT NULL AND length(m.text) > 0
        THEN 'Text present'
        ELSE 'No text'
    END AS Text_Status,
    a.ROWID AS Attachment_ID,
    a.mime_type AS MIME_Type,
    a.total_bytes AS Size_Bytes
FROM message AS m
JOIN message_attachment_join AS maj
    ON maj.message_id = m.ROWID
JOIN attachment AS a
    ON a.ROWID = maj.attachment_id
ORDER BY m.date;
SQL
Message_ID  UTC_Time             From_Me  Service   Handle_ID  Text_Status   Attachment_ID  MIME_Type         Size_Bytes
----------  -------------------  -------  --------  ---------  ------------  -------------  ----------------  ----------
15          2020-03-27 18:41:13  0        iMessage  13         Text present  1                                770534
36          2020-04-12 12:56:45  1        iMessage  13         Text present  3              image/jpeg        676542
37          2020-04-12 12:58:01  0        iMessage  13         Text present  4              image/jpeg        368536
38          2020-04-12 13:16:19  1        iMessage  13         Text present  5              application/pdf   846528
39          2020-04-12 13:21:15  0        iMessage  13         Text present  6              application/pdf   7073725
45          2020-04-16 11:45:50  1        iMessage  13         Text present  7                                1598

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Attachment_ID,
    filename AS Stored_Filename,
    mime_type AS MIME_Type,
    total_bytes AS Size_Bytes,
    transfer_state AS Transfer_State
FROM attachment
ORDER BY ROWID;
SQL
```

Attachment_ID	Stored_Filename	MIME_Type	Size_Bytes	Transfer_State
1	~/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment		770534	5
2	/var/tmp/com.apple.messages/com.apple.MobileSMS/Media/E54EFEAC-40A3-45D3-88BD-84B3E0A576FC/IMG_0008.JPG	image/jpeg	676542	0
3	~/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG	image/jpeg	676542	5
4	~/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg	image/jpeg	368536	5
5	~/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf	application/pdf	846528	5
6	~/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf	application/pdf	7073725	5
7	~/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-ADC0		1598	5

```
.headers on
.mode column

SELECT
    ROWID AS Attachment_ID,
    filename AS Stored_Filename,
    mime_type AS MIME_Type,
    total_bytes AS Size_Bytes,
    transfer_state AS Transfer_State
FROM attachment
ORDER BY ROWID;
SQL
```

Attachment_ID	Stored_Filename	MIME_Type	Size_Bytes	Transfer_State
1	~/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment		770534	5
2	/var/tmp/com.apple.messages/com.apple.MobileSMS/Media/E54EFEAC-40A3-45D3-88BD-84B3E0A576FC/IMG_0008.JPG	image/jpeg	676542	0
3	~/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG	image/jpeg	676542	5
4	~/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg	image/jpeg	368536	5
5	~/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up ?.pdf	application/pdf	846528	5
6	~/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf	application/pdf	7073725	5
7	~/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-ADC00A191D32/069EBD0E-FDA6-486A-AA02-49E369691551.pluginPayloadAttachment		1598	5

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"

echo "===== SMS ATTACHMENT DIRECTORY ====="
sudo find "$ROOT/private/var/mobile/Library/SMS/Attachments" \
-type f -printf '%p (%s bytes)\n' 2>/dev/null | sort

echo
echo "===== SEARCH FOR IDENTIFIED ATTACHMENTS ====="
sudo find "$ROOT" -type f \( \
-iname 'IMG_0008.JPG' -o \
-iname 'FullSizeRender.jpeg' -o \
-iname 'How was an iPhone set up?.pdf' -o \
-iname 'iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf' \
\) -printf '%p (%s bytes)\n' 2>/dev/null
===== SMS ATTACHMENT DIRECTORY =====
[sudo] password for kali:
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/._IMG_0008.JPG (1188 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG (676542 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg (368536 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/._How was an iPhone set up?.pdf (176 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf (846528 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf (7073725 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment (770534 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-ADC00A191D32/069EBD0E-FDA6-486A-AA02-49E369691551.pluginPayloadAttachment (1598 bytes)

===== SEARCH FOR IDENTIFIED ATTACHMENTS =====
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf (7073725 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/How was an iPhone set up?.pdf (846528 bytes
```

```
sudo find "$ROOT" -type f \( \
-iname 'IMG_0008.JPG' -o \
-iname 'FullSizeRender.jpeg' -o \
-iname 'How was an iPhone set up?.pdf' -o \
-iname 'iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf' \
\) -printf '%p (%s bytes)\n' 2>/dev/null
==== SMS ATTACHMENT DIRECTORY ====
[sudo] password for kali:
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/._IMG_0008.JPG (1188 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG (676542 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg (368536 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/._How was an iPhone set up?.pdf (176 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf (846528 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf (7073725 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment (770534 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-ADC00A191D32/069EBD0E-FDA6-486A-AA02-49E369691551.pluginPayloadAttachment (1598 bytes)

==== SEARCH FOR IDENTIFIED ATTACHMENTS ====
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf (7073725 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/How was an iPhone set up?.pdf (846528 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf (7073725 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg (368536 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf (846528 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG (676542 bytes)
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0008.JPG (676542 bytes)

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"

sudo sha256sum \
"$ROOT/private/var/mobile/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment" \
"$ROOT/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG" \
"$ROOT/private/var/mobile/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg" \
"$ROOT/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf" \
"$ROOT/private/var/mobile/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf" \
"$ROOT/private/var/mobile/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-ADC00A191D32/069EBD0E-FDA6-486A-AA02-49E369691551.pluginPayloadAttachment" \
"$ROOT/private/var/mobile/Media/DCIM/100APPLE/IMG_0008.JPG" \
"$ROOT/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/How was an iPhone set up?.pdf" \
"$ROOT/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf" \
| tee output/hashes/SMS_Attachment_SHA256.txt
51520a944c2e287a425080a97e1e78d902811d8a56ff4984f85ac5facc2df6ff  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/cb/11/F1E68FA1-1F02-4544-A1C2-E9FC69565B4D/AF54E2D6-D6C5-4652-ABED-58C6D0662D0E.pluginPayloadAttachment
df817f1c02f2c57c749a6a2d86e6beaafeb2186703745650c6bd784075757edf  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/10/00/A0159B39-9B33-401F-B5F2-87638BBCF97B/IMG_0008.JPG
1eb7fcc25d433fe5bf2783fec140c0088b2fae117bc15473ce7c6ca1e1827b4d  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/54/04/9A5EF4E2-87A7-4923-ABD7-218D94989A09/FullSizeRender.jpeg
82118655004fa94dc9257988cab0e02b7efda5fada929cbd92d04def11a136b6  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/68/08/430548CB-3157-418F-9B0E-3E677784E7A4/How was an iPhone set up?.pdf
255ac0cac61a3f701852bdd76a31f39e95b3760f04d53eb9e10a45cc2464cb3e  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/97/07/476F1BF4-4FE5-466A-AD8C-605576C5F16D/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf
f99ae173d40052f2907877a85afe9af32f875b346e4db1783ea94da48409bb65  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/Attachments/db/11/DDDBCB8D-6279-49B0-85B1-ADC00A191D32/069EBD0E-FDA6-486A-AA02-49E369691551.pluginPayloadAttachment
df817f1c02f2c57c749a6a2d86e6beaafeb2186703745650c6bd784075757edf  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/DCIM/100APPLE/IMG_0008.JPG
82118655004fa94dc9257988cab0e02b7efda5fada929cbd92d04def11a136b6  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/How was an iPhone set up?.pdf
255ac0cac61a3f701852bdd76a31f39e95b3760f04d53eb9e10a45cc2464cb3e  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Mobile Documents/com~apple~CloudDocs/Downloads/iOS_Bug_Reporting_for_Forensic_Purposes_1.2.pdf

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    ROWID AS Chat_ID,
    guid AS Chat_GUID,
    style AS Chat_Style,
    chat_identifier AS Chat_Identifier,
    service_name AS Service,
    display_name AS Display_Name
FROM chat
ORDER BY ROWID;
SQL
Chat_ID  Chat_GUID                          Chat_Style  Chat_Identifier          Service   Display_Name
-------  --------------------------------  ----------  -----------------------  --------  ------------
1        SMS;-;22000                        45          22000                    SMS
2        SMS;-;+18634205597                 45          +18634205597             SMS
3        SMS;-;+13176444906                 45          +13176444906             SMS
4        SMS;-;85760                        45          85760                    SMS
5        SMS;-;+17026604496                 45          +17026604496             SMS
6        SMS;-;32665                        45          32665                    SMS
7        SMS;-;+12074242620                 45          +12074242620             SMS
8        SMS;-;+13233208923                 45          +13233208923             SMS
9        SMS;-;47543                        45          47543                    SMS
10       SMS;-;+12349008110                 45          +12349008110             SMS
11       SMS;-;732873                       45          732873                   SMS
12       SMS;-;+13214065930                 45          +13214065930             SMS
13       iMessage;-;joshua.hickman1@me.com  45          joshua.hickman1@me.com   iMessage
14       SMS;-;+18334030436                 45          +18334030436             SMS
15       SMS;-;78156                        45          78156                    SMS
16       SMS;-;44398                        45          44398                    SMS
17       SMS;-;+19195790479                 45          +19195790479             SMS

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    c.ROWID AS Chat_ID,
    'Chat-' || c.ROWID AS Conversation_Reference,
    c.service_name AS Service,
    COUNT(m.ROWID) AS Message_Count,
    datetime(
        MIN(m.date) / 1000000000 + 978307200,
        'unixepoch'
    ) AS First_Message_UTC,
    datetime(
        MAX(m.date) / 1000000000 + 978307200,
        'unixepoch'
    ) AS Last_Message_UTC
FROM chat c
LEFT JOIN chat_message_join cmj
    ON cmj.chat_id = c.ROWID
LEFT JOIN message m
    ON m.ROWID = cmj.message_id
GROUP BY c.ROWID, c.service_name
ORDER BY c.ROWID;
SQL
```

Chat_ID	Conversation_Reference	Service	Message_Count	First_Message_UTC	Last_Message_UTC
1	Chat-1	SMS	2	2020-03-22 01:10:59	2020-04-07 01:05:15
2	Chat-2	SMS	1	2020-03-22 15:16:16	2020-03-22 15:16:16
3	Chat-3	SMS	1	2020-03-22 19:21:40	2020-03-22 19:21:40
4	Chat-4	SMS	1	2020-03-23 01:06:38	2020-03-23 01:06:38
5	Chat-5	SMS	1	2020-03-23 01:38:03	2020-03-23 01:38:03
6	Chat-6	SMS	14	2020-03-23 17:41:36	2020-04-15 17:19:51
7	Chat-7	SMS	1	2020-03-24 01:21:34	2020-03-24 01:21:34
8	Chat-8	SMS	1	2020-03-24 01:27:40	2020-03-24 01:27:40

```
    COUNT(m.ROWID) AS Message_Count,
    datetime(
        MIN(m.date) / 1000000000 + 978307200,
        'unixepoch'
    ) AS First_Message_UTC,
    datetime(
        MAX(m.date) / 1000000000 + 978307200,
        'unixepoch'
    ) AS Last_Message_UTC
FROM chat c
LEFT JOIN chat_message_join cmj
    ON cmj.chat_id = c.ROWID
LEFT JOIN message m
    ON m.ROWID = cmj.message_id
GROUP BY c.ROWID, c.service_name
ORDER BY c.ROWID;
SQL
```

Chat_ID	Conversation_Reference	Service	Message_Count	First_Message_UTC	Last_Message_UTC
1	Chat-1	SMS	2	2020-03-22 01:10:59	2020-04-07 01:05:15
2	Chat-2	SMS	1	2020-03-22 15:16:16	2020-03-22 15:16:16
3	Chat-3	SMS	1	2020-03-22 19:21:40	2020-03-22 19:21:40
4	Chat-4	SMS	1	2020-03-23 01:06:38	2020-03-23 01:06:38
5	Chat-5	SMS	1	2020-03-23 01:38:03	2020-03-23 01:38:03
6	Chat-6	SMS	14	2020-03-23 17:41:36	2020-04-15 17:19:51
7	Chat-7	SMS	1	2020-03-24 01:21:34	2020-03-24 01:21:34
8	Chat-8	SMS	1	2020-03-24 01:27:40	2020-03-24 01:27:40
9	Chat-9	SMS	1	2020-03-24 01:52:23	2020-03-24 01:52:23
10	Chat-10	SMS	1	2020-03-25 00:47:38	2020-03-25 00:47:38
11	Chat-11	SMS	1	2020-03-25 00:49:24	2020-03-25 00:49:24
12	Chat-12	SMS	1	2020-03-26 01:11:36	2020-03-26 01:11:36
13	Chat-13	iMessage	16	2020-03-27 18:41:13	2020-04-16 12:59:29
14	Chat-14	SMS	1	2020-04-05 00:50:39	2020-04-05 00:50:39
15	Chat-15	SMS	1	2020-04-06 01:22:44	2020-04-06 01:22:44
16	Chat-16	SMS	1	2020-04-12 02:08:31	2020-04-12 02:08:31
17	Chat-17	SMS	3	2020-04-12 16:14:26	2020-04-15 16:20:41

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

mouse pointer outside or press Ctrl+Alt

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sqlite3 -readonly working/messages/sms.db <<'SQL'
.headers on
.mode column

SELECT
    m.ROWID AS Message_ID,
    cmj.chat_id AS Chat_ID,
    datetime(
        m.date / 1000000000 + 978307200,
        'unixepoch'
    ) AS Timestamp_UTC,
    CASE
        WHEN m.is_from_me = 1 THEN 'Outgoing'
        WHEN m.is_from_me = 0 THEN 'Incoming'
        ELSE 'Unknown'
    END AS Direction,
    m.service AS Service,
    length(m.text) AS Text_Length,
    m.cache_has_attachments AS Attachment_Flag
FROM message m
JOIN chat_message_join cmj
    ON cmj.message_id = m.ROWID
WHERE cmj.chat_id IN (6, 13)
ORDER BY m.date;
SQL
Message_ID  Chat_ID  Timestamp_UTC        Direction  Service   Text_Length  Attachment_Flag
----------  -------  -------------------  ---------  --------  -----------  ---------------
6           6        2020-03-23 17:41:36  Incoming   SMS       100          0
12          6        2020-03-25 17:54:08  Incoming   SMS       101          0
14          6        2020-03-26 20:55:46  Incoming   SMS       87           0
15          13       2020-03-27 18:41:13  Incoming   iMessage  65           1
16          6        2020-03-28 18:18:43  Incoming   SMS       87           0
17          6        2020-03-30 08:15:36  Incoming   SMS       101          0
18          6        2020-03-31 11:45:04  Incoming   SMS       101          0
19          6        2020-04-01 17:39:22  Incoming   SMS       100          0
```

```
    ON cmj.message_id = m.ROWID
WHERE cmj.chat_id IN (6, 13)
ORDER BY m.date;
SQL
```

Message_ID	Chat_ID	Timestamp_UTC	Direction	Service	Text_Length	Attachment_Flag
6	6	2020-03-23 17:41:36	Incoming	SMS	100	0
12	6	2020-03-25 17:54:08	Incoming	SMS	101	0
14	6	2020-03-26 20:55:46	Incoming	SMS	87	0
15	13	2020-03-27 18:41:13	Incoming	iMessage	65	1
16	6	2020-03-28 18:18:43	Incoming	SMS	87	0
17	6	2020-03-30 08:15:36	Incoming	SMS	101	0
18	6	2020-03-31 11:45:04	Incoming	SMS	101	0
19	6	2020-04-01 17:39:22	Incoming	SMS	100	0
20	6	2020-04-02 20:05:48	Incoming	SMS	87	0
21	6	2020-04-04 18:44:25	Incoming	SMS	100	0
24	6	2020-04-06 13:38:30	Incoming	SMS	100	0
26	6	2020-04-07 13:59:59	Incoming	SMS	101	0
27	6	2020-04-08 18:31:44	Incoming	SMS	101	0
28	6	2020-04-09 19:03:12	Incoming	SMS	101	0
29	13	2020-04-11 18:36:12	Incoming	iMessage		0
30	13	2020-04-11 18:36:24	Incoming	iMessage		0
31	13	2020-04-11 18:40:51	Incoming	iMessage		0
32	13	2020-04-11 18:44:14	Incoming	iMessage		0
34	13	2020-04-12 12:54:21	Outgoing	iMessage	44	0
35	13	2020-04-12 12:55:23	Incoming	iMessage	69	0
36	13	2020-04-12 12:56:45	Outgoing	iMessage	1	1
37	13	2020-04-12 12:58:01	Incoming	iMessage	1	1
38	13	2020-04-12 13:16:19	Outgoing	iMessage	43	1
39	13	2020-04-12 13:21:15	Incoming	iMessage	38	1
41	13	2020-04-12 16:15:00	Outgoing	iMessage	50	0
44	6	2020-04-15 17:19:51	Incoming	SMS	102	0
45	13	2020-04-16 11:45:50	Outgoing	iMessage	102	1
46	13	2020-04-16 12:17:58	Outgoing	iMessage	41	0
47	13	2020-04-16 12:56:49	Incoming	iMessage	44	0
48	13	2020-04-16 12:59:29	Incoming	iMessage	51	0

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
    ON cmj.message_id = m.ROWID
WHERE cmj.chat_id IN (6, 13)
ORDER BY m.date;
SQL
```

Message_ID	Chat_ID	Timestamp_UTC	Direction	Service	Text_Length	Attachment_Flag
6	6	2020-03-23 17:41:36	Incoming	SMS	100	0
12	6	2020-03-25 17:54:08	Incoming	SMS	101	0
14	6	2020-03-26 20:55:46	Incoming	SMS	87	0
15	13	2020-03-27 18:41:13	Incoming	iMessage	65	1
16	6	2020-03-28 18:18:43	Incoming	SMS	87	0
17	6	2020-03-30 08:15:36	Incoming	SMS	101	0
18	6	2020-03-31 11:45:04	Incoming	SMS	101	0
19	6	2020-04-01 17:39:22	Incoming	SMS	100	0
20	6	2020-04-02 20:05:48	Incoming	SMS	87	0
21	6	2020-04-04 18:44:25	Incoming	SMS	100	0
24	6	2020-04-06 13:38:30	Incoming	SMS	100	0
26	6	2020-04-07 13:59:59	Incoming	SMS	101	0
27	6	2020-04-08 18:31:44	Incoming	SMS	101	0
28	6	2020-04-09 19:03:12	Incoming	SMS	101	0
29	13	2020-04-11 18:36:12	Incoming	iMessage		0
30	13	2020-04-11 18:36:24	Incoming	iMessage		0
31	13	2020-04-11 18:40:51	Incoming	iMessage		0
32	13	2020-04-11 18:44:14	Incoming	iMessage		0
34	13	2020-04-12 12:54:21	Outgoing	iMessage	44	0
35	13	2020-04-12 12:55:23	Incoming	iMessage	69	0
36	13	2020-04-12 12:56:45	Outgoing	iMessage	1	1
37	13	2020-04-12 12:58:01	Incoming	iMessage	1	1
38	13	2020-04-12 13:16:19	Outgoing	iMessage	43	1
39	13	2020-04-12 13:21:15	Incoming	iMessage	38	1
41	13	2020-04-12 16:15:00	Outgoing	iMessage	50	0
44	6	2020-04-15 17:19:51	Incoming	SMS	102	0
45	13	2020-04-16 11:45:50	Outgoing	iMessage	102	1
46	13	2020-04-16 12:17:58	Outgoing	iMessage	41	0
47	13	2020-04-16 12:56:49	Incoming	iMessage	44	0
48	13	2020-04-16 12:59:29	Incoming	iMessage	51	0

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
8     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B65796564417263686976657D1080954726F6F748001A20B0C55246E75
                    6C6C5F101C636F6D2E6170706C652E44617461636C6173732E5562697175
                    69747908111A24293237494C5153565C0000000000000101000000000000
                    000D0000000000000000000000000000007B

9     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B65796564417263686976657D1080954726F6F748001A20B0C55246E75
                    6C6C5F101F636F6D2E6170706C652E44617461636C6173732E436F726552
                    6F7574696E6508111A24293237494C5153565C0000000000000101000000
                    000000000D0000000000000000000000000000007E

10    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B65796564417263686976657D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E536D617274
                    44656661756C747308111A24293237494C5153565C0000000000000101000
                    0000000000000D0000000000000000000000000000080

11    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B65796564417263686976657D1080954726F6F748001A20B0C55246E75
                    6C6C5F101D636F6D2E6170706C652E44617461636C6173732E52656D696E
                    6465727308111A24293237494C5153565C0000000000000101000000000
                    00000D0000000000000000000000000000007C

12    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B65796564417263686976657D1080954726F6F748001A20B0C55246E75
                    6C6C5F1022636F6D2E6170706C652E44617461636C6173732E436F6E7461
                    6374735365617263808111A24293237494C5153565C000000000000010
                    0000000000000000D0000000000000000000000000000081

13    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B65796564417263686976657D1080954726F6F748001A20B0C55246E75
```

```
13    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101C636F6D2E6170706C652E44617461636C6173732E436F6E7461
                    63747308111A24293237494C5153565C0000000000000101000000000000
                    000D0000000000000000000000000000007B

14    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101C636F6D2E6170706C652E44617461636C6173732E4B65795661
                    6C756508111A24293237494C5153565C0000000000000101000000000000
                    000D0000000000000000000000000000007B

15    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101A636F6D2E6170706C652E44617461636C6173732E53746F636B
                    7308111A24293237494C5153565C0000000000000101000000000000000D
                    0000000000000000000000000000079

16    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1020636F6D2E6170706C652E44617461636C6173732E436C6F7564
                    4465736B746F7008111A24293237494C5153565C0000000000001010000
                    000000000000D0000000000000000000000000000007F

17    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1025636F6D2E6170706C652E44617461636C6173732E434B446174
                    6162617365536572766963650811A24293237494C5153565C0000000000
                    000101000000000000000D0000000000000000000000000000000084

18    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1018636F6D2E6170706C652E44617461636C6173732E486F6D6508
```

```
18    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B65796564172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1018636F6D2E6170706C652E44617461636C6173732E486F6D6508
                    111A24293237494C5153565C00000000000001010000000000000D0000
                    0000000000000000000000000077

19    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B65796564172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101B636F6D2E6170706C652E44617461636C6173732E53686F6562
                    6F7808111A24293237494C5153565C00000000000001010000000000000D
                    0D0000000000000000000000000000007A

20    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B65796564172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1018636F6D2E6170706C652E44617461636C6173732E5369726908
                    111A24293237494C5153565C00000000000001010000000000000D0000
                    0000000000000000000000000077

21    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B65796564172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E4465766963
                    654C6F6361746F7208111A24293237494C5153565C000000000000010100
                    0000000000000D0000000000000000000000000000000080

22    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B65796564172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1018636F6D2E6170706C652E44617461636C6173732E4E65777308
                    111A24293237494C5153565C00000000000001010000000000000D0000
                    0000000000000000000000000077

23    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B65796564172636869766572D1080954726F6F748001A20B0C55246E75
```

```
23    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101D636F6D2E6170706C652E44617461636C6173732E5368617265
                    4D656E7508111A24293237494C5153565C0000000000001010000000000
                    00000D0000000000000000000000000000007C

24    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1029636F6D2E6170706C652E44617461636C6173732E50726F7465
                    63746564436C6F756453746F7261676508111A24293237494C5153565C00
                    0000000000010100000000000000D0000000000000000000000000000000
                    88

25    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101F636F6D2E6170706C652E44617461636C6173732E4D65646961
                    53747265616D08111A24293237494C5153565C0000000000001010000000
                    000000000D0000000000000000000000000000007E

26    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101D636F6D2E6170706C652E44617461636C6173732E43616C656E
                    6461727308111A24293237494C5153565C0000000000001010000000000
                    00000D0000000000000000000000000000007C

27    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1020636F6D2E6170706C652E44617461636C6173732E4B65796368
                    61696E53796E6308111A24293237494C5153565C0000000000001010000
                    0000000000D0000000000000000000000000000007F

28    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120001 86A05F100F4E53
```

```
                    000000000000D000000000000000000000000000000007F

28    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101A636F6D2E6170706C652E44617461636C6173732E50686F746F
                    7308111A24293237494C5153565C0000000000000101000000000000000D
                    0000000000000000000000000000000079

29    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1023636F6D2E6170706C652E44617461636C6173732E434B446576
                    6963655365727669636508111A24293237494C5153565C00000000000001
                    010000000000000000D000000000000000000000000000000082

30    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E5368617265
                    6453747265616D7308111A24293237494C5153565C00000000000010100
                    0000000000000D000000000000000000000000000000080

31    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1022636F6D2E6170706C652E44617461636C6173732E434B536861
                    7265536572766963650811A24293237494C5153565C0000000000000101
                    0000000000000D000000000000000000000000000000081

32    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1019636F6D2E6170706C652E44617461636C6173732E4E6F746573
                    08111A24293237494C5153565C0000000000000101000000000000000D00
                    0000000000000000000000000000078

33    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
```

```
33    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101E636F6D2E6170706C652E44617461636C6173732E4E6574776F
                    726B56504E08111A24293237494C5153565C0000000000000010100000000
                    0000000D0000000000000000000000000000007D

34    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101A636F6D2E6170706C652E44617461636C6173732E4261636B75
                    7008111A24293237494C5153565C000000000000010100000000000000D
                    00000000000000000000000000000079

35    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E5368617265
                    4C6F636174696F6E08111A24293237494C5153565C000000000000010100
                    00000000000000D0000000000000000000000000000080

36    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101F636F6D2E6170706C652E44617461636C6173732E436C6F7564
                    50686F746F7308111A24293237494C5153565C0000000000000101000000
                    000000000D0000000000000000000000000000007E

37    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E434B436F64
                    6553657276696365508111A24293237494C5153565C000000000000010100
                    00000000000000D0000000000000000000000000000080

38    blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
```

```
                    000000000D00000000000000000000000000007E

37      blob        62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E434B436F64
                    6553657276696365081111A24293237494C5153565C000000000000010100
                    0000000000000D0000000000000000000000000000000080

38      blob        62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1020636F6D2E6170706C652E44617461636C6173732E417661696C
                    6162696C69747908111A24293237494C5153565C0000000000000010100000
                    0000000000D0000000000000000000000000000007F

39      blob        62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101F636F6D2E6170706C652E44617461636C6173732E5365617263
                    6850617274790811A24293237494C5153565C00000000000010100000000
                    000000000D0000000000000000000000000000007E

40      blob        62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1018636F6D2E6170706C652E44617461636C6173732E69776D6208
                    111A24293237494C5153565C000000000000010100000000000000D0000
                    0000000000000000000000077

41      blob        62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1024636F6D2E6170706C652E44617461636C6173732E434B4D6574
                    7269637353657276696365081111A24293237494C5153565C000000000000
                    0101000000000000000D0000000000000000000000000000000083

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DB="working/accounts3/Accounts3.sqlite"

echo "═══ ACCOUNT PROPERTY OWNERSHIP ═══"

sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Property_Ownership.txt
.headers on
.mode column

SELECT
    p.ZOWNER AS account_pk,
    t.ZACCOUNTTYPEDESCRIPTION AS account_type,
    p.ZKEY AS property_key,
    length(p.ZVALUE) AS value_bytes,
    typeof(p.ZVALUE) AS storage_type
FROM ZACCOUNTPROPERTY AS p
LEFT JOIN ZACCOUNT AS a ON p.ZOWNER = a.Z_PK
LEFT JOIN ZACCOUNTTYPE AS t ON a.ZACCOUNTTYPE = t.Z_PK
ORDER BY p.ZOWNER, p.ZKEY;
SQL
═══ ACCOUNT PROPERTY OWNERSHIP ═══
account_pk  account_type      property_key                                  value_bytes  storage_type
──────────  ────────────────  ────────────────────────────────────────────  ───────────  ────────────
1           iTunes Store      cookies                                       2261         blob
1           iTunes Store      dsid                                          139          blob
1           iTunes Store      eligibleServices                              252          blob
1           iTunes Store      firstName                                     138          blob
1           iTunes Store      isActiveLockerAccount                         138          blob
1           iTunes Store      isManagedAppleID                              138          blob
1           iTunes Store      lastName                                      138          blob
1           iTunes Store      mergedPrivacyAcknowledgements                 138          blob
1           iTunes Store      privacyAcknowledgement                        576          blob
1           iTunes Store      scope                                         139          blob
1           iTunes Store      storefrontID                                  149          blob
2           iTunes Store      accountCredits                                138          blob
2           iTunes Store      accountFlags                                  915          blob
```

```
═══ ACCOUNT PROPERTY OWNERSHIP ═══
```

account_pk	account_type	property_key	value_bytes	storage_type
1	iTunes Store	cookies	2261	blob
1	iTunes Store	dsid	139	blob
1	iTunes Store	eligibleServices	252	blob
1	iTunes Store	firstName	138	blob
1	iTunes Store	isActiveLockerAccount	138	blob
1	iTunes Store	isManagedAppleID	138	blob
1	iTunes Store	lastName	138	blob
1	iTunes Store	mergedPrivacyAcknowledgements	138	blob
1	iTunes Store	privacyAcknowledgement	576	blob
1	iTunes Store	scope	139	blob
1	iTunes Store	storefrontID	149	blob
2	iTunes Store	accountCredits	138	blob
2	iTunes Store	accountFlags	915	blob
2	iTunes Store	altDSID	186	blob
2	iTunes Store	automaticDownloadKinds	219	blob
2	iTunes Store	availableServiceTypes	139	blob
2	iTunes Store	cloudLibraryStateReason	280	blob
2	iTunes Store	cookies	6441	blob
2	iTunes Store	didAgreeToTerms	138	blob
2	iTunes Store	dsid	146	blob
2	iTunes Store	eligibleServices	276	blob
2	iTunes Store	enabledServiceTypes	139	blob
2	iTunes Store	firstName	145	blob
2	iTunes Store	freeDownloadsPasswordSetting	139	blob
2	iTunes Store	iCloudFamily	138	blob
2	iTunes Store	inGoodStanding	138	blob
2	iTunes Store	isActiveLockerAccount	138	blob
2	iTunes Store	isManagedAppleID	138	blob
2	iTunes Store	isNewCustomer	138	blob
2	iTunes Store	kind	139	blob
2	iTunes Store	lastAuthenticateCredentialSource	139	blob
2	iTunes Store	lastAuthenticated	223	blob
2	iTunes Store	lastAuthenticationServerResponse	3304	blob
2	iTunes Store	lastName	142	blob
2	iTunes Store	musicPrivateListeningEnabled	138	blob
2	iTunes Store	paidPurchasesPasswordSetting	139	blob

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ python3 - <<'PY' | tee output/inventory/Accounts3_Data_Classes_Decoded.txt
import sqlite3
import plistlib

db = "working/accounts3/Accounts3.sqlite"
con = sqlite3.connect(f"file:{db}?mode=ro", uri=True)

print("Z_PK | DECODED DATA CLASS")

for pk, blob in con.execute("SELECT Z_PK, ZNAME FROM ZDATACLASS ORDER BY Z_PK"):
    try:
        value = plistlib.loads(blob)
        print(f"{pk} | {value}")
    except Exception as e:
        print(f"{pk} | Decode error: {e}")

con.close()
PY
Z_PK | DECODED DATA CLASS
1 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Bookmarks']}
2 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Health']}
3 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Notifications']}
4 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Messages']}
5 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Content']}
6 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Mail']}
7 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.ServerDocuments']}
8 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Ubiquity']}
9 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.CoreRoutine']}
10 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.SmartDefaults']}
11 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Reminders']}
12 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.ContactsSearch']}
13 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Contacts']}
14 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.KeyValue']}
15 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.Stocks']}
16 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.CloudDesktop']}
17 | {'$version': 100000, '$archiver': 'NSKeyedArchiver', '$top': {'root': UID(1)}, '$objects': ['$null', 'com.apple.Dataclass.CKDatabaseService']}
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2
ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"

sudo find "$ROOT/private/var/mobile" -type f \( \
-iname "sms.db" -o -iname "AddressBook.sqlitedb" \
-o -iname "CallHistory.storedata" -o -iname "History.db" \
-o -iname "Calendar.sqlitedb" -o -iname "Photos.sqlite" \
-o -iname "knowledgeC.db" -o -iname "healthdb_secure.sqlite" \
-o -iname "Cache.db" \
\) -printf '%p | %s bytes\n' | sort \
| tee output/inventory/Key_iOS_Database_Locations.txt
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/0B93E4DC-3E6E-443E-81C3-61292FDA6091/Library/Caches/com.kik.chat/Cache
.db | 1146880 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/0F51CB71-2E7B-4B9B-BB63-CEBE2D398AE8/Library/Caches/com.apple.AppleMed
iaServices/Cache.db | 2387968 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/0F51CB71-2E7B-4B9B-BB63-CEBE2D398AE8/Library/Caches/com.apple.podcasts
/Cache.db | 446464 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/0F51CB71-2E7B-4B9B-BB63-CEBE2D398AE8/Library/Caches/com.apple.podcasts
/com.apple.AMPStoreKit.JS/Cache.db | 65536 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/142EF15D-F4D2-4737-909C-60AA005A4743/Library/Caches/com.apple.AppleAcc
ount/Cache.db | 73728 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/142EF15D-F4D2-4737-909C-60AA005A4743/Library/Caches/com.apple.AppleMed
iaServices/Cache.db | 57344 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/142EF15D-F4D2-4737-909C-60AA005A4743/Library/Caches/com.apple.Preferen
ces/Cache.db | 81920 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/1A31CE72-5345-4BD5-9A74-D137FF9FEC70/Library/Caches/com.apple.mobilema
il/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/1A366D76-74D8-46EC-91C0-B1F8378D8828/Library/Caches/com.zhiliaoapp.mus
ically/Cache.db | 61440 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/237E0C71-5961-459B-B8AC-831364AB4679/Library/Caches/com.hammerandchise
l.discord/Cache.db | 405504 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/2C0EA118-A71B-4CFD-AF2E-7FE132AC59DF/Library/Caches/com.tinginteractiv
e.usms/Cache.db | 1916928 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/2E5D09C9-1142-4332-86FC-E71DC449C32B/Library/Caches/ch.protonmail.prot
onmail/Cache.db | 4096 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/30A884F2-32EA-45FC-ACD5-7639FE9CACDA/Library/Caches/com.skout.SKOUT/Ca
che.db | 4096 bytes
```

```
onmail/Cache.db | 4096 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/30A884F2-32EA-45FC-ACD5-7639FE9CACDA/Library/Caches/com.skout.SKOUT/Cache.db | 4096 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/30A884F2-32EA-45FC-ACD5-7639FE9CACDA/Library/Caches/com.skout.SKOUT/nsurlcache/Cache.db | 0 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/30A884F2-32EA-45FC-ACD5-7639FE9CACDA/Library/Caches/com.skout.SKOUT/TMGNetworkCache/Cache.db | 225280 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/30CC1B22-FC02-4F71-A2DF-C78A6D50098C/Library/Caches/co.babypenguin.imo/Cache.db | 81920 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/32BDEBF7-0383-4D53-A927-7BE6BE516B50/Library/Caches/com.apple.AppleAccount/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/32BDEBF7-0383-4D53-A927-7BE6BE516B50/Library/Caches/com.apple.AppleMediaServices/Cache.db | 897024 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/32BDEBF7-0383-4D53-A927-7BE6BE516B50/Library/Caches/com.apple.AppStore/Cache.db | 1761280 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/454F2341-011D-45FD-AED9-FEA01B82658D/Library/Caches/com.reddit.Reddit/Cache.db | 180224 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/454F2341-011D-45FD-AED9-FEA01B82658D/Library/Caches/com.reddit.Reddit/imagedownload/Cache.db | 675840 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/45831D5F-1CB8-4F41-A20D-C3451D0E6459/Library/Caches/net.whatsapp.WhatsApp/Cache.db | 57344 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/4AB3E649-0D19-4DAA-91DC-7AED82FD53BB/Library/Caches/com.apple.Music/Cache.db | 765952 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/5611BB53-9A74-4EEA-B7A3-72290FEAF095/Library/Caches/com.mewe/Cache.db | 188416 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/575389B0-AF53-4365-8DEA-A099AE56814B/Library/Caches/com.signify.hue.blue/Cache.db | 69632 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/5EDB4A63-F50B-4449-8244-2A74FB935FB3/Library/Caches/com.fitbit.FitbitMobile/Cache.db | 204800 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/659C2309-A3EF-4EC9-8CEF-B281D80F0D7B/Library/Caches/com.belkin.plugin/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/6C653DCC-A39C-484E-8457-2C3CAAFF9AD7/Library/Caches/ph.telegra.Telegraph/Cache.db | 114688 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/6DD40ADC-30FE-452B-B002-A15CAAF04795/Library/Caches/com.adhoclabs.burner/Cache.db | 528384 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/71B2EBD8-B6DF-4118-908C-7B2976A3019E/Library/Caches/com.burbn.instagram/IGCache/Cache.db | 0 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/79B6E76A-FEB5-4A32-92F6-444FAEE57A5E/Library/Caches/com.apple.mobilephone/Cache.db | 53248 bytes
```

```
ount/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/32BDEBF7-0383-4D53-A927-7BE6BE516B50/Library/Caches/com.apple.AppleMediaServices/Cache.db | 897024 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/32BDEBF7-0383-4D53-A927-7BE6BE516B50/Library/Caches/com.apple.AppStore/Cache.db | 1761280 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/454F2341-011D-45FD-AED9-FEA01B82658D/Library/Caches/com.reddit.Reddit/Cache.db | 180224 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/454F2341-011D-45FD-AED9-FEA01B82658D/Library/Caches/com.reddit.Reddit/imagedownload/Cache.db | 675840 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/45831D5F-1CB8-4F41-A20D-C3451D0E6459/Library/Caches/net.whatsapp.WhatsApp/Cache.db | 57344 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/4AB3E649-0D19-4DAA-91DC-7AED82FD53BB/Library/Caches/com.apple.Music/Cache.db | 765952 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/5611BB53-9A74-4EEA-B7A3-72290FEAF095/Library/Caches/com.mewe/Cache.db | 188416 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/575389B0-AF53-4365-8DEA-A099AE56814B/Library/Caches/com.signify.hue.blue/Cache.db | 69632 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/5EDB4A63-F50B-4449-8244-2A74FB935FB3/Library/Caches/com.fitbit.FitbitMobile/Cache.db | 204800 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/659C2309-A3EF-4EC9-8CEF-B281D80F0D7B/Library/Caches/com.belkin.plugin/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/6C653DCC-A39C-484E-8457-2C3CAAFF9AD7/Library/Caches/ph.telegra.Telegraph/Cache.db | 114688 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/6DD40ADC-30FE-452B-B002-A15CAAF04795/Library/Caches/com.adhoclabs.burner/Cache.db | 528384 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/71B2EBD8-B6DF-4118-908C-7B2976A3019E/Library/Caches/com.burbn.instagram/IGCache/Cache.db | 0 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/79B6E76A-FEB5-4A32-92F6-444FAEE57A5E/Library/Caches/com.apple.mobilephone/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/7CC43357-2BAA-4DF0-8472-8CE77567E9D9/Library/Caches/com.burbn.threads/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/842C9702-4CA9-4A73-A315-62A1F4FD7537/Library/Caches/com.apple.ctcategories.service/Cache.db | 61440 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/842C9702-4CA9-4A73-A315-62A1F4FD7537/Library/Caches/com.apple.mobilesafari/Cache.db | 77824 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/842C9702-4CA9-4A73-A315-62A1F4FD7537/Library/Caches/com.apple.SafariViewService/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Containers/Data/Application/842C9702-4CA9-4A73-A315-62A1F4FD7537/Library/Caches/com.apple.WebKit.Networking/com.apple.SafariViewService.2.PrivateBrowsing/Cache.db | 53248 bytes
```

```
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.remindd/com.apple.dataaccessd.E9B5703B-F844-4845-AD3D-08DE58806F82/Cache.
db | 4096 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.Safari.SafeBrowsing/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.ScreenTimeAgent/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.Search.framework/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.Security/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.siri-distributed-evaluation/Cache.db | 81920 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.softwareupdateservicesd/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.springboard/Cache.db | 61440 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.tipsd/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.apple.triald/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/com.saurik.Cydia/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/kjc.loader/Cache.db | 61440 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/OTACrashCopier/Cache.db | 53248 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/rtcreportingd/Cache.db | 90112 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/sharedCaches/com.apple.iTunesStore.NSURLCache/Cache.db | 1380352 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/sharedCaches/com.apple.Radio.RadioRequestURLCache/Cache.db | 61440 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Caches/sharedCaches/com.apple.WatchListKit.NSURLCache/Cache.db | 4096 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Calendar/Calendar.sqlitedb | 712704 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CallHistoryDB/CallHistory.storedata | 73728 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/com.apple.internal.ck/history.db | 45056 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db | 14213120 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Health/healthdb_secure.sqlite | 10002432 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Safari/History.db | 98304 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/sms.db | 356352 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Media/PhotoData/Photos.sqlite | 2334720 bytes
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ sudo find "$ROOT/private/var" -type f \( \
-iname "*location*.db" -o -iname "*routine*.db" \
-o -iname "*knowledge*.db" \
\) -printf '%p | %s bytes\n' | sort \
| tee output/inventory/Location_Routine_Database_Candidates.txt
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Assistant/knowledgeC.db | 217088 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/CoreDuet/Knowledge/knowledgeC.db | 14213120 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/DeviceRegistry/8DDD238A-F8E1-41D2-B0BC-D2756E655890/CompanionSync/com.apple.private.alloy.
location.auth.db | 81920 bytes
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/DeviceRegistry/8DDD238A-F8E1-41D2-B0BC-D2756E655890/CompanionSync/com.apple.private.alloy.
location.wifitilesync.db | 81920 bytes
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ CURRENT DIRECTORY ═══"
pwd

echo "═══ CHECK EVIDENCE ROOT ═══"
ls -ld working/filesystem/Volumes/JOSH/NoTar-13-4-1

echo "═══ LOCATE SMS DATABASE ═══"
sudo find working/filesystem -type f -name "sms.db" -printf '%p | %s bytes\n'
═══ CURRENT DIRECTORY ═══
/home/kali/SBT-DF205-Case2
═══ CHECK EVIDENCE ROOT ═══
drwxr-xr-x 14 99 99 4096 Apr 18  2020 working/filesystem/Volumes/JOSH/NoTar-13-4-1
═══ LOCATE SMS DATABASE ═══
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/sms.db | 356352 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ sudo ls -la "working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/"
total 3176
drwx------   6 99 99    4096 Nov  4 03:45 .
drwx--x--x 127 99 99    4096 Nov  4 03:55 ..
drwx------  15 99 99    4096 Apr 16  2020 Attachments
drwx------   2 99 99    4096 Mar 22  2020 CloudKitMetaData
-rw-r--r--   1 99 99   18437 Apr 16  2020 com.apple.messages.geometrycache_v3.plist
drwxr-xr-x   3 99 99    4096 Apr 16  2020 Drafts
drwxrwxrwx   3 99 99    4096 Apr 12  2020 PluginMetaDataCache
-rw-r--r--   1 99 99   32768 Apr 12  2020 prewarm.db
-rw-r--r--   1 99 99   32768 Apr 12  2020 prewarm.db-shm
-rw-r--r--   1 99 99    4152 Apr 12  2020 prewarm.db-wal
-rw-r--r--   1 99 99  356352 Apr 16  2020 sms.db
-rw-r--r--   1 99 99   32768 Apr 16  2020 sms.db-shm
-rw-r--r--   1 99 99 2743952 Apr 16  2020 sms.db-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

mkdir -p working/messages

ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"
SMS="$ROOT/private/var/mobile/Library/SMS"

sudo cp "$SMS/sms.db" "$SMS/sms.db-shm" "$SMS/sms.db-wal" working/messages/

sudo chown kali:kali working/messages/*
ls -lh working/messages/
total 3.0M
-rw-r--r-- 1 kali kali 348K Nov  4 14:36 sms.db
-rw-r--r-- 1 kali kali  32K Nov  4 14:36 sms.db-shm
-rw-r--r-- 1 kali kali 2.7M Nov  4 14:36 sms.db-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"
SMS="$ROOT/private/var/mobile/Library/SMS"

echo "===== ORIGINAL SOURCE HASHES ====="
sudo sha256sum "$SMS/sms.db" "$SMS/sms.db-shm" "$SMS/sms.db-wal" \
| tee output/hashes/SMS_Source_SHA256.txt

echo
echo "===== WORKING COPY HASHES ====="
sha256sum working/messages/sms.db working/messages/sms.db-shm working/messages/sms.db-wal \
| tee output/hashes/SMS_Working_SHA256.txt
===== ORIGINAL SOURCE HASHES =====
67e847fdd1e51dddaae79576ecee15a8f7855370d0b158e86b1f9769bbf5c1f7  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/sms.db
4a11b6d595f73f858cc0531bbe8752719d53123d03f551d4f4a508ecdcb4b3a3  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/sms.db-shm
995c71555e8895c4ef010f2894d9f32f4e623bf0487e92e3b426c6b7f2f8b1f8  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/SMS/sms.db-wal

===== WORKING COPY HASHES =====
67e847fdd1e51dddaae79576ecee15a8f7855370d0b158e86b1f9769bbf5c1f7  working/messages/sms.db
4a11b6d595f73f858cc0531bbe8752719d53123d03f551d4f4a508ecdcb4b3a3  working/messages/sms.db-shm
995c71555e8895c4ef010f2894d9f32f4e623bf0487e92e3b426c6b7f2f8b1f8  working/messages/sms.db-wal

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
_server_change_token_blob BLOB, ck_record_id TEXT, original_guid TEXT UNIQUE NOT NULL, sr_ck_sync_state INTEGER DEFAULT 0, sr_ck_server_change_token_blob BLOB, sr
_ck_record_id TEXT)
chat|CREATE TABLE chat (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NOT NULL, style INTEGER, state INTEGER, account_id TEXT, properties BLOB, chat_i
dentifier TEXT, service_name TEXT, room_name TEXT, account_login TEXT, is_archived INTEGER DEFAULT 0, last_addressed_handle TEXT, display_name TEXT, group_id TEXT
, is_filtered INTEGER, successful_query INTEGER, engram_id TEXT, server_change_token TEXT, ck_sync_state INTEGER DEFAULT 0, original_group_id TEXT, last_read_mess
age_timestamp INTEGER DEFAULT 0, sr_server_change_token TEXT, sr_ck_sync_state INTEGER DEFAULT 0, cloudkit_record_id TEXT, sr_cloudkit_record_id TEXT, last_addres
sed_sim_id TEXT, is_blackholed INTEGER DEFAULT 0)
chat_handle_join|CREATE TABLE chat_handle_join (chat_id INTEGER REFERENCES chat (ROWID) ON DELETE CASCADE, handle_id INTEGER REFERENCES handle (ROWID) ON DELETE C
ASCADE, UNIQUE(chat_id, handle_id))
chat_message_join|CREATE TABLE chat_message_join (chat_id INTEGER REFERENCES chat (ROWID) ON DELETE CASCADE, message_id INTEGER REFERENCES message (ROWID) ON DELE
TE CASCADE, message_date INTEGER DEFAULT 0, PRIMARY KEY (chat_id, message_id))
deleted_messages|CREATE TABLE deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid TEXT NOT NULL)
handle|CREATE TABLE handle (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, id TEXT NOT NULL, country TEXT, service TEXT NOT NULL, uncanonicalized_id TEXT, person
_centric_id TEXT, UNIQUE (id, service) )
kvtable|CREATE TABLE kvtable (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, key TEXT UNIQUE NOT NULL, value BLOB NOT NULL)
message|CREATE TABLE message (ROWID INTEGER PRIMARY KEY AUTOINCREMENT, guid TEXT UNIQUE NOT NULL, text TEXT, replace INTEGER DEFAULT 0, service_center TEXT, handl
e_id INTEGER DEFAULT 0, subject TEXT, country TEXT, attributedBody BLOB, version INTEGER DEFAULT 0, type INTEGER DEFAULT 0, service TEXT, account TEXT, account_gu
id TEXT, error INTEGER DEFAULT 0, date INTEGER, date_read INTEGER, date_delivered INTEGER, is_delivered INTEGER DEFAULT 0, is_finished INTEGER DEFAULT 0, is_emote
 INTEGER DEFAULT 0, is_from_me INTEGER DEFAULT 0, is_empty INTEGER DEFAULT 0, is_delayed INTEGER DEFAULT 0, is_auto_reply INTEGER DEFAULT 0, is_prepared INTEGER D
EFAULT 0, is_read INTEGER DEFAULT 0, is_system_message INTEGER DEFAULT 0, is_sent INTEGER DEFAULT 0, has_dd_results INTEGER DEFAULT 0, is_service_message INTEGER
DEFAULT 0, is_forward INTEGER DEFAULT 0, was_downgraded INTEGER DEFAULT 0, is_archive INTEGER DEFAULT 0, cache_has_attachments INTEGER DEFAULT 0, cache_roomnames
TEXT, was_data_detected INTEGER DEFAULT 0, was_deduplicated INTEGER DEFAULT 0, is_audio_message INTEGER DEFAULT 0, is_played INTEGER DEFAULT 0, date_played INTEGE
R, item_type INTEGER DEFAULT 0, other_handle INTEGER DEFAULT 0, group_title TEXT, group_action_type INTEGER DEFAULT 0, share_status INTEGER DEFAULT 0, share_direc
tion INTEGER DEFAULT 0, is_expirable INTEGER DEFAULT 0, expire_state INTEGER DEFAULT 0, message_action_type INTEGER DEFAULT 0, message_source INTEGER DEFAULT 0, a
ssociated_message_guid TEXT, associated_message_type INTEGER DEFAULT 0, balloon_bundle_id TEXT, payload_data BLOB, expressive_send_style_id TEXT, associated_messa
ge_range_location INTEGER DEFAULT 0, associated_message_range_length INTEGER DEFAULT 0, time_expressive_send_played INTEGER, message_summary_info BLOB, ck_sync_st
ate INTEGER DEFAULT 0, ck_record_id TEXT, ck_record_change_tag TEXT, destination_caller_id TEXT, sr_ck_sync_state INTEGER DEFAULT 0, sr_ck_record_id TEXT, sr_ck_r
ecord_change_tag TEXT, is_corrupt INTEGER DEFAULT 0, reply_to_guid TEXT, sort_id INTEGER, is_spam INTEGER DEFAULT 0)
message_attachment_join|CREATE TABLE message_attachment_join (message_id INTEGER REFERENCES message (ROWID) ON DELETE CASCADE, attachment_id INTEGER REFERENCES at
tachment (ROWID) ON DELETE CASCADE, UNIQUE(message_id, attachment_id))
message_processing_task|CREATE TABLE message_processing_task (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid TEXT NOT NULL, task_flags INTEGER NOT NULL )
sqlite_sequence|CREATE TABLE sqlite_sequence(name,seq)
sqlite_stat1|CREATE TABLE sqlite_stat1(tbl,idx,stat)
sync_deleted_attachments|CREATE TABLE sync_deleted_attachments (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid TEXT NOT NULL, recordID TEXT )
sync_deleted_chats|CREATE TABLE sync_deleted_chats (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid TEXT NOT NULL, recordID TEXT,timestamp INTEGER)
sync_deleted_messages|CREATE TABLE sync_deleted_messages (ROWID INTEGER PRIMARY KEY AUTOINCREMENT UNIQUE, guid TEXT NOT NULL, recordID TEXT )

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
 Birth: 2026-11-04 03:46:41.958720471 -0500

===== COPY DATABASE AND SIDECARS =====

===== SET WORKING-COPY OWNERSHIP =====

===== WORKING COPY FILES =====
total 680
-rw-r--r-- 1 kali kali 241664 Apr 16  2020 Accounts3.sqlite
-rw-r--r-- 1 kali kali  32768 Apr 16  2020 Accounts3.sqlite-shm
-rw-r--r-- 1 kali kali 420272 Apr 16  2020 Accounts3.sqlite-wal

┌──(kali㉿kali)-[~/SBI-DF205-Case2]
```

```
└─$ cd ~/SBT-DF205-Case2

SRC="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts"
DEST="working/accounts3"

echo "===== SOURCE HASHES ====="
sudo sha256sum "$SRC/Accounts3.sqlite" \
               "$SRC/Accounts3.sqlite-wal" \
               "$SRC/Accounts3.sqlite-shm" \
| tee output/hashes/Accounts3_Source_SHA256.txt

echo
echo "===== WORKING COPY HASHES ====="
sha256sum "$DEST/Accounts3.sqlite" \
          "$DEST/Accounts3.sqlite-wal" \
          "$DEST/Accounts3.sqlite-shm" \
| tee output/hashes/Accounts3_Working_SHA256.txt
===== SOURCE HASHES =====
6b6386c241edd9850d608349b3d8fbacedcf59a64534959cc456021e9a477de2  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Acc
ounts/Accounts3.sqlite
48440fe69c765a886eafe612bdec9f5feb1b9fc343e8b9f65109403b5979a14c  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Acc
ounts/Accounts3.sqlite-wal
a90f6956cafbd5f21914600e85ca1c68ca4564716e73b8b923405a693c4f166c  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Acc
ounts/Accounts3.sqlite-shm

===== WORKING COPY HASHES =====
6b6386c241edd9850d608349b3d8fbacedcf59a64534959cc456021e9a477de2  working/accounts3/Accounts3.sqlite
48440fe69c765a886eafe612bdec9f5feb1b9fc343e8b9f65109403b5979a14c  working/accounts3/Accounts3.sqlite-wal
a90f6956cafbd5f21914600e85ca1c68ca4564716e73b8b923405a693c4f166c  working/accounts3/Accounts3.sqlite-shm

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DB="working/accounts3/Accounts3.sqlite"

echo "═══ SQLITE VERSION ═══"
sqlite3 --version

echo
echo "═══ TABLES ═══"
sqlite3 -readonly "$DB" ".tables" \
| tee output/inventory/Accounts3_Tables.txt

echo
echo "═══ TABLE AND COLUMN DEFINITIONS ═══"
sqlite3 -readonly "$DB" ".schema" \
| tee output/inventory/Accounts3_Schema.txt
═══ SQLITE VERSION ═══
3.46.1 2024-08-13 09:16:08 c9c2ab54ba1f5f46360f1b4f35d849cd3f080e6fc2b6c60e91b16c63f69aalt1 (64-bit)

═══ TABLES ═══
ZACCESSOPTIONSKEY          Z_2ENABLEDDATACLASSES
ZACCOUNT                   Z_2PROVISIONEDDATACLASSES
ZACCOUNTPROPERTY           Z_4SUPPORTEDDATACLASSES
ZACCOUNTTYPE               Z_4SYNCABLEDATACLASSES
ZAUTHORIZATION             Z_METADATA
ZCREDENTIALITEM            Z_MODELCACHE
ZDATACLASS                 Z_PRIMARYKEY
Z_1OWNINGACCOUNTTYPES

═══ TABLE AND COLUMN DEFINITIONS ═══
CREATE TABLE ZACCESSOPTIONSKEY ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZNAME VARCHAR );
```

```
═══ TABLE AND COLUMN DEFINITIONS ═══
CREATE TABLE ZACCESSOPTIONSKEY ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZNAME VARCHAR );
CREATE TABLE Z_1OWNINGACCOUNTTYPES ( Z_1ACCESSKEYS INTEGER, Z_4OWNINGACCOUNTTYPES INTEGER, PRIMARY KEY (Z_1ACCESSKEYS, Z_4OWNINGACCOUNTTYPES)
 );
CREATE TABLE ZACCOUNT ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZACTIVE INTEGER, ZAUTHENTICATED INTEGER, ZSUPPORTSAUTHENTICAT
ION INTEGER, ZVISIBLE INTEGER, ZACCOUNTTYPE INTEGER, ZPARENTACCOUNT INTEGER, ZDATE TIMESTAMP, ZLASTCREDENTIALRENEWALREJECTIONDATE TIMESTAMP,
ZACCOUNTDESCRIPTION VARCHAR, ZAUTHENTICATIONTYPE VARCHAR, ZCREDENTIALTYPE VARCHAR, ZIDENTIFIER VARCHAR, ZOWNINGBUNDLEID VARCHAR, ZUSERNAME VA
RCHAR, ZDATACLASSPROPERTIES BLOB );
CREATE TABLE Z_2ENABLEDDATACLASSES ( Z_2ENABLEDACCOUNTS INTEGER, Z_7ENABLEDDATACLASSES INTEGER, PRIMARY KEY (Z_2ENABLEDACCOUNTS, Z_7ENABLEDDA
TACLASSES) );
CREATE TABLE Z_2PROVISIONEDDATACLASSES ( Z_2PROVISIONEDACCOUNTS INTEGER, Z_7PROVISIONEDDATACLASSES INTEGER, PRIMARY KEY (Z_2PROVISIONEDACCOUN
TS, Z_7PROVISIONEDDATACLASSES) );
CREATE TABLE ZACCOUNTPROPERTY ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZOWNER INTEGER, ZKEY VARCHAR, ZVALUE BLOB );
CREATE TABLE ZACCOUNTTYPE ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZENCRYPTACCOUNTPROPERTIES INTEGER, ZOBSOLETE INTEGER, ZSU
PPORTSAUTHENTICATION INTEGER, ZSUPPORTSMULTIPLEACCOUNTS INTEGER, ZVISIBILITY INTEGER, ZACCOUNTTYPEDESCRIPTION VARCHAR, ZCREDENTIALPROTECTIONP
OLICY VARCHAR, ZCREDENTIALTYPE VARCHAR, ZIDENTIFIER VARCHAR, ZOWNINGBUNDLEID VARCHAR );
CREATE TABLE Z_4SUPPORTEDDATACLASSES ( Z_4SUPPORTEDTYPES INTEGER, Z_7SUPPORTEDDATACLASSES INTEGER, PRIMARY KEY (Z_4SUPPORTEDTYPES, Z_7SUPPORT
EDDATACLASSES) );
CREATE TABLE Z_4SYNCABLEDATACLASSES ( Z_4SYNCABLETYPES INTEGER, Z_7SYNCABLEDATACLASSES INTEGER, PRIMARY KEY (Z_4SYNCABLETYPES, Z_7SYNCABLEDAT
ACLASSES) );
CREATE TABLE ZAUTHORIZATION ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZACCOUNTTYPE INTEGER, ZBUNDLEID VARCHAR, ZGRANTEDPERMIS
SIONS VARCHAR, ZOPTIONS BLOB );
CREATE TABLE ZCREDENTIALITEM ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZPERSISTENT INTEGER, ZEXPIRATIONDATE TIMESTAMP, ZACCOU
NTIDENTIFIER VARCHAR, ZSERVICENAME VARCHAR );
CREATE TABLE ZDATACLASS ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZNAME BLOB );
CREATE INDEX Z_1OWNINGACCOUNTTYPES_Z_4OWNINGACCOUNTTYPES_INDEX ON Z_1OWNINGACCOUNTTYPES (Z_4OWNINGACCOUNTTYPES, Z_1ACCESSKEYS);
CREATE INDEX ZACCOUNT_ZACCOUNTTYPE_INDEX ON ZACCOUNT (ZACCOUNTTYPE);
CREATE INDEX ZACCOUNT_ZPARENTACCOUNT_INDEX ON ZACCOUNT (ZPARENTACCOUNT);
CREATE INDEX Z_2ENABLEDDATACLASSES_Z_7ENABLEDDATACLASSES_INDEX ON Z_2ENABLEDDATACLASSES (Z_7ENABLEDDATACLASSES, Z_2ENABLEDACCOUNTS);
CREATE INDEX Z_2PROVISIONEDDATACLASSES_Z_7PROVISIONEDDATACLASSES_INDEX ON Z_2PROVISIONEDDATACLASSES (Z_7PROVISIONEDDATACLASSES, Z_2PROVISIONE
```

```
TACLASSES) );
CREATE TABLE Z_2PROVISIONEDDATACLASSES ( Z_2PROVISIONEDACCOUNTS INTEGER, Z_7PROVISIONEDDATACLASSES INTEGER, PRIMARY KEY (Z_2PROVISIONEDACCOUN
TS, Z_7PROVISIONEDDATACLASSES) );
CREATE TABLE ZACCOUNTPROPERTY ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZOWNER INTEGER, ZKEY VARCHAR, ZVALUE BLOB );
CREATE TABLE ZACCOUNTTYPE ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZENCRYPTACCOUNTPROPERTIES INTEGER, ZOBSOLETE INTEGER, ZSU
PPORTSAUTHENTICATION INTEGER, ZSUPPORTSMULTIPLEACCOUNTS INTEGER, ZVISIBILITY INTEGER, ZACCOUNTTYPEDESCRIPTION VARCHAR, ZCREDENTIALPROTECTIONP
OLICY VARCHAR, ZCREDENTIALTYPE VARCHAR, ZIDENTIFIER VARCHAR, ZOWNINGBUNDLEID VARCHAR );
CREATE TABLE Z_4SUPPORTEDDATACLASSES ( Z_4SUPPORTEDTYPES INTEGER, Z_7SUPPORTEDDATACLASSES INTEGER, PRIMARY KEY (Z_4SUPPORTEDTYPES, Z_7SUPPORT
EDDATACLASSES) );
CREATE TABLE Z_4SYNCABLEDATACLASSES ( Z_4SYNCABLETYPES INTEGER, Z_7SYNCABLEDATACLASSES INTEGER, PRIMARY KEY (Z_4SYNCABLETYPES, Z_7SYNCABLEDAT
ACLASSES) );
CREATE TABLE ZAUTHORIZATION ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZACCOUNTTYPE INTEGER, ZBUNDLEID VARCHAR, ZGRANTEDPERMIS
SIONS VARCHAR, ZOPTIONS BLOB );
CREATE TABLE ZCREDENTIALITEM ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZPERSISTENT INTEGER, ZEXPIRATIONDATE TIMESTAMP, ZACCOU
NTIDENTIFIER VARCHAR, ZSERVICENAME VARCHAR );
CREATE TABLE ZDATACLASS ( Z_PK INTEGER PRIMARY KEY, Z_ENT INTEGER, Z_OPT INTEGER, ZNAME BLOB );
CREATE INDEX Z_1OWNINGACCOUNTTYPES_Z_4OWNINGACCOUNTTYPES_INDEX ON Z_1OWNINGACCOUNTTYPES (Z_4OWNINGACCOUNTTYPES, Z_1ACCESSKEYS);
CREATE INDEX ZACCOUNT_ZACCOUNTTYPE_INDEX ON ZACCOUNT (ZACCOUNTTYPE);
CREATE INDEX ZACCOUNT_ZPARENTACCOUNT_INDEX ON ZACCOUNT (ZPARENTACCOUNT);
CREATE INDEX Z_2ENABLEDDATACLASSES_Z_7ENABLEDDATACLASSES_INDEX ON Z_2ENABLEDDATACLASSES (Z_7ENABLEDDATACLASSES, Z_2ENABLEDACCOUNTS);
CREATE INDEX Z_2PROVISIONEDDATACLASSES_Z_7PROVISIONEDDATACLASSES_INDEX ON Z_2PROVISIONEDDATACLASSES (Z_7PROVISIONEDDATACLASSES, Z_2PROVISIONE
DACCOUNTS);
CREATE INDEX ZACCOUNTPROPERTY_ZOWNER_INDEX ON ZACCOUNTPROPERTY (ZOWNER);
CREATE INDEX Z_4SUPPORTEDDATACLASSES_Z_7SUPPORTEDDATACLASSES_INDEX ON Z_4SUPPORTEDDATACLASSES (Z_7SUPPORTEDDATACLASSES, Z_4SUPPORTEDTYPES);
CREATE INDEX Z_4SYNCABLEDATACLASSES_Z_7SYNCABLEDATACLASSES_INDEX ON Z_4SYNCABLEDATACLASSES (Z_7SYNCABLEDATACLASSES, Z_4SYNCABLETYPES);
CREATE INDEX ZAUTHORIZATION_ZACCOUNTTYPE_INDEX ON ZAUTHORIZATION (ZACCOUNTTYPE);
CREATE TABLE Z_PRIMARYKEY (Z_ENT INTEGER PRIMARY KEY, Z_NAME VARCHAR, Z_SUPER INTEGER, Z_MAX INTEGER);
CREATE TABLE Z_METADATA (Z_VERSION INTEGER PRIMARY KEY, Z_UUID VARCHAR(255), Z_PLIST BLOB);
CREATE TABLE Z_MODELCACHE (Z_CONTENT BLOB);

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DB="working/accounts3/Accounts3.sqlite"

echo "═══ MAIN TABLE ROW COUNTS ═══"
sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Row_Counts.txt
.headers on
.mode column

SELECT 'ZACCOUNT' AS table_name, COUNT(*) AS rows FROM ZACCOUNT
UNION ALL SELECT 'ZACCOUNTTYPE', COUNT(*) FROM ZACCOUNTTYPE
UNION ALL SELECT 'ZAUTHORIZATION', COUNT(*) FROM ZAUTHORIZATION
UNION ALL SELECT 'ZCREDENTIALITEM', COUNT(*) FROM ZCREDENTIALITEM
UNION ALL SELECT 'ZACCOUNTPROPERTY', COUNT(*) FROM ZACCOUNTPROPERTY
UNION ALL SELECT 'ZDATACLASS', COUNT(*) FROM ZDATACLASS;
SQL
═══ MAIN TABLE ROW COUNTS ═══
table_name        rows
────────────────  ────
ZACCOUNT          18
ZACCOUNTTYPE      53
ZAUTHORIZATION    0
ZCREDENTIALITEM   0
ZACCOUNTPROPERTY  149
ZDATACLASS        41

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ echo "═══ ACCOUNT TYPE DEFINITIONS ═══"
sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Account_Types.txt
.headers on
.mode column

SELECT
    Z_PK,
    ZACCOUNTTYPEDESCRIPTION,
    ZIDENTIFIER,
    ZOWNINGBUNDLEID,
    ZCREDENTIALTYPE,
    ZSUPPORTSAUTHENTICATION,
    ZSUPPORTSMULTIPLEACCOUNTS
FROM ZACCOUNTTYPE
ORDER BY Z_PK;
SQL
═══ ACCOUNT TYPE DEFINITIONS ═══
Z_PK  ZACCOUNTTYPEDESCRIPTION  ZIDENTIFIER                          ZOWNINGBUNDLEID      ZCREDENTIALTYPE         ZSUPPORTSAUTHENTICATION
ZSUPPORTSMULTIPLEACCOUNTS
────  ──────────────────────  ──────────────────────────────────────  ──────────────────  ──────────────────────  ──────────────────────
──────────────────────
1     Hotmail                 com.apple.account.Hotmail             com.apple.accountsd  oauth2                  1
1
2     Facebook                com.apple.facebook                    com.apple.accountsd  oauth2                  1
0
3     macOS Server            com.apple.account.osxserver           com.apple.accountsd  password                0
1
4     Tudou                   com.apple.tudou                       com.apple.accountsd                          0
0
5     Vimeo                   com.apple.vimeo                       com.apple.accountsd  oauth                   1
```

```
═══ ACCOUNT TYPE DEFINITIONS ═══
Z_PK  ZACCOUNTTYPEDESCRIPTION  ZIDENTIFIER                          ZOWNINGBUNDLEID      ZCREDENTIALTYPE         ZSUPPORTSAUTHENTICATION
ZSUPPORTSMULTIPLEACCOUNTS
────  ──────────────────────   ──────────────────────────────────    ──────────────────   ──────────────────      ──────────────────────
──────────────────────
1     Hotmail                  com.apple.account.Hotmail             com.apple.accountsd  oauth2                  1
1
2     Facebook                 com.apple.facebook                    com.apple.accountsd  oauth2                  1
0
3     macOS Server             com.apple.account.osxserver           com.apple.accountsd  password                0
1
4     Tudou                    com.apple.tudou                       com.apple.accountsd                          0
0
5     Vimeo                    com.apple.vimeo                       com.apple.accountsd  oauth                   1
0
6     Sina Weibo               com.apple.sinaweibo                   com.apple.accountsd  oauth                   1
1
7     LinkedIn                 com.apple.linkedin                    com.apple.accountsd  oauth                   1
0
8     POP                      com.apple.account.POP                 com.apple.accountsd  password                1
1
9     LinkedIn Legacy          com.apple.account.linkedInLegacy      com.apple.accountsd  oauth                   1
0
10    iCloud                   com.apple.account.AppleAccount        com.apple.accountsd  token                   1
1
11    Messages                 com.apple.account.Madrid              com.apple.accountsd  token                   1
1
12    AppleID                  com.apple.account.AppleID             com.apple.accountsd  appleid-tokens          1
1
13    Flickr                   com.apple.flickr                      com.apple.accountsd  oauth                   1
0
14    CalDAV Legacy            com.apple.account.CalDAVLegacy        com.apple.accountsd  password                1
```

```
1
13    Flickr                  com.apple.flickr                      com.apple.accountsd  oauth                 1
0
14    CalDAV Legacy           com.apple.account.CalDAVLegacy        com.apple.accountsd  password              1
1
15    Apple ID                com.apple.account.AppleIDAuthentication  com.apple.accountsd  appleid-authentication  1
1
16    Facebook Legacy         com.apple.account.facebookLegacy      com.apple.accountsd  oauth2                1
0
17    IMAPMail                com.apple.account.IMAPMail            com.apple.accountsd  password              1
1
18    Single Sign-on          com.apple.account.SingleSignOn        com.apple.accountsd  password              1
1
19    Device Locator          com.apple.account.DeviceLocator       com.apple.accountsd  token                 0
0
20    Exchange                com.apple.account.Exchange            com.apple.accountsd  password              1
1
21    FaceTime                com.apple.account.FaceTime            com.apple.accountsd  token                 1
1
22    iTunes Store (Sandbox)  com.apple.account.iTunesStore.sandbox  com.apple.accountsd  password              1
1
23    IMAPNotes               com.apple.account.IMAPNotes           com.apple.accountsd  password              1
1
24    Find My Friends         com.apple.account.FindMyFriends       com.apple.accountsd  token                 0
0
25    iTunes Store            com.apple.account.iTunesStore         com.apple.accountsd  password              1
1
26    SubscribedCalendar      com.apple.account.SubscribedCalendar  com.apple.accountsd  password              1
1
27    Aol                     com.apple.account.aol                 com.apple.accountsd  oauth2                1
1
28    IMAP                    com.apple.account.IMAP                com.apple.accountsd  password              1
```

```
27    Aol                     com.apple.account.aol                com.apple.accountsd  oauth2              1
1
28    IMAP                    com.apple.account.IMAP               com.apple.accountsd  password            1
1
29    Yahoo                   com.apple.account.Yahoo              com.apple.accountsd  oauth2              1
1
30    163                     com.apple.account.163                com.apple.accountsd  password            0
1
31    BookmarkDAV             com.apple.account.BookmarkDAV        com.apple.accountsd  password            1
1
32    126                     com.apple.account.126                com.apple.accountsd  password            0
1
33    CalDAV                  com.apple.account.CalDAV             com.apple.accountsd  password            1
1
34    On My Device            com.apple.account.OnMyDevice         com.apple.accountsd                      0
0
35    CardDAV Legacy          com.apple.account.CardDAVLegacy      com.apple.accountsd  password            1
1
36    SMTP                    com.apple.account.SMTP               com.apple.accountsd  password            1
1
37    Twitter                 com.apple.twitter                    com.apple.accountsd  oauth               1
1
38    Tencent Weibo           com.apple.account.tencentweibo       com.apple.accountsd  oauth2              1
0
39    MSO                     com.apple.account.mso                com.apple.accountsd  token               1
1
40    Messages                com.apple.account.IdentityServices   com.apple.accountsd  token               1
1
41    Yahoo IM                com.apple.account.yahooim            com.apple.accountsd  password            0
1
42    Gmail                   com.apple.account.Google             com.apple.accountsd  oauth2              1
1
```

```
1
40    Messages                com.apple.account.IdentityServices      com.apple.accountsd  token                1
1
41    Yahoo IM                com.apple.account.yahooim               com.apple.accountsd  password             0
1
42    Gmail                   com.apple.account.Google                com.apple.accountsd  oauth2               1
1
43    CardDAV                 com.apple.account.CardDAV               com.apple.accountsd  password             1
1
44    IDMS                    com.apple.account.idms                  com.apple.accountsd  token                0
1
45    LDAP                    com.apple.account.LDAP                  com.apple.accountsd  password             1
1
46    CloudKit                com.apple.account.CloudKit              com.apple.accountsd  token                1
1
47    QQ                      com.apple.account.qq                    com.apple.accountsd  password             0
1
48    Yelp                    com.apple.account.yelp                  com.apple.accountsd  oauth                0
0
49    Holiday Calendar        com.apple.account.HolidayCalendar       com.apple.accountsd  password             0
0
50    Jabber                  com.apple.account.jabber                com.apple.accountsd  password             0
1
51    Game Center             com.apple.account.GameCenter            com.apple.accountsd  token                1
1
52    Youku                   com.apple.youku                         com.apple.accountsd                       0
0
53    Remote Management       com.apple.account.RemoteManagement      com.apple.accountsd  token                1
1

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DB="working/accounts3/Accounts3.sqlite"

echo "═══ ACCOUNT RECORDS WITH TYPE LABELS ═══"
sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Account_Records.txt
.headers on
.mode column
.nullvalue NULL

SELECT
    a.Z_PK AS account_pk,
    t.ZACCOUNTTYPEDESCRIPTION AS account_type,
    a.ZACTIVE AS active,
    a.ZAUTHENTICATED AS authenticated,
    a.ZVISIBLE AS visible,
    a.ZDATE AS stored_date,
    a.ZACCOUNTDESCRIPTION AS account_description,
    a.ZOWNINGBUNDLEID AS owning_bundle
FROM ZACCOUNT AS a
LEFT JOIN ZACCOUNTTYPE AS t
    ON a.ZACCOUNTTYPE = t.Z_PK
ORDER BY a.Z_PK;
SQL
═══ ACCOUNT RECORDS WITH TYPE LABELS ═══
account_pk  account_type      active  authenticated  visible  stored_date       account_description   owning_bundle
──────────  ────────────      ──────  ─────────────  ───────  ──────────        ──────────────────    ─────────────
1           iTunes Store      0       1              1        606519591.912371  Local                 locationd
2           iTunes Store      1       1              1        606520062.043928  NULL                  com.apple.purplebuddy
3           Messages          1       1              1        606520062.507476  NULL                  com.apple.identityservicesd
```

```
Session  Actions  Edit  View  Help
    a.ZDATE AS stored_date,
    a.ZACCOUNTDESCRIPTION AS account_description,
    a.ZOWNINGBUNDLEID AS owning_bundle
FROM ZACCOUNT AS a
LEFT JOIN ZACCOUNTTYPE AS t
    ON a.ZACCOUNTTYPE = t.Z_PK
ORDER BY a.Z_PK;
SQL
═══ ACCOUNT RECORDS WITH TYPE LABELS ═══
```

account_pk	account_type	active	authenticated	visible	stored_date	account_description	owning_bundle
1	iTunes Store	0	1	1	606519591.912371	Local	locationd
2	iTunes Store	1	1	1	606520062.043928	NULL	com.apple.purplebuddy
3	Messages	1	1	1	606520062.507476	NULL	com.apple.identityservicesd
4	iCloud	1	1	1	606520077.068197	iCloud	com.apple.purplebuddy
5	Find My Friends	1	1	1	606520075.27839	NULL	com.apple.accounts.accountsd
6	Device Locator	1	1	1	606520075.243605	NULL	com.apple.accounts.accountsd
7	CloudKit	1	1	1	606520062.363132	NULL	com.apple.accounts.accountsd
8	IMAPNotes	1	1	1	606520075.446426	NULL	com.apple.accounts.accountsd
9	CalDAV	1	1	1	606520075.3066	NULL	com.apple.accounts.accountsd
10	CardDAV	1	1	1	606520075.373321	NULL	com.apple.accounts.accountsd
11	IDMS	1	1	1	606520077.847509	NULL	com.apple.AuthKit
12	Apple ID	1	1	1	606520078.027195	NULL	com.apple.AuthKit
13	Holiday Calendar	1	0	0	606520156.473805	Holiday Calendar	dataaccessd
14	Game Center	1	1	1	606520787.089834	thisisdfir@gmail.com	com.apple.AuthKit
15	CalDAV	1	1	1	606532289.45302	NULL	com.apple.accounts.accountsd
16	IMAPNotes	1	1	1	606532289.541797	NULL	com.apple.accounts.accountsd
17	CardDAV	1	1	1	606532289.508673	NULL	com.apple.accounts.accountsd
18	Gmail	1	1	1	606532289.572603	Gmail	com.apple.Preferences

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

DB="working/accounts3/Accounts3.sqlite"

echo "===== ACCOUNT TIMESTAMPS IN UTC ====="
sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Account_Timestamps_UTC.txt
.headers on
.mode column

SELECT
    Z_PK AS account_pk,
    datetime('2001-01-01 00:00:00', '+' || ZDATE || ' seconds') || ' UTC' AS stored_date_utc,
    ZACCOUNTDESCRIPTION AS description
FROM ZACCOUNT
ORDER BY ZDATE;
SQL
===== ACCOUNT TIMESTAMPS IN UTC =====
account_pk  stored_date_utc          description
----------  -----------------------  --------------------
1           2020-03-21 21:39:51 UTC  Local
2           2020-03-21 21:47:42 UTC
7           2020-03-21 21:47:42 UTC
3           2020-03-21 21:47:42 UTC
6           2020-03-21 21:47:55 UTC
5           2020-03-21 21:47:55 UTC
9           2020-03-21 21:47:55 UTC
10          2020-03-21 21:47:55 UTC
8           2020-03-21 21:47:55 UTC
4           2020-03-21 21:47:57 UTC  iCloud
11          2020-03-21 21:47:57 UTC
12          2020-03-21 21:47:58 UTC
13          2020-03-21 21:49:16 UTC  Holiday Calendar
14          2020-03-21 21:59:47 UTC  thisisdfir@gmail.com
15          2020-03-22 01:11:29 UTC
17          2020-03-22 01:11:29 UTC
16          2020-03-22 01:11:29 UTC
18          2020-03-22 01:11:29 UTC  Gmail
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ echo "═══ ACCOUNT PROPERTY KEY SUMMARY ═══"
sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Property_Key_Summary.txt
.headers on
.mode column

SELECT
    ZKEY AS property_key,
    COUNT(*) AS occurrences,
    COUNT(DISTINCT ZOWNER) AS account_owners,
    GROUP_CONCAT(DISTINCT typeof(ZVALUE)) AS value_storage_types
FROM ZACCOUNTPROPERTY
GROUP BY ZKEY
ORDER BY ZKEY;
SQL
═══ ACCOUNT PROPERTY KEY SUMMARY ═══
```

property_key	occurrences	account_owners	value_storage_types
AKFollowUpAccountRefreshTimestamp	1	1	blob
AuthenticationScheme	1	1	blob
CalDAVAccountVersion	2	2	blob
CalDAVMainPrincipalUID	2	2	blob
CalDAVMobileAccountCollectionSetName	1	1	blob
CalDAVMobileAccountIsWritable	2	2	blob
CalDAVMobileAccountSearchPropertySetKey_CoreDAV	2	2	blob
CalDAVMobileAccountServerVersion	2	2	blob
CalDAVPrincipals	2	2	blob
Class	1	1	blob
DAAccountDidAutodiscover	4	4	blob
DAAccountHost	4	4	blob
DAAccountPort	4	4	blob
DAAccountPrincipalPath	2	2	blob
DAAccountUseSSL	2	2	blob
DAAccountUseTrustedSSLCertificate	5	5	blob
DSID	1	1	blob
DeviceTrustRevoked	1	1	blob
FullUserName	1	1	blob
GKCredentialScope-5	1	1	blob
GKCredentialScope-5-mod-date	1	1	blob

GKCredentialScope-5	1	1	blob
GKCredentialScope-5-mod-date	1	1	blob
GKPlayerInternal	1	1	blob
InRegistrationExpirationDate	1	1	blob
LastEmailAliasesSyncDate	1	1	blob
MFServerSSLCertificateIsTrusted	1	1	blob
SSLEnabled	1	1	blob
StoreArchiveOnServer	1	1	blob
StoreDraftsOnServer	1	1	blob
StoreSentMessagesOnServer	1	1	blob
StoreTrashOnServer	1	1	blob
WarmUpVerificationTimeStamp	1	1	blob
account-exists	1	1	blob
accountClass	1	1	blob
accountCredits	1	1	blob
accountFlags	1	1	blob
additionalInfo	1	1	blob
aliases	1	1	blob
altDSID	5	5	blob
altDSID-5	1	1	blob
altDSID-5-mod-date	1	1	blob
apple-id	1	1	blob
appleIDAliases	1	1	blob
appleId	1	1	blob
auth-mode	1	1	blob
auth-token-receipt-date	1	1	blob
automaticDownloadKinds	1	1	blob
availableServiceTypes	1	1	blob
bundleRef	1	1	blob
cloudDocsMigrationComplete	1	1	blob
cloudLibraryStateReason	1	1	blob
com.apple.ak.checkin-allowed	1	1	blob
cookies	2	2	blob
deviceCount	1	1	blob
deviceCountFetchDate	1	1	blob
didAgreeToTerms	1	1	blob
dsid	4	4	blob
eligibleServices	2	2	blob
email-address	1	1	blob

```
email-address                          1          1          blob
enabledServiceTypes                    1          1          blob
familyEligible                         1          1          blob
firstName                              4          4          blob
freeDownloadsPasswordSetting           1          1          blob
handles                                1          1          blob
hasOptionalTerms                       1          1          blob
iCloudFamily                           1          1          blob
iCloudNotificationLastRegisterDate     1          1          blob
inGoodStanding                         1          1          blob
invitation-context                     1          1          blob
isActiveLockerAccount                  2          2          blob
isManagedAppleID                       3          3          blob
isNewCustomer                          1          1          blob
isSandboxAcct                          1          1          blob
kind                                   1          1          blob
lastAuthenticateCredentialSource       1          1          blob
lastAuthenticated                      1          1          blob
lastAuthenticationServerResponse       1          1          blob
lastKnownQuota                         1          1          blob
lastName                               4          4          blob
mergedPrivacyAcknowledgements          1          1          blob
musicPrivateListeningEnabled           1          1          blob
notesMigrated                          1          1          blob
paidPurchasesPasswordSetting           1          1          blob
personID                               1          1          blob
playerID                               1          1          blob
primaryAccount                         1          1          blob
primaryEmail                           1          1          blob
primaryEmailVerified                   1          1          blob
privacyAcknowledgement                 3          3          blob
profile-id                             1          1          blob
protocolVersion                        1          1          blob
regionInfo                             1          1          blob
registerSuccessCriteria                1          1          blob
remindersDidFinishMigrationToCloudKit  1          1          blob
remindersIsUsingCloudKit               1          1          blob
remindersMigrated                      1          1          blob
repair-state                           1          1          blob
```

```
remindersIsUsingCloudKit                1               1               blob
remindersMigrated                       1               1               blob
repair-state                            1               1               blob
repairState                             1               1               blob
retaining-services                      1               1               blob
scope                                   2               2               blob
security-level                          1               1               blob
self-handle                             1               1               blob
storefrontID                            2               2               blob
userDisabledICK                         1               1               blob
usesCloudDocs                           1               1               blob
validPayment                            1               1               blob
vettedPrimaryEmail                      1               1               blob

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ echo "═══ DATA CLASS DEFINITIONS ═══"
sqlite3 -readonly "$DB" <<'SQL' | tee output/inventory/Accounts3_Data_Classes.txt
.headers on
.mode column

SELECT
    Z_PK,
    typeof(ZNAME) AS storage_type,
    CASE
        WHEN typeof(ZNAME) = 'text' THEN ZNAME
        ELSE hex(ZNAME)
    END AS data_class_name
FROM ZDATACLASS
ORDER BY Z_PK;
SQL
═══ DATA CLASS DEFINITIONS ═══
Z_PK  storage_type  data_class_name
────  ────────────  ──────────────────────────────────────────────────────────────
1     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101D636F6D2E6170706C652E44617461636C6173732E426F6F6B6D
                    61726B7308111A24293237494C5153565C0000000000000101000000000000
                    00000D0000000000000000000000000000007C

2     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101A636F6D2E6170706C652E44617461636C6173732E4865616C74
                    6808111A24293237494C5153565C0000000000000101000000000000000D
                    0000000000000000000000000000000079

3     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A6563747312000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E4E6F746966
```

```
3     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1021636F6D2E6170706C652E44617461636C6173732E4E6F746966
                    69636174696F6E7308111A24293237494C5153565C00000000000010100
                    0000000000000D0000000000000000000000000000000080

4     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101C636F6D2E6170706C652E44617461636C6173732E4D65737361
                    67657308111A24293237494C5153565C00000000000001010000000000000
                    000D0000000000000000000000000000007B

5     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F101B636F6D2E6170706C652E44617461636C6173732E436F6E7465
                    6E7408111A24293237494C5153565C00000000000001010000000000000000
                    0D0000000000000000000000000000007A

6     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1018636F6D2E6170706C652E44617461636C6173732E4D61696C08
                    111A24293237494C5153565C00000000000001010000000000000000D0000
                    0000000000000000000000077

7     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
                    6C6C5F1023636F6D2E6170706C652E44617461636C6173732E5365727665
                    72446F63756D656E747308111A24293237494C5153565C00000000000001
                    010000000000000000D0000000000000000000000000000000082

8     blob          62706C6973743030D4010203040506070A582476657273696F6E59246172
                    6368697665725424746F7058246F626A65637473120000186A05F100F4E53
                    4B657965644172636869766572D1080954726F6F748001A20B0C55246E75
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

cat >> output/Handling_Log.csv <<'EOF'
SBT-DF205-C2,2026-11-04,Charles Uke,TAR extraction,13-4-1.tar,Extracted to working/filesystem using sudo tar; SCHILY.fflags metadata warnings were displayed,Completed
SBT-DF205-C2,2026-11-04,Charles Uke,TAR hash verification,13-4-1.tar,Rechecked SHA-256 and matched recorded baseline,Successful
EOF

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ SYSTEM VERSION PLIST ═══"
sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f -name "SystemVersion.plist" -print

echo
echo "═══ DEVICE / BUILD MANIFEST FILES ═══"
sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f \( -iname "*BuildManifest*" -o -iname "*DeviceInfo*" \) -print \
| head -40
═══ SYSTEM VERSION PLIST ═══
[sudo] password for kali:
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/CoreServices/SystemVersion.plist

═══ DEVICE / BUILD MANIFEST FILES ═══
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/MobileAsset/Assets/com_apple_MobileAsset_MobileAccessoryUpdate_haywire/e74091e55c8bd6b13e4ca2415e6ce006cc76ac5d.asset/AssetData/BuildManifest.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/MobileAsset/Assets/com_apple_MobileAsset_MobileAccessoryUpdate_haywire/71dda23b88d4d7c9823665f9d2374c23246c2505.asset/AssetData/BuildManifest.plist
```

```
were displayed,Completed
SBT-DF205-C2,2026-11-04,Charles Uke,TAR hash verification,13-4-1.tar,Rechecked SHA-256 and matched recorded baseline,Successful
EOF

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ SYSTEM VERSION PLIST ═══"
sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f -name "SystemVersion.plist" -print

echo
echo "═══ DEVICE / BUILD MANIFEST FILES ═══"
sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f \( -iname "*BuildManifest*" -o -iname "*DeviceInfo*" \) -print \
| head -40
═══ SYSTEM VERSION PLIST ═══
[sudo] password for kali:
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/CoreServices/SystemVersion.plist

═══ DEVICE / BUILD MANIFEST FILES ═══
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/MobileAsset/Assets/com_apple_MobileAsset_MobileAccessoryUpdate_haywire/e74091e55c8bd6b13e4ca2415e6ce006cc76ac5d.asset/AssetData/BuildManifest.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/MobileAsset/Assets/com_apple_MobileAsset_MobileAccessoryUpdate_haywire/71dda23b88d4d7c9823665f9d2374c23246c2505.asset/AssetData/BuildManifest.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/MediaRemote/MediaRemoteTelevisionDeviceInfoIdentifier.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Bundle/Application/76A331D9-9A26-4DDD-9FDF-EAEF497EADC0/WeMo_Universal.app/DeviceInformation.plist

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

PLIST="working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/CoreServices/SystemVersion.plist"

echo "═══ SYSTEM VERSION PLIST METADATA ═══"
sudo stat "$PLIST"

echo
echo "═══ SYSTEM VERSION PLIST CONTENTS ═══"
sudo python3 -c '
import plistlib
from pprint import pprint

path = "'"$PLIST"'"
with open(path, "rb") as f:
    data = plistlib.load(f)
pprint(data)
' | tee output/inventory/SystemVersion_Plist.txt
═══ SYSTEM VERSION PLIST METADATA ═══
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/CoreServices/SystemVersion.plist
  Size: 573             Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 4751726     Links: 1
Access: (0444/-r--r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:41:20.607462391 -0500
Modify: 2020-04-02 23:16:07.000000000 -0400
Change: 2026-11-04 03:41:20.607462391 -0500
 Birth: 2026-11-04 03:41:20.607462391 -0500

═══ SYSTEM VERSION PLIST CONTENTS ═══
{'BuildID': '9695BE08-7547-11EA-B2DF-F0563A0B6C5E',
```

```
echo
echo "═══ SYSTEM VERSION PLIST CONTENTS ═══"
sudo python3 -c '
import plistlib
from pprint import pprint

path = "'"$PLIST"'"
with open(path, "rb") as f:
    data = plistlib.load(f)
pprint(data)
' | tee output/inventory/SystemVersion_Plist.txt
═══ SYSTEM VERSION PLIST METADATA ═══
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/CoreServices/SystemVersion.plist
  Size: 573             Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 4751726     Links: 1
Access: (0444/-r--r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:41:20.607462391 -0500
Modify: 2020-04-02 23:16:07.000000000 -0400
Change: 2026-11-04 03:41:20.607462391 -0500
 Birth: 2026-11-04 03:41:20.607462391 -0500

═══ SYSTEM VERSION PLIST CONTENTS ═══
{'BuildID': '9695BE08-7547-11EA-B2DF-F0563A0B6C5E',
 'ProductBuildVersion': '17E262',
 'ProductCopyright': '1983-2020 Apple Inc.',
 'ProductName': 'iPhone OS',
 'ProductVersion': '13.4.1',
 'SystemImageID': '747CABD1-394F-4F90-92E9-CC6B5E1699DE'}

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"

echo "═══ MOBILEGESTALT FILES ═══"
sudo find "$ROOT" -type f \
\( -iname "*MobileGestalt*" -o -iname "*DeviceInfo*" -o -iname "*lockdown*" \) \
-print
═══ MOBILEGESTALT FILES ═══
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/DataClassMigrators/MobileGestaltMigrator.migrator/MobileGestaltMigrator
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/UserEventPlugins/MobileGestaltEvents.plugin/MobileGestaltEvents
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/LaunchDaemons/com.apple.mobile.lockdown.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/LaunchDaemons/com.apple.mobilegestalt.xpc.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/usr/libexec/com.apple.automation.defaultslockdownserviced
working/filesystem/Volumes/JOSH/NoTar-13-4-1/usr/libexec/MobileGestaltHelper
working/filesystem/Volumes/JOSH/NoTar-13-4-1/usr/libexec/lockdownd
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/MediaRemote/MediaRemoteTelevisionDeviceInfoIdentifier.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/logs/lockdownd.log
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/logs/lockdown_service.log
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/logs/lockdownd.log.1
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Bundle/Application/76A331D9-9A26-4DDD-9FDF-EAEF497EADC0/WeMo_Universal.app
/DeviceInformation.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/._systemgroup.com.apple.mobilegestaltcache
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Library/Caches
/com.apple.MobileGestalt.plist
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Library/Caches
/._com.apple.MobileGestalt.plist

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ >....

if isinstance(data, dict):
    print("Top-level keys:", ", ".join(map(str, data.keys())))

terms = (
    "ProductType",
    "DeviceClass",
    "HardwareModel",
    "ModelNumber",
    "BoardId",
    "BoardID",
    "MarketingName",
    "DeviceName",
)

def walk(obj, location="root"):
    if isinstance(obj, dict):
        for key, value in obj.items():
            key_text = str(key)
            if any(term.lower() in key_text.lower() for term in terms):
                if isinstance(value, (str, int, float, bool)):
                    print(f"{location}.{key_text} = {value!r}")
                else:
                    print(f"{location}.{key_text} = [{type(value).__name__}]")
            walk(value, f"{location}.{key_text}")
    elif isinstance(obj, list):
        for index, value in enumerate(obj):
            walk(value, f"{location}[{index}]")
```

```
def walk(obj, location="root"):
    if isinstance(obj, dict):
        for key, value in obj.items():
            key_text = str(key)
            if any(term.lower() in key_text.lower() for term in terms):
                if isinstance(value, (str, int, float, bool)):
                    print(f"{location}.{key_text} = {value!r}")
                else:
                    print(f"{location}.{key_text} = [{type(value).__name__}]")
            walk(value, f"{location}.{key_text}")
    elif isinstance(obj, list):
        for index, value in enumerate(obj):
            walk(value, f"{location}[{index}]")

walk(data)
PY
==== MOBILEGESTALT FILE METADATA ====
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Librar
y/Caches/com.apple.MobileGestalt.plist
  Size: 6585            Blocks: 16         IO Block: 4096   regular file
Device: 8,1     Inode: 7622551     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:49:04.161207020 -0500
Modify: 2020-04-13 15:15:45.000000000 -0400
Change: 2026-11-04 03:49:04.161207020 -0500
 Birth: 2026-11-04 03:49:04.161207020 -0500

==== MOBILEGESTALT IDENTIFICATION PROPERTIES ====
Top-level structure: dict
Top-level keys: CacheUUID, CacheData, CacheExtra, CacheVersion
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ >....
ches/com.apple.MobileGestalt.plist"

sudo python3 - "$MG" <<'PY' | tee output/inventory/MobileGestalt_Structure.txt
import plistlib
import sys

with open(sys.argv[1], "rb") as f:
    data = plistlib.load(f)

def describe(value, path="root", depth=0):
    indent = "  " * depth
    print(f"{indent}{path}: {type(value).__name__}", end="")

    if isinstance(value, dict):
        print(f", {len(value)} entries")
        for key, child in value.items():
            describe(child, f"{path}.{key}", depth + 1)
    elif isinstance(value, list):
        print(f", {len(value)} items")
        for i, child in enumerate(value[:10]):
            describe(child, f"{path}[{i}]", depth + 1)
    elif isinstance(value, bytes):
        print(f", {len(value)} bytes")
    elif isinstance(value, (str, int, float, bool)):
        print(f" = {value!r}")
    else:
        print()

for key, value in data.items():
```

```
for key, value in data.items():
    describe(value, key, 0)
PY
CacheUUID: str = '42ED04EB-D206-4950-9271-721381A9DD2F'
CacheData: bytes, 4176 bytes
CacheExtra: dict, 43 entries
  CacheExtra.oBbtJ8x+s1q0OkaiocPuog: bytes, 24 bytes
  CacheExtra.96GRvvjuBKkU4HzNsYcHPA: str = '12.5.0'
  CacheExtra.1Rm/mWYEI5ttaC0dJ3sHBQ: bytes, 8 bytes
  CacheExtra.TZ/0j62wM3D0CuRt+Nc/Lw: bytes, 20 bytes
  CacheExtra.xUHcyT2/HE8oi/4LaOI+Sw: str = 'GUID_partition_scheme'
  CacheExtra.NaA/zJV7myg2w4YNmSe4yQ: str = '4345'
  CacheExtra.DViRIxZ/ZwO007CLcEYvZw: str = ''
  CacheExtra.QbQzuIbef01P4JeoL9EmKg: float = 1.0
  CacheExtra.gBw7IWiBnLHaA+lBrZBgWw: int = 999
  CacheExtra.pB5sZVvnp+QjZQtt2KfQvA: str = 'mav10'
  CacheExtra.qwXfFvH5jPXPxrny0XuGtQ: str = '9695BE08-7547-11EA-B2DF-F0563A0B6C5E'
  CacheExtra.+3Uf0Pm5F8Xy7Onyvko0vA: str = 'iPhone'
  CacheExtra.LeSRsiLoJCMhjn6nd6GWbQ: str = 'iBoot-5540.102.4'
  CacheExtra.Nzu4E/VsXjEIa83CkRdZrQ: str = 'sha1'
  CacheExtra.JhEU414EIaDvAz8ki5DSqw: str = '#e5bdb5'
  CacheExtra.9MZ5AdH43csAUajl/dU+IQ: list, 1 items
    CacheExtra.9MZ5AdH43csAUajl/dU+IQ[0]: int = 1
  CacheExtra.Z/dqyWS6OZTRy10UcmUAhw: str = 'iPhone SE'
  CacheExtra.zHeENZu+wbg7PUprwNwBWg: str = 'LL/A'
  CacheExtra.91LyMcx4z1w3SGVeqteMnA: bytes, 64 bytes
  CacheExtra.rkqlwPcRHwixY4gapPjanw: str = 'iPhone'
  CacheExtra.IMLaTlxS7ITtwfbRfPYWuA: str = 'A'
  CacheExtra.4qfpxrvLtWillIHpIsVgMA: str = '747CABD1-394F-4F90-92E9-CC6B5E1699DE'
  CacheExtra.D0cJ8r7U5zve6uA6QbOiLA: str = 'MLXL2'
  CacheExtra.h63QSdBCiT/z0WU6rdQv6Q: str = 'LL'
```

```
CacheExtra.9MZ5AdH43csAUajl/dU+IQ: list, 1 items
  CacheExtra.9MZ5AdH43csAUajl/dU+IQ[0]: int = 1
CacheExtra.Z/dqyWS6OZTRy10UcmUAhw: str = 'iPhone SE'
CacheExtra.zHeENZu+wbg7PUprwNwBWg: str = 'LL/A'
CacheExtra.91LyMcx4z1w3SGVeqteMnA: bytes, 64 bytes
CacheExtra.rkqlwPcRHwixY4gapPjanw: str = 'iPhone'
CacheExtra.IMLaTlxS7ITtwfbRfPYWuA: str = 'A'
CacheExtra.4qfpxrvLtWillIHpIsVgMA: str = '747CABD1-394F-4F90-92E9-CC6B5E1699DE'
CacheExtra.D0cJ8r7U5zve6uA6QbOiLA: str = 'MLXL2'
CacheExtra.h63QSdBCiT/z0WU6rdQv6Q: str = 'LL'
CacheExtra.qNNddlUK+B/YlooNoymwgA: str = '13.4.1'
CacheExtra.k7QIBwZJJOVw+Sej/8h8VA: str = 'arm64'
CacheExtra.mZfUC7qo4pURNhyMHZ62RQ: str = '17E262'
CacheExtra.ivIu8YTDnBSrYv/SN4G8Ag: str = 'iPhone OS'
CacheExtra.4W7X4OWHjri5PGaAGsCWxw: int = 51
CacheExtra.emXA9B552rnSoI7xXE91DA: float = 1.0
CacheExtra.97JDvERpVwO+GHtthIh7hA: str = 'A1662'
CacheExtra.NUYAz1eq3Flzt7ZQxXC/ng: float = 1.0
CacheExtra.oPeik/9e8lQWMszEjbPzng: dict, 10 entries
  CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceIdiom: str = 'phone'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceSubType: int = 568
  CacheExtra.oPeik/9e8lQWMszEjbPzng.GraphicsFeatureSetFallbacks: str = 'APPLE3v1:APPLE2:APPLE1:GLES2,0'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.CompatibleDeviceFallback: str = '0'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDisplayGamut: str = 'sRGB'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceProductDescription: str = 'iPhone SE'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDynamicDisplayMode: str = '0'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.GraphicsFeatureSetClass: str = 'APPLE3'
  CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceScaleFactor: int = 2
  CacheExtra.oPeik/9e8lQWMszEjbPzng.DevicePerformanceMemoryClass: int = 2
CacheExtra.9s45ldrCC1WF+7b6C4H2BA: str = 'iPhone'
CacheExtra.HXTqT3UXOKuTEklxz+wMAA: bytes, 16 bytes
CacheExtra.nSo8opze5rFk+EdBoR6tBw: list, 5 items
```

```
    CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceIdiom: str = 'phone'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceSubType: int = 568
    CacheExtra.oPeik/9e8lQWMszEjbPzng.GraphicsFeatureSetFallbacks: str = 'APPLE3v1:APPLE2:APPLE1:GLES2,0'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.CompatibleDeviceFallback: str = '0'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDisplayGamut: str = 'sRGB'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceProductDescription: str = 'iPhone SE'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDynamicDisplayMode: str = '0'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.GraphicsFeatureSetClass: str = 'APPLE3'
    CacheExtra.oPeik/9e8lQWMszEjbPzng.ArtworkDeviceScaleFactor: int = 2
    CacheExtra.oPeik/9e8lQWMszEjbPzng.DevicePerformanceMemoryClass: int = 2
  CacheExtra.9s45ldrCC1WF+7b6C4H2BA: str = 'iPhone'
  CacheExtra.HXTqT3UXOKuTEklxz+wMAA: bytes, 16 bytes
  CacheExtra.nSo8opze5rFk+EdBoR6tBw: list, 5 items
    CacheExtra.nSo8opze5rFk+EdBoR6tBw[0]: str = 'CU'
    CacheExtra.nSo8opze5rFk+EdBoR6tBw[1]: str = 'IR'
    CacheExtra.nSo8opze5rFk+EdBoR6tBw[2]: str = 'SD'
    CacheExtra.nSo8opze5rFk+EdBoR6tBw[3]: str = 'SY'
    CacheExtra.nSo8opze5rFk+EdBoR6tBw[4]: str = 'KP'
  CacheExtra.AoKnINTLPoKML3ctoP0AZg: dict, 3 entries
    CacheExtra.AoKnINTLPoKML3ctoP0AZg.buffer-compression: bool = False
    CacheExtra.AoKnINTLPoKML3ctoP0AZg.universal-buffer-compression: bool = False
    CacheExtra.AoKnINTLPoKML3ctoP0AZg.media-compression: bool = False
  CacheExtra.5pYKlGnYYBzGvAlIU8RjEQ: str = 's8003'
  CacheExtra.h9jDsbgj7xIVeIQ8S3/X3Q: str = 'iPhone8,4'
  CacheExtra.mumHZHMLEfAuTkkd28fHlQ: str = '#c8caca'
  CacheExtra./YYygAofPDbhrwToVsXdeA: str = 'N69AP'
  CacheExtra.k+KTni1jrwErpcDMEnn3aw: str = '1170'
  CacheExtra.c7fCSBIbX1mFaRoKT5zTIw: str = 'TDK'
CacheVersion: str = '17E262'

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

MG="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Library/C
ches/com.apple.MobileGestalt.plist"

sha256sum "$MG" | tee output/hashes/MobileGestalt_SHA256.txt
sha256sum: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Li
rary/Caches/com.apple.MobileGestalt.plist: Permission denied

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

PTP="working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist"

echo "===== PTP DEVICE INFO METADATA ====="
sudo stat "$PTP"

echo
echo "===== PTP DEVICE INFO CONTENTS ====="
sudo python3 - "$PTP" <<'PY' | tee output/inventory/PTP_Device_Info.txt
import plistlib
import sys
from pprint import pprint

with open(sys.argv[1], "rb") as f:
    data = plistlib.load(f)

pprint(data)
PY
===== PTP DEVICE INFO METADATA =====
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

PTP="working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist"

echo "═══ PTP DEVICE INFO METADATA ═══"
sudo stat "$PTP"

echo
echo "═══ PTP DEVICE INFO CONTENTS ═══"
sudo python3 - "$PTP" <<'PY' | tee output/inventory/PTP_Device_Info.txt
import plistlib
import sys
from pprint import pprint

with open(sys.argv[1], "rb") as f:
    data = plistlib.load(f)

pprint(data)
PY
═══ PTP DEVICE INFO METADATA ═══
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist
  Size: 858             Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 4753667     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:43:10.487919935 -0500
Modify: 2020-04-02 23:15:37.000000000 -0400
Change: 2026-11-04 03:43:10.487919935 -0500
 Birth: 2026-11-04 03:43:10.487919935 -0500

═══ PTP DEVICE INFO CONTENTS ═══
```

```
PY
═══ PTP DEVICE INFO METADATA ═══
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/System/Library/ApplePTP/PTPDeviceInfoDataset.plist
  Size: 858             Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 4753667     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:43:10.487919935 -0500
Modify: 2020-04-02 23:15:37.000000000 -0400
Change: 2026-11-04 03:43:10.487919935 -0500
 Birth: 2026-11-04 03:43:10.487919935 -0500

═══ PTP DEVICE INFO CONTENTS ═══
{'CaptureFormats': [12289, 14337, 14347, 14349],
 'DevicePropertiesSupported': [54018.0, 55815],
 'DeviceVersion': '3.0',
 'EventsSupported': [16385,
                     16386,
                     16387,
                     16388,
                     16389,
                     16390,
                     50948,
                     50949,
                     50950,
                     50951,
                     50952],
 'FunctionalMode': 0,
 'ImageFormats': [12288,
                  12289,
                  12290,
                  12295,
                  12296,
```

```
                      50952],
 'FunctionalMode': 0,
 'ImageFormats': [12288,
                  12289,
                  12290,
                  12295,
                  12296,
                  12297,
                  12298,
                  12299,
                  12300,
                  12301,
                  14337,
                  14347,
                  14349],
 'Manufacturer': 'Apple Inc.',
 'Model': 'Apple Mobile Device',
 'OperationsSupported': [4097,
                         4098,
                         4099,
                         4100,
                         4101,
                         4102,
                         4103,
                         4104,
                         4105,
                         4106,
                         4107,
                         4116,
                         4117,
                         4123,
                         4131,
```

```
Session  Actions  Edit  View  Help
                        4106,
                        4107,
                        4116,
                        4117,
                        4123,
                        4131,
                        36865,
                        36866,
                        36867,
                        36868,
                        36869,
                        36870,
                        36871,
                        36872,
                        36873,
                        36874,
                        36875,
                        36876,
                        38657,
                        38658,
                        38913,
                        38914,
                        38915,
                        38917],
 'SerialNumber': 'BC92D16C-E3EC-5532-BEAC-2B26FECE2233',
 'StandardVersion': 110,
 'VendorExtensionDesc': 'Device has no vendor extensions',
 'VendorExtensionID': 6,
 'VendorExtensionVersion': 100}

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case2

MG="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Library/Caches/com.apple.MobileGestalt.plist"

sudo sha256sum "$MG" | tee output/hashes/MobileGestalt_SHA256.txt
[sudo] password for kali:
38dbec6334f6b7f395bfc4e3d11e0af94f26e0d132d6f16a12f167603d110051  working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Shared/SystemGroup/systemgroup.com.apple.mobilegestaltcache/Library/Caches/com.apple.MobileGestalt.plist

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

ROOT="working/filesystem/Volumes/JOSH/NoTar-13-4-1"

echo "═══ ACCOUNTS3 DATABASES ═══"
sudo find "$ROOT" -type f -name "Accounts3*" -print

echo
echo "═══ ACCOUNTS DATABASE FILES ═══"
sudo find "$ROOT" -type f \
\( -iname "*account*.sqlite" -o -iname "*account*.db" -o -iname "*account*.sqlite3" \) \
-print | head -80
═══ ACCOUNTS3 DATABASES ═══
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/VerifiedBackup/Accounts3.sqlite
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite-shm

═══ ACCOUNTS DATABASE FILES ═══
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/VerifiedBackup/Accounts3.sqlite
working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/com.apple.iTunesCloud/ICDelegateAccountStore.db

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

SRC="working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts"
DEST="working/accounts3"

mkdir -p "$DEST"

echo "===== SOURCE FILE METADATA ====="
sudo stat "$SRC/Accounts3.sqlite" \
          "$SRC/Accounts3.sqlite-wal" \
          "$SRC/Accounts3.sqlite-shm" \
| tee output/inventory/Accounts3_Source_Metadata.txt

echo
echo "===== COPY DATABASE AND SIDECARS ====="
sudo cp -a "$SRC/Accounts3.sqlite" \
           "$SRC/Accounts3.sqlite-wal" \
           "$SRC/Accounts3.sqlite-shm" \
           "$DEST/"

echo
echo "===== SET WORKING-COPY OWNERSHIP ====="
sudo chown kali:kali "$DEST/Accounts3.sqlite" \
                     "$DEST/Accounts3.sqlite-wal" \
                     "$DEST/Accounts3.sqlite-shm"

echo
echo "===== WORKING COPY FILES ====="
ls -l "$DEST"
===== SOURCE FILE METADATA =====
```

```
===== SOURCE FILE METADATA =====
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite
  Size: 241664          Blocks: 472        IO Block: 4096   regular file
Device: 8,1     Inode: 4754347     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:46:41.726732305 -0500
Modify: 2020-04-16 09:34:05.000000000 -0400
Change: 2026-11-04 03:46:41.958720471 -0500
 Birth: 2026-11-04 03:46:41.726732305 -0500
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite-wal
  Size: 420272          Blocks: 824        IO Block: 4096   regular file
Device: 8,1     Inode: 4754349     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:46:41.958720471 -0500
Modify: 2020-04-16 12:01:02.000000000 -0400
Change: 2026-11-04 03:46:41.962720267 -0500
 Birth: 2026-11-04 03:46:41.958720471 -0500
  File: working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/mobile/Library/Accounts/Accounts3.sqlite-shm
  Size: 32768           Blocks: 64         IO Block: 4096   regular file
Device: 8,1     Inode: 4754348     Links: 1
Access: (0644/-rw-r--r--)  Uid: (   99/ UNKNOWN)   Gid: (   99/ UNKNOWN)
Access: 2026-11-04 03:46:41.958720471 -0500
Modify: 2020-04-16 11:52:57.000000000 -0400
Change: 2026-11-04 03:46:41.958720471 -0500
 Birth: 2026-11-04 03:46:41.958720471 -0500

===== COPY DATABASE AND SIDECARS =====

===== SET WORKING-COPY OWNERSHIP =====

===== WORKING COPY FILES =====
total 680
```

```
┌──(kali㉿kali)-[~]
└─$ echo "═══ SYSTEM INFORMATION ═══"
date -u
uname -a

echo
echo "═══ DISK SPACE ═══"
df -h /

echo
echo "═══ HOME DIRECTORY ═══"
df -h ~
═══ SYSTEM INFORMATION ═══
Tue Nov  3 08:49:52 PM UTC 2026
Linux kali 6.12.38+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.38-1kali1 (2025-08-12) x86_64 GNU/Linux

═══ DISK SPACE ═══
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        79G   66G  9.2G  88% /

═══ HOME DIRECTORY ═══
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        79G   66G  9.2G  88% /

┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ ~/SBT-DF205-Case2
zsh: no such file or directory: /home/kali/SBT-DF205-Case2

┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF205-Case2

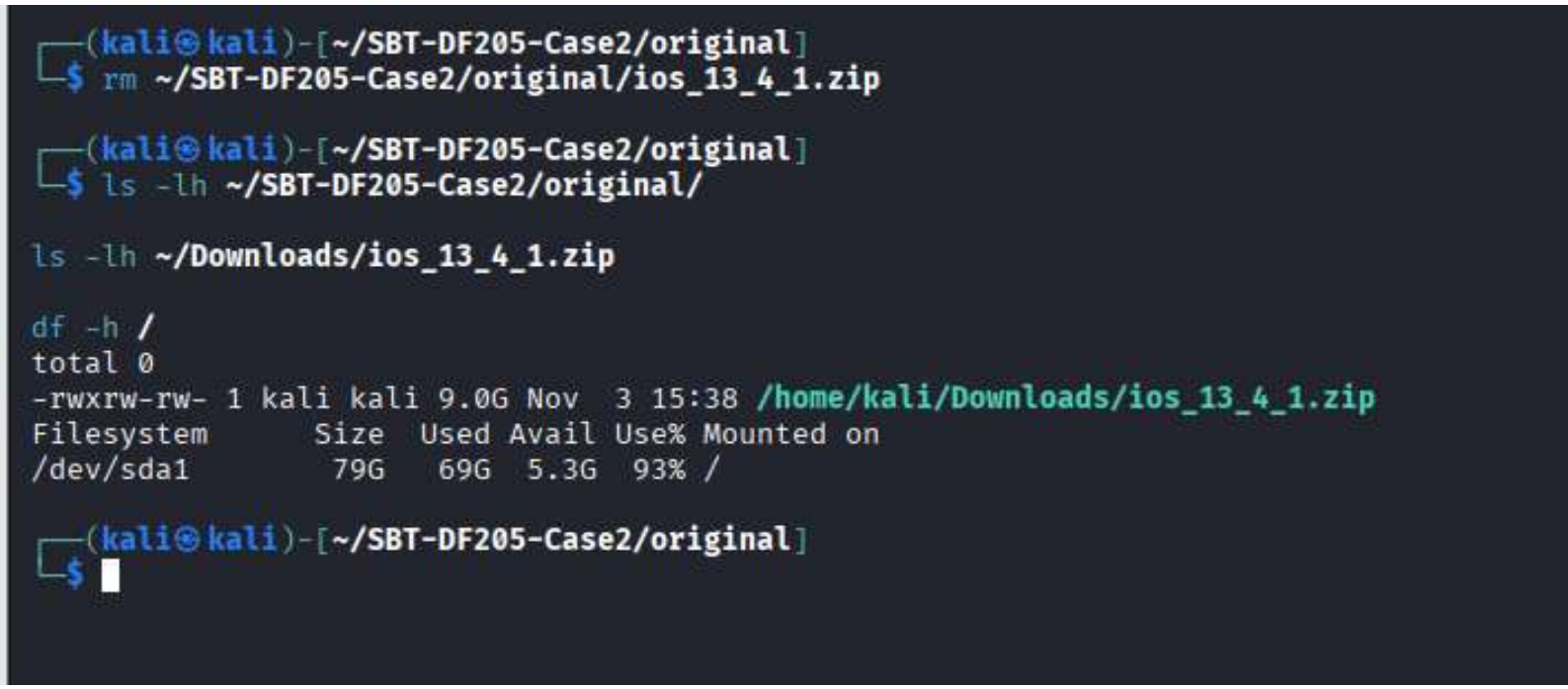
cd ~/SBT-DF205-Case2

mkdir -p original
mkdir -p working
mkdir -p output
mkdir -p output/hashes
mkdir -p output/inventory
mkdir -p output/messages
mkdir -p output/safari
mkdir -p output/media
mkdir -p output/knowledgec
mkdir -p output/health
mkdir -p output/location
mkdir -p output/timeline
mkdir -p screenshots
mkdir -p notes
mkdir -p report
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ tree -d ~/SBT-DF205-Case2
/home/kali/SBT-DF205-Case2
├── notes
├── original
├── output
│   ├── hashes
│   ├── health
│   ├── inventory
│   ├── knowledgec
│   ├── location
│   ├── media
│   ├── messages
│   ├── safari
│   └── timeline
├── report
├── screenshots
└── working

16 directories

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```


┌──(kali㉿kali)-[~/SBT-DF205-Case2/original]
└─$ rm ~/SBT-DF205-Case2/original/ios_13_4_1.zip
┌──(kali㉿kali)-[~/SBT-DF205-Case2/original]
└─$ ls -lh ~/SBT-DF205-Case2/original/
ls -lh ~/Downloads/ios_13_4_1.zip
df -h /
total 0
-rwxrw-rw- 1 kali kali 9.0G Nov 3 15:38 /home/kali/Downloads/ios_13_4_1.zip
Filesystem Size Used Avail Use% Mounted on
/dev/sda1 79G 69G 5.3G 93% /
┌──(kali㉿kali)-[~/SBT-DF205-Case2/original]
└─$

```
┌──(kali㉿kali)-[~]
└─$ sudo parted /dev/sda
GNU Parted 3.6
Using /dev/sda
Welcome to GNU Parted! Type 'help' to view a list of commands.
(parted) resizepart 1 100%
Warning: Partition /dev/sda1 is being used. Are you sure you want to continue?
Yes/No? yes
(parted) quit
Information: You may need to update /etc/fstab.


┌──(kali㉿kali)-[~]
└─$ sudo resize2fs /dev/sda1
resize2fs 1.47.2 (1-Jan-2025)
Filesystem at /dev/sda1 is mounted on /; on-line resizing required
old_desc_blocks = 11, new_desc_blocks = 23
The filesystem on /dev/sda1 is now 47211776 (4k) blocks long.


┌──(kali㉿kali)-[~]
└─$ echo "===== PARTITION LAYOUT ====="
lsblk

echo
echo "===== FILESYSTEM CAPACITY ====="
df -h /

echo
echo "===== PARTITION BOUNDARIES ====="
sudo parted /dev/sda print free
```

```
┌──(kali㉿kali)-[~]
└─$ echo "═══ PARTITION LAYOUT ═══"
lsblk

echo
echo "═══ FILESYSTEM CAPACITY ═══"
df -h /

echo
echo "═══ PARTITION BOUNDARIES ═══"
sudo parted /dev/sda print free
═══ PARTITION LAYOUT ═══
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 180.1G  0 disk
└─sda1   8:1    0 180.1G  0 part /

═══ FILESYSTEM CAPACITY ═══
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G   69G  100G  41% /

═══ PARTITION BOUNDARIES ═══
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sda: 193GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
        1024B   1049kB  1048kB           Free Space
 1      1049kB  193GB   193GB   primary  ext4         boot
```

```
lsblk

echo
echo "==== FILESYSTEM CAPACITY ===="
df -h /

echo
echo "==== PARTITION BOUNDARIES ===="
sudo parted /dev/sda print free
==== PARTITION LAYOUT ====
NAME   MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda      8:0    0 180.1G  0 disk
└─sda1   8:1    0 180.1G  0 part /

==== FILESYSTEM CAPACITY ====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G   69G  100G  41% /

==== PARTITION BOUNDARIES ====
Model: VMware, VMware Virtual S (scsi)
Disk /dev/sda: 193GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

Number  Start   End     Size    Type     File system  Flags
        1024B   1049kB  1048kB           Free Space
 1      1049kB  193GB   193GB   primary  ext4         boot


┌──(kali㉿kali)-[~]
└─$
```

```
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case2

cp ~/Downloads/ios_13_4_1.zip original/

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ ls -lh original/ios_13_4_1.zip

stat -c '%n %s bytes' original/ios_13_4_1.zip
-rwxrw-r-- 1 kali kali 9.0G Nov  3 17:01 original/ios_13_4_1.zip
original/ios_13_4_1.zip 9629286492 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ sha256sum ~/Downloads/ios_13_4_1.zip
761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551  /home/kali/Downloads/ios_13_4_1.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ sha256sum original/ios_13_4_1.zip
761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551  original/ios_13_4_1.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ sha256sum original/ios_13_4_1.zip \
| tee output/hashes/ZIP_SHA256.txt
761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551  original/ios_13_4_1.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cat output/hashes/ZIP_SHA256.txt
761854a669c38da6ac0a908afe77763d54ccb83044ff123350890d408c154551  original/ios_13_4_1.zip

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ unzip -l original/ios_13_4_1.zip \
| tee output/inventory/ZIP_Contents.txt
Archive:  original/ios_13_4_1.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
        0  2020-04-19 19:52   iOS 13.4.1 Extraction/
  8916857  2020-04-19 19:34   iOS 13.4.1 Extraction/iOS13-4-1-ImageCreation.pdf
        0  2020-04-16 13:03   iOS 13.4.1 Extraction/iTunes Backup/
420400511  2020-04-16 11:50   iOS 13.4.1 Extraction/iTunes Backup/c623fbd7e91b041e07a68f8523f53a35973e475d.zip
        0  2020-04-16 13:45   iOS 13.4.1 Extraction/Extraction Logs/
     2289  2020-04-16 13:45   iOS 13.4.1 Extraction/Extraction Logs/activity_log.txt
     1784  2020-04-16 13:45   iOS 13.4.1 Extraction/Extraction Logs/image_info.txt
        0  2020-04-16 11:48   iOS 13.4.1 Extraction/Sysdiagnose Logs/
235760752  2020-04-16 11:45   iOS 13.4.1 Extraction/Sysdiagnose Logs/sysdiagnose_2020.04.16_11-44-04-0400_iPhone-OS_iPhone_17E262.tar.gz
        0  2020-04-19 19:52   iOS 13.4.1 Extraction/Extraction/
17223813120  2020-04-18 22:20   iOS 13.4.1 Extraction/Extraction/13-4-1.tar
---------                     -------
17888895313                   11 files

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ grep -F "Extraction/13-4-1.tar" \
output/inventory/ZIP_Contents.txt
17223813120  2020-04-18 22:20   iOS 13.4.1 Extraction/Extraction/13-4-1.tar

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

unzip -t original/ios_13_4_1.zip \
| tee output/inventory/ZIP_Integrity_Test.txt
Archive:  original/ios_13_4_1.zip
    testing: iOS 13.4.1 Extraction/   OK
    testing: iOS 13.4.1 Extraction/iOS13-4-1-ImageCreation.pdf   OK
    testing: iOS 13.4.1 Extraction/iTunes Backup/   OK
    testing: iOS 13.4.1 Extraction/iTunes Backup/c623fbd7e91b041e07a68f8523f53a35973e475d.zip   OK
    testing: iOS 13.4.1 Extraction/Extraction Logs/   OK
    testing: iOS 13.4.1 Extraction/Extraction Logs/activity_log.txt   OK
    testing: iOS 13.4.1 Extraction/Extraction Logs/image_info.txt   OK
    testing: iOS 13.4.1 Extraction/Sysdiagnose Logs/   OK
    testing: iOS 13.4.1 Extraction/Sysdiagnose Logs/sysdiagnose_2020.04.16_11-44-04-0400_iPhone-OS_iPhone_17E262.tar.gz   OK
    testing: iOS 13.4.1 Extraction/Extraction/   OK
    testing: iOS 13.4.1 Extraction/Extraction/13-4-1.tar   OK
No errors detected in compressed data of original/ios_13_4_1.zip.

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G   78G   91G  47% /

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

unzip -p original/ios_13_4_1.zip \
"iOS 13.4.1 Extraction/Extraction/13-4-1.tar" \
> original/13-4-1.tar

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ ls -lh original/13-4-1.tar

stat -c '%n %s bytes' original/13-4-1.tar
-rw-rw-r-- 1 kali kali 17G Nov  3 17:41 original/13-4-1.tar
original/13-4-1.tar 17223813120 bytes

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ sha256sum original/13-4-1.tar \
| tee output/hashes/TAR_SHA256.txt
991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb  original/13-4-1.tar

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cat output/hashes/TAR_SHA256.txt
991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb  original/13-4-1.tar

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ tar -tvf original/13-4-1.tar \
| head -40 \
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ tar -tvf original/13-4-1.tar \
| head -40 \
| tee output/inventory/TAR_Contents_First40.txt
drwxr-xr-x _unknown/_unknown 0 2020-04-18 16:03 Volumes/JOSH/NoTar-13-4-1/
drwx------ _unknown/_unknown 0 2020-04-02 23:14 Volumes/JOSH/NoTar-13-4-1/.ba/
---------- _unknown/_unknown 0 2020-04-02 23:14 Volumes/JOSH/NoTar-13-4-1/.file
drwx------ _unknown/_unknown 0 2020-04-02 23:14 Volumes/JOSH/NoTar-13-4-1/.mb/
-rw-r--r-- _unknown/_unknown 0 2020-04-02 23:18 Volumes/JOSH/NoTar-13-4-1/.Trashes
drwxrwxr-x _unknown/_unknown 0 2020-04-18 18:59 Volumes/JOSH/NoTar-13-4-1/Applications/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:10 Volumes/JOSH/NoTar-13-4-1/bin/
drwxrwxr-t _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/cores/
dr-xr-xr-x _unknown/_unknown 0 2020-04-16 11:52 Volumes/JOSH/NoTar-13-4-1/dev/
drwxrwxr-x _unknown/_unknown 0 2020-04-02 23:14 Volumes/JOSH/NoTar-13-4-1/Developer/
lrwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/etc → private/etc
drwxrwxr-x _unknown/_unknown 0 2020-04-13 15:10 Volumes/JOSH/NoTar-13-4-1/Library/
drwxr-xr-x _unknown/_unknown 0 2020-04-18 22:07 Volumes/JOSH/NoTar-13-4-1/private/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/sbin/
drwxr-xr-x _unknown/_unknown 0 2020-04-02 23:14 Volumes/JOSH/NoTar-13-4-1/System/
lrwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/tmp → private/var/tmp
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:14 Volumes/JOSH/NoTar-13-4-1/usr/
lrwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/var → private/var
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/bin/
drwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/include/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/lib/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/libexec/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:14 Volumes/JOSH/NoTar-13-4-1/usr/local/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/sbin/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/share/
drwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/
```

```
lrwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/tmp → private/var/tmp
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:14 Volumes/JOSH/NoTar-13-4-1/usr/
lrwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/var → private/var
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/bin/
drwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/include/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/lib/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/libexec/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:14 Volumes/JOSH/NoTar-13-4-1/usr/local/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/sbin/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/share/
drwxr-xr-x _unknown/_unknown 0 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/
drwxr-xr-x _unknown/_unknown 0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/
-rw-r--r-- _unknown/_unknown 26409624 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/Mav10.Release.bbfw
drwxr-xr-x _unknown/_unknown        0 2020-04-13 15:11 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/ramdisk/
-rw-r--r-- _unknown/_unknown 104579099 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/ramdisk/arm64SURamDisk.dmg
-rw-r--r-- _unknown/_unknown      6528 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/applelogo@2x~iphone.im4p
-rw-r--r-- _unknown/_unknown      9449 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterycharging0@2x~iphone.
im4p
-rw-r--r-- _unknown/_unknown     34731 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterycharging1@2x~iphone.
im4p
-rw-r--r-- _unknown/_unknown     82272 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batteryfull@2x~iphone.im4p
-rw-r--r-- _unknown/_unknown     43523 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterylow0@2x~iphone.im4p
-rw-r--r-- _unknown/_unknown      2141 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/batterylow1@2x~iphone.im4p
-rw-r--r-- _unknown/_unknown      3676 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/glyphplugin@1136~iphone-lig
htning.im4p
-rw-r--r-- _unknown/_unknown    197755 2020-04-02 23:17 Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/recoverymode@1136~iphone-li
ghtning.im4p

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF205-Case2

mkdir -p working/filesystem

df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G   94G   75G  56% /

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ tar -tvf original/13-4-1.tar \
| awk '
{
    size=$3
    if (size ~ /^[0-9]+$/) total+=size
    count++
}
END {
    printf "Entries listed: %d\n", count
    printf "Total regular-file bytes (approx.): %.0f\n", total
    printf "Total size (GiB): %.2f\n", total/1024/1024/1024
}' | tee output/inventory/TAR_Size_Summary.txt
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
Entries listed: 351466
Total regular-file bytes (approx.): 16971247691
Total size (GiB): 15.81

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ tar -tf original/13-4-1.tar | head -30
Volumes/JOSH/NoTar-13-4-1/
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ tar -tf original/13-4-1.tar | head -30
Volumes/JOSH/NoTar-13-4-1/
Volumes/JOSH/NoTar-13-4-1/.ba/
Volumes/JOSH/NoTar-13-4-1/.file
Volumes/JOSH/NoTar-13-4-1/.mb/
Volumes/JOSH/NoTar-13-4-1/.Trashes
Volumes/JOSH/NoTar-13-4-1/Applications/
Volumes/JOSH/NoTar-13-4-1/bin/
Volumes/JOSH/NoTar-13-4-1/cores/
Volumes/JOSH/NoTar-13-4-1/dev/
Volumes/JOSH/NoTar-13-4-1/Developer/
Volumes/JOSH/NoTar-13-4-1/etc
Volumes/JOSH/NoTar-13-4-1/Library/
Volumes/JOSH/NoTar-13-4-1/private/
Volumes/JOSH/NoTar-13-4-1/sbin/
Volumes/JOSH/NoTar-13-4-1/System/
Volumes/JOSH/NoTar-13-4-1/tmp
Volumes/JOSH/NoTar-13-4-1/usr/
Volumes/JOSH/NoTar-13-4-1/var
Volumes/JOSH/NoTar-13-4-1/usr/bin/
Volumes/JOSH/NoTar-13-4-1/usr/include/
Volumes/JOSH/NoTar-13-4-1/usr/lib/
Volumes/JOSH/NoTar-13-4-1/usr/libexec/
Volumes/JOSH/NoTar-13-4-1/usr/local/
Volumes/JOSH/NoTar-13-4-1/usr/sbin/
Volumes/JOSH/NoTar-13-4-1/usr/share/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/firmware/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/all_flash/
Volumes/JOSH/NoTar-13-4-1/usr/standalone/update/Mav10.Release.bbfw
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

tar --numeric-owner --same-permissions \
-xpf original/13-4-1.tar \
-C working/filesystem
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
tar: Volumes/JOSH/NoTar-13-4-1/private/var/tmp/.req/com.apple.checkra1nd.plist: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/tmp/.opt/com.apple.checkra1nd.plist: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/root/Library/Application Support/DumpPanic/LegacyDataProcessed: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery: Cannot mkdir: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery: Cannot mkdir: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery/Library: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery/Library: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery/Library/Preferences: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery/Library: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/MobileSoftwareUpdate/Hardware/Battery/Library/Preferences/com.apple.batteryhealthdata.plist: Canno
t open: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/icutz: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/localtime: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz: Cannot mkdir: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz_latest: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/zoneinfo: Cannot open: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz: Cannot mkdir: Permission denied
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0/icutz: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0/versions.plist: Cannot open: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0: Cannot mkdir: No such file or directory
tar: Volumes/JOSH/NoTar-13-4-1/private/var/db/timezone/tz/2019c.1.0/zoneinfo: Cannot mkdir: No such file or directory
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ echo "═══ EXTRACTED FILESYSTEM ═══"
ls -ld working/filesystem/Volumes/JOSH/NoTar-13-4-1

echo
echo "═══ DISK SPACE AFTER EXTRACTION ═══"
df -h /

echo
echo "═══ DATABASE FILE COUNT ═══"
find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f \( -name "*.sqlite" -o -name "*.sqlite3" -o -name "*.db" \) \
| wc -l
═══ EXTRACTED FILESYSTEM ═══
drwxr-xr-x 14 kali kali 4096 Apr 18  2020 working/filesystem/Volumes/JOSH/NoTar-13-4-1

═══ DISK SPACE AFTER EXTRACTION ═══
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G  111G   58G  66% /

═══ DATABASE FILE COUNT ═══
find: ‘working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Data/System/B4EB95A3-0436-4B63-B3E8-F30F0DF4F3DB/Documents’: Permis
sion denied
find: ‘working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Data/System/B4EB95A3-0436-4B63-B3E8-F30F0DF4F3DB/Library’: Permissi
on denied
find: ‘working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Data/System/B4EB95A3-0436-4B63-B3E8-F30F0DF4F3DB/tmp’: Permission d
enied
find: ‘working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Data/System/DC9FD525-3574-4A52-82A9-BC99FE60EDA1/Documents’: Permis
sion denied
find: ‘working/filesystem/Volumes/JOSH/NoTar-13-4-1/private/var/containers/Data/System/DC9FD525-3574-4A52-82A9-BC99FE60EDA1/Library’: Permissi
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

sudo tar --numeric-owner --same-permissions \
-xpf original/13-4-1.tar \
-C working/filesystem
[sudo] password for kali:
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ echo "===== ROOT DIRECTORY ====="
ls -ld working/filesystem/Volumes/JOSH/NoTar-13-4-1

echo
echo "===== DISK SPACE ====="
df -h /

echo
echo "===== DATABASE COUNT ====="
sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f \( -name "*.sqlite" -o -name "*.sqlite3" -o -name "*.db" \) \
| wc -l
===== ROOT DIRECTORY =====
drwxr-xr-x 14 99 99 4096 Apr 18  2020 working/filesystem/Volumes/JOSH/NoTar-13-4-1

===== DISK SPACE =====
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G  111G   58G  66% /

===== DATABASE COUNT =====
```

```
-xpf original/13-4-1.tar \
-C working/filesystem
[sudo] password for kali:
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'
tar: Ignoring unknown extended header keyword 'SCHILY.fflags'

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ echo "═══ ROOT DIRECTORY ═══"
ls -ld working/filesystem/Volumes/JOSH/NoTar-13-4-1

echo
echo "═══ DISK SPACE ═══"
df -h /

echo
echo "═══ DATABASE COUNT ═══"
sudo find working/filesystem/Volumes/JOSH/NoTar-13-4-1 \
-type f \( -name "*.sqlite" -o -name "*.sqlite3" -o -name "*.db" \) \
| wc -l
═══ ROOT DIRECTORY ═══
drwxr-xr-x 14 99 99 4096 Apr 18  2020 working/filesystem/Volumes/JOSH/NoTar-13-4-1

═══ DISK SPACE ═══
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1       177G  111G   58G  66% /

═══ DATABASE COUNT ═══
[sudo] password for kali:
858

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$ cd ~/SBT-DF205-Case2

echo "═══ BASELINE TAR HASH ═══"
cat output/hashes/TAR_SHA256.txt

echo
echo "═══ CURRENT TAR HASH ═══"
sha256sum original/13-4-1.tar | tee output/hashes/TAR_SHA256_Recheck.txt
═══ BASELINE TAR HASH ═══
991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb  original/13-4-1.tar

═══ CURRENT TAR HASH ═══
991727d92145a08623e998c008cc2a05f086be722ee2945815d0eb46a78f04fb  original/13-4-1.tar

┌──(kali㉿kali)-[~/SBT-DF205-Case2]
└─$
```